

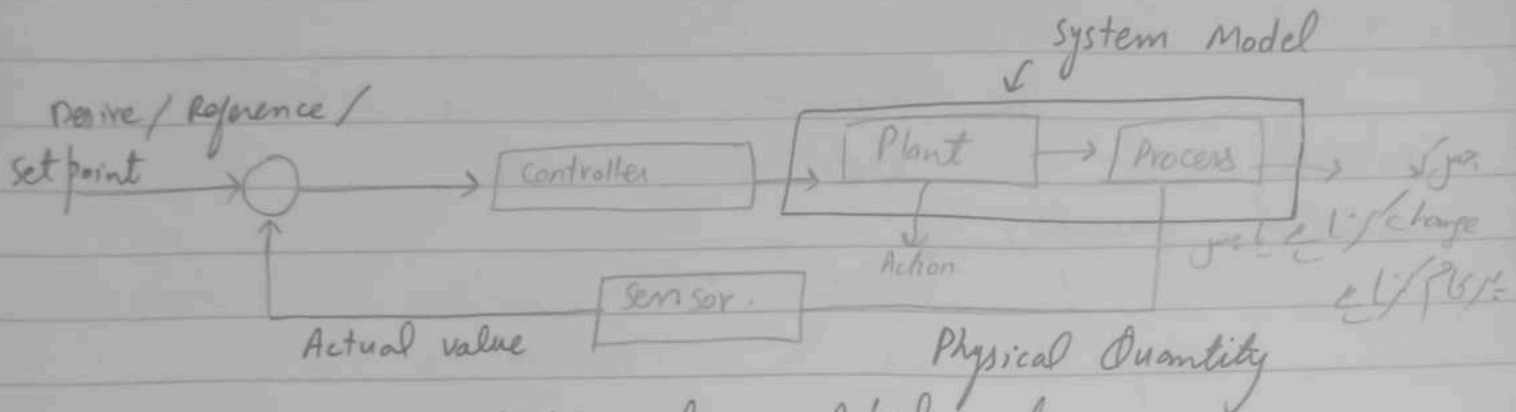
Book :-

Control System Engineering

Day - 1

Define:-

Is a mechanism that we develop to manipulate the input and achieve a desired object.

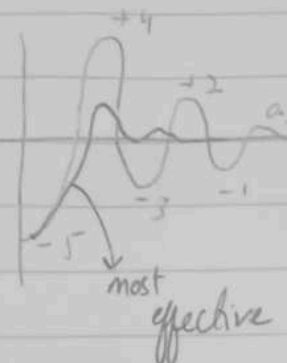


$$e = \text{error} = \text{Desire value} - \text{Actual value}$$

Our main task is to zero the error.

Reference / Desire	Actual Value	error
25°C	30°C	-5°C
25°C	21°C	+4°C
25°C	28°C	-3°C
25°C	23°C	+2°C
25°C	26°C	-1°C
25°C	24.5°C	+0.5°C
25°C	25°C	0

∴ sampling time
1 min

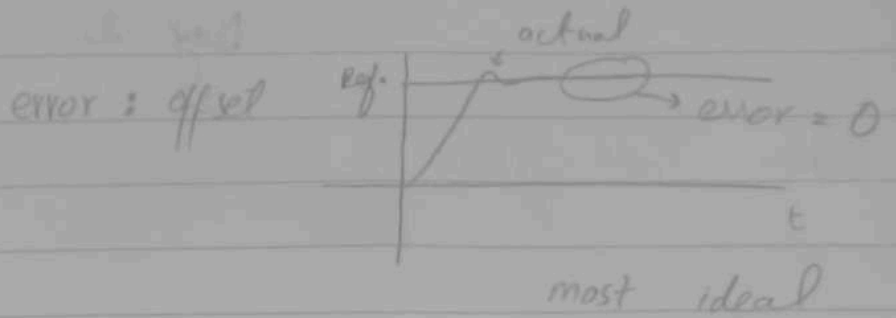


Controlled variable:

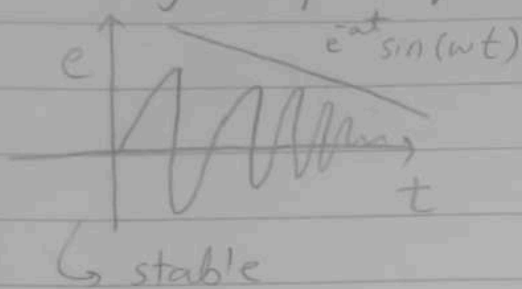
Manipulated variable:

Elevator control (significance of time ^{comes} for response control)

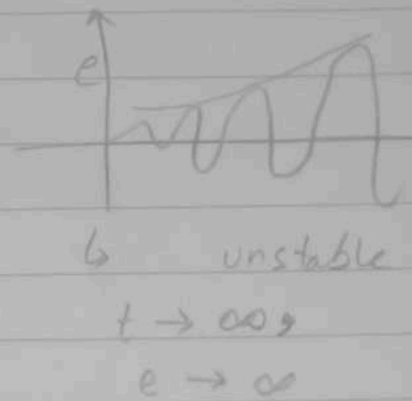
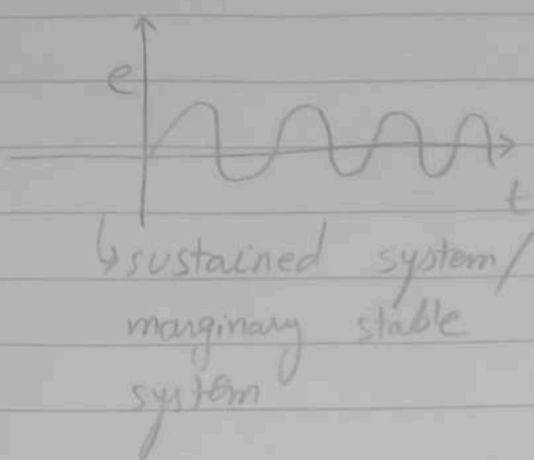
OS $\rightarrow$ overshoot
 t_r = Rise time
 t_s = settling time
 e_{ss} = steady state error



Stability of system :-



$t \rightarrow \infty, e = \text{zero value}$



$t \rightarrow \infty, e \rightarrow \infty$

Objective to make Feedback system :-

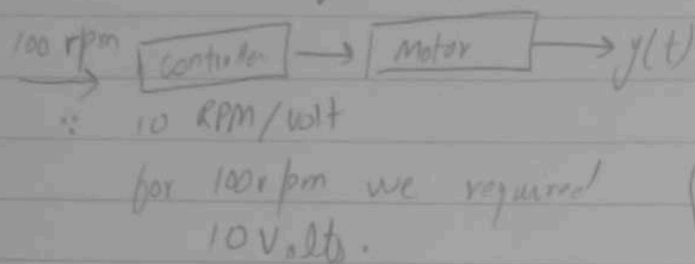
- i) ensure stability
- ii) error minimization
- iii) minimize the overshoot, rise time, settling time, e_{ss}

$\rightarrow$ Transient response and steady state are directly proportional to each other.

$\rightarrow t_r$ will always less than t_s .

Day 2

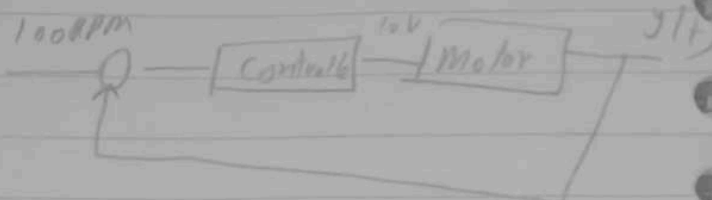
Open loop system



There is no feedback system in this loop.

Closed loop system

" we can't consider the distraction due to the environment. (for both)



- when load is applied the speed is slow so the controller is fed back with the error value.
- Feedback has a ability that it automatically remove its error.

Response :-

⇒ Total Response of a system = Natural Response + Forced Response

↳ depends on the dynamics of the system structure and their components

↳ depends on the type of the input applied.

$R(t) \rightarrow \boxed{h(t)} \rightarrow C(t) \rightarrow$ in time domain

$R(s) \rightarrow \boxed{H(s)} \rightarrow C(s) \rightarrow$ in s domain

Air war → air force system



Mon Tue Wed Thu Fri Sat Sun

Date: / /

Input

2

Output

$$3 \sin(\omega t) + 2 \rightarrow$$

natural response

force response

∵ due to input

is constant in this case

the force response will be also constant cause of input nature.

If there is a dynamic system it represent by the differential equation (d/dt). And it can be controlled.

Step inputs

impulse



step



RAMP



parabolic



In matlab we give these inputs to our system to check is it correct or not.

traits

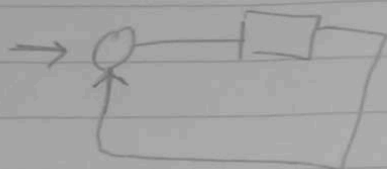
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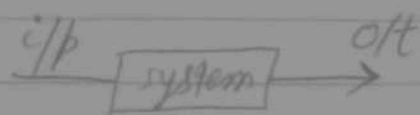
Control system Design Process:

- i) Defining various parts of physical into mathematical term
- ii) Draw a block diagram
- iii) Derive a mathematic model (in freq/ time domain) ← Laplace transfer function
- iv) Reduce the block diagram for simplicity
- v) Analyze the traits of a system (stability, os, tr, ts, tp, etc)
- vi) Design the system

⇒ Our reduced block diagram will be



Revision of Laplace Transfer



$r(t)$ $h(t)$ $c(t)$

$$c(t) = r(t) * h(t) \rightarrow 1 \text{ way}$$

Transfer this in Laplace transform

$$C(s) = R(s) \cdot H(s)$$

$$H(s) = \frac{C(s)}{R(s)} = \frac{\text{o/p}}{\text{i/p}}$$

Laplace Transform

$$f(t) \xLeftrightarrow{\quad} F(s)$$

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

$$\mathcal{L}^{-1}\{F(s)\} = f(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) e^{st} ds$$



Day - 3

B.

Partial Fraction Examples

$$1) F_1(s) = \frac{2}{(s+1)(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$2 = A(s+2) + B(s+1)$$

$$\text{put } s = -2$$

$$2 = A(0) + B(-2+1)$$

$$\boxed{B = -2}$$

$$\text{put } s = -1$$

$$2 = A(-1+2) + B(0)$$

$$\boxed{2 = A}$$

$$F_1(s) = \frac{2}{s+1} - \frac{2}{s+2}$$

Taking Inverse Laplace

$$f_1(t) = 2e^{-t} - 2e^{-2t}$$



$$2) F_2(s) = \frac{s^3 + 2s^2 + 6s + 7}{s^2 + 3s + 2}$$

$$s^2 + 3s + 2$$

$$s - 1$$

$$s^2 + 3s + 2 \overline{) s^3 + 2s^2 + 6s + 7}$$

$$- s^3 + 3s^2 + 2s$$

$$- s^2 + 4s + 7$$

$$+ s^2 - 3s - 2$$

$$7s + 9$$

$$F_2(s) = s - 1 + \frac{7s + 9}{s^2 + 3s + 2}$$

$$F_2(s) = s - 1 + \frac{7s + 9}{(s + 1)(s + 2)}$$

$$\frac{7s + 9}{(s + 1)(s + 2)} = \frac{A}{(s + 1)} + \frac{B}{(s + 2)}$$

$$7s + 9 = A(s + 2) + B(s + 1)$$

$$\text{put } s = -1$$

$$1A = 2$$

$$\text{put } s = -2$$

$$-B = -5$$

$$B = 5$$

$$\frac{7s + 9}{(s + 1)(s + 2)} = \frac{2}{s + 1} + \frac{5}{s + 2}$$

$$\therefore F_2(s) = s - 1 + \frac{2}{s + 1} + \frac{5}{s + 2}$$

Taking inverse Laplace

$$f_2(t) = \delta(t) - \delta(t) + 2e^{-t} + 5e^{-2t}$$

$$\mathcal{L}\{\delta(t)\} = 1$$

$$\mathcal{L}\{\dot{\delta}(t)\} = sF(s) - 0 \quad \leftarrow \text{using differential property}$$

$$= s(1) - 0$$

$$\mathcal{L}\{\delta(t)\} = s$$

If there is "s" and higher power of "s" in numerator then its the differential property.



$$3) F_3(s) = \frac{2}{(s+1)(s+2)^2}$$

$$\frac{2}{(s+1)(s+2)^2} = \frac{A}{(s+1)} + \frac{B}{(s+2)} + \frac{C}{(s+2)^2}$$

$$2 = A(s+2)^2 + B(s+1)(s+2) + C(s+1)$$

put $s = -1$

$$2 = A(-1+2)^2 + B(0) + C(0)$$

$$\boxed{A = 2}$$

put $s = -2$

$$2 = A(0) + B(0) + C(-1)$$

$$\boxed{C = -2}$$

put $s = 0$

$$2 = 4A + 2B + C$$

$$2 = 4(2) + 2B - 2$$

$$2 + 2 = 8 + 2B$$

$$-4 = 2B$$

$$\boxed{B = -2}$$

$$F_3(s) = \frac{2}{(s+1)} - \frac{2}{(s+2)} - \frac{2}{(s+2)^2}$$

Taking inverse Laplace

$$f_3(t) = 2e^{-t} - 2e^{-2t} - 2e^{-2t}t$$

$$\left\{ \begin{array}{l} t \xrightarrow{L} \frac{1}{s^2} \\ e^{-at}t \xrightarrow{L} \frac{1}{(s+a)^2} \\ e^{-2t}t \rightarrow \frac{1}{(s+2)^2} \end{array} \right.$$



~~4) $F_4(s)$~~

$$\frac{4}{(s+2)^3} \Rightarrow 2e^{-2t}t^2$$

$$t^2 = \frac{2}{s^3}$$

$$e^{-2t}t^2 = \frac{2}{(s+2)^3}$$

$$\frac{10}{(s+8)^5}$$

$$t^4 = \frac{24}{s^5}$$

$$e^{-8t} = \frac{24}{(s+8)^5}$$

$$\frac{10}{(s+8)^5} = \frac{1}{2 \cdot 4} e^{-8t} t^4$$

$$4) F_4(s) = \frac{s}{s^2+4}$$

$$= \frac{s}{s^2+(2)^2}$$

Taking inverse Laplace

$$f_4(t) = \cos(2t)$$

$$5) F_5(s) = \frac{10}{s^2+16}$$

$$= 2.5 \left(\frac{4}{s^2+4^2} \right)$$

$$F_5(s) = 2.5 \sin(4t)$$

$$6) F_6(s) = \frac{2}{s^2+2s+5} \quad (\text{for complex root we use this method})$$

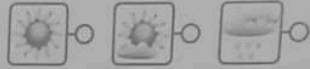
$$= \frac{2}{(s^2+2s+1)+4}$$

$$= \frac{2}{(s+1)^2+4}$$

$$= e^{-t} \sin(2t)$$

$$\frac{2}{s^2+(2)^2} \xrightarrow{L} \sin(2t)$$

$$\frac{2}{(s+1)^2+4} \xrightarrow{L} e^{-t} \sin(2t)$$



$$7) F_7(s) = \frac{s+2}{s^2+4s+5}$$

$$= \frac{s+2}{(s^2+4s+4)+1}$$

$$= \frac{s+2}{(s+2)^2+1}$$

$$\boxed{\frac{s+2}{s^2+4s+5} = e^{-2t} \cos t}$$

$$8) F_8(s) = \frac{2}{s^2+8}$$

$$= \frac{2}{(s^2) + (2\sqrt{2})^2}$$

$$= \frac{\sqrt{2} \sin(2\sqrt{2}t)}{2}$$

$$9) F_9(s) = \frac{s+5}{s^2+4s+20}$$

$$= \frac{s+5}{(s^2+4s+4)+16}$$

$$= \frac{s+5}{(s+2)^2 + (4)^2}$$

$$= \frac{s+2+3}{(s+2)^2 + (4)^2}$$

$$= \frac{s+2}{(s+2)^2 + (4)^2} + \frac{3}{(s+2)^2 + (4)^2} \times \frac{4}{4}$$

Taking inverse Laplace

$$= e^{-2t} \cos(4t) + \frac{3}{4} e^{-2t} \sin(4t) \text{ answer.}$$

$$10) F(s) = \frac{3}{s(s^2+2s+5)}$$

$$\frac{3}{s(s^2+2s+5)} = \frac{A}{s} + \frac{Bs+C}{s^2+2s+5}$$

$$3 = A(s^2+2s+5) + (Bs+C)s$$

put $s=0$

$$3 = 5A$$

$$\boxed{A = \frac{3}{5}}$$

put $s=1$

$$3 = 8A + B + C$$

$$3 = \frac{24}{5} + B + C$$

$$B+C = \frac{-9}{5}$$

put $s=-1$

$$3 = 4A + B - C$$

$$3 = \frac{12}{5} + B - C$$

$$B - C = \frac{3}{5}$$

$$A = \frac{3}{5}$$

$$B = -\frac{3}{5}$$

$$C = -\frac{6}{5}$$



$$F(s) = \frac{3}{s} \left(\frac{1}{s} \right) + \frac{-3/s - 6/s}{s^2 + 2s + 5}$$

$$= \frac{3/s}{s} - \frac{3}{s} \left(\frac{s+2}{s^2 + 2s + 5} \right)$$

$$= \frac{3/s}{s} - \frac{3}{s} \left(\frac{s+2}{(s+1)^2 + (2)^2} \right)$$

$$= \frac{-3/s}{s} - \frac{3}{s} \left(\frac{s+1}{(s+1)^2 + (2)^2} + \frac{1 \times 2}{2((s+1)^2 + (2)^2)} \right)$$

Taking Laplace inverse

$$= \frac{-3}{s} u(t) - \frac{3}{s} \left(e^{-t} \cos 2t + \frac{1}{2} e^{-t} \sin 2t \right)$$



Remainder of example 10

$$f(t) = \frac{3}{5} - \frac{3}{5} \left[e^{-t} \cos 2t + 0.5 e^{-t} \sin 2t \right]$$

$$= \frac{3}{5} - \frac{3}{5} e^{-t} (\cos 2t + 0.5 \sin 2t)$$

$$= \frac{3}{5} - \frac{3}{5} e^{-t} [M \cos(2t - \phi)] \quad \text{--- (A)}$$

$$M \cos \phi = 1$$

$$M \sin \phi = 0.5$$

$$\tan \phi = \frac{\sin \phi}{\cos \phi}$$

$$\tan \phi = \frac{0.5}{1}$$

$$\phi = \tan^{-1}(0.5)$$

$$\boxed{\phi = 26.565^\circ}$$

$$\boxed{M = 1.118}$$

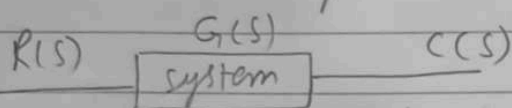
Put in eq (A)

$$= \frac{3}{5} - \frac{3}{5} e^{-t} [1.118 \cos(2t - 26.565^\circ)]$$

$$= 0.6 - 0.6708 e^{-t} \cos(2t - 26.6^\circ)$$

Transfer Function:-

A mathematical representation of the relationship between the input & output expressed as the ratio of the Laplace transform of the output to the Laplace transform of the input



$$G(s) = \frac{C(s)}{R(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0} \Rightarrow \frac{\text{effect}}{\text{cause}}$$

1- $\dot{c}(t) + 2c(t) = 1(t)$

initial condition $c(0) = 0$, $\dot{c}(0) = 0$

Find $G(s) = \frac{c(s)}{1(s)}$

Taking Laplace on b/s

$$sC(s) - \overset{0}{c(0)} + 2C(s) = R(s)$$

Note

"inverse Laplace of the transfer function is called 'Impulse response'."

$$sC(s) + 2C(s) = R(s) \Rightarrow (s+2)C(s) = R(s)$$

$$\frac{C(s)}{R(s)} = \frac{1}{s+2}$$

$$G(s) = \frac{1}{s+2}$$

$$G(s) = \frac{1}{s+2}$$

$\downarrow$

$$g(t) = e^{-2t} \text{ (impulse response)}$$



② $\ddot{c}(t) + 2\dot{c}(t) = u(t)$

$$s^2 C(s) + s \cancel{c(0)} - \dot{c}(0) + s C(s) - \cancel{c(0)} = R(s)$$

$$s^2 C(s) + s C(s)$$

$$C(s) (s^2 + s) = R(s)$$

$$\frac{R(s)}{C(s)} = \frac{1}{s^2 + s}$$

$$G(s) = \frac{1}{s^2 + s} \Rightarrow G(s) = \frac{1}{s(s+1)} \Rightarrow \frac{0.5}{s} - \frac{0.5}{s+1}$$

$L^{-1} \downarrow$

$$g(t) = 0.5u(t) + 0.5e^{-t}u(t)$$

$$g(t) = 0.5 - 0.5e^{-t}$$

$$\ddot{c} + 3\dot{c} + 7c = \ddot{x} + 4\dot{x} + 3x$$

③ $C(s) [s^3 + 3s^2 + 7s + 5] = R(s) [s^2 + 4s + 3]$

$$\frac{C(s)}{R(s)} = \frac{s^2 + 4s + 3}{s^3 + 3s^2 + 7s + 5}$$

$$G_1(s) = \frac{(s+1)(s+3)}{(s+1)(s^2 + 2s + 5)}$$

$$G_1(s) = \frac{s+3}{s^2 + 2s + 5}$$

$$G_1(s) = \frac{s+3}{(s^2 + 2s + 1) + 4}$$

$$G(s) = \frac{(s+1)+3}{(s+1)^2 + (2)^2}$$

$$G(s) = \frac{(s+1)}{(s+1)^2 + (2)^2} + \frac{2}{(s+1)^2 + (2)^2}$$

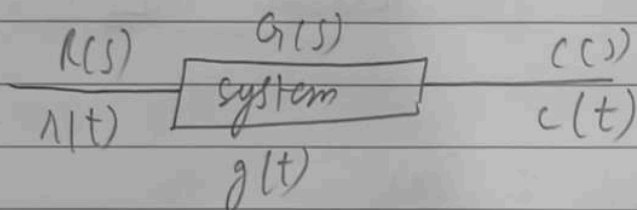
$$g(t) = e^{-t} \cos(2t) + e^{-t} \sin(2t)$$

$$4 - \frac{C(s)}{R(s)} = \frac{2s+1}{s^2+6s+2}$$

$$C(s)(s^2+6s+2) = R(s)(2s+1)$$

$$\begin{aligned} \dot{C}(t) + \ddot{C}(t) + C(t) &= 2\dot{R}(t) + R(t) \\ \Rightarrow \boxed{\ddot{C}(t) + \dot{C}(t) + C(t) = 2\dot{R}(t) + R(t)} \end{aligned}$$

$$5 - \frac{C(s)}{R(s)} = \frac{s}{(s+4)(s+8)}$$



find unit ramp response of this system.
 $r(t) = t \Rightarrow R(s) = \frac{1}{s^2}$

$$C(s) = \left(\frac{s}{(s+4)(s+8)} \right) R(s)$$

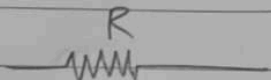
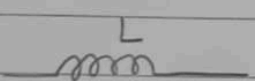


$$C(s) = \left(\frac{s}{(s+4)(s+8)} \right) \left(\frac{1}{s^2} \right)$$

$$C(s) = \frac{1}{s(s+4)(s+8)} \Rightarrow \frac{A}{s} + \frac{B}{s+4} + \frac{C}{s+8}$$

$$C(t) = A + B e^{-4t} + C e^{-8t}$$

Electrical Network Transfer function:

Components	V-I relationship	Laplace	$\frac{V(s)}{I(s)}$ ← Cause Effect Impedance =
	$i(t) = C \dot{V}(t)$	$V(s) = \frac{1}{Cs} I(s)$	$\frac{1}{Cs}$
	$V(t) = R \cdot i(t)$	$V(s) = R \cdot I(s)$	R
	$V(t) = L \dot{i}(t)$	$V(s) = (Ls) I(s)$	Ls

$$i(t) = C \dot{V}(t)$$

$$I(s) = C \cdot s V(s)$$

$$\boxed{V(s) = \frac{1}{Cs} I(s)}$$

$$V(t) = L \dot{i}(t)$$

$$V(s) = L s I(s)$$

Important $\Rightarrow$ How to solve an electric network

$\rightarrow$ Redraw the electronic circuit replacing $V(t) \rightarrow V(s)$
 $i(t) \rightarrow I(s)$

$\rightarrow$ Replace all PASSIVE component (R, L, C) with their S-domain impedance

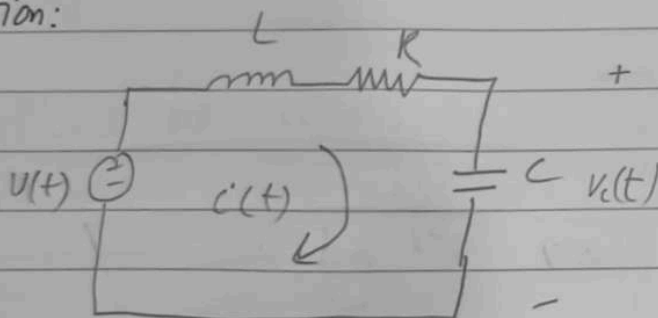
$$R \rightarrow R$$

$$L \rightarrow Ls$$

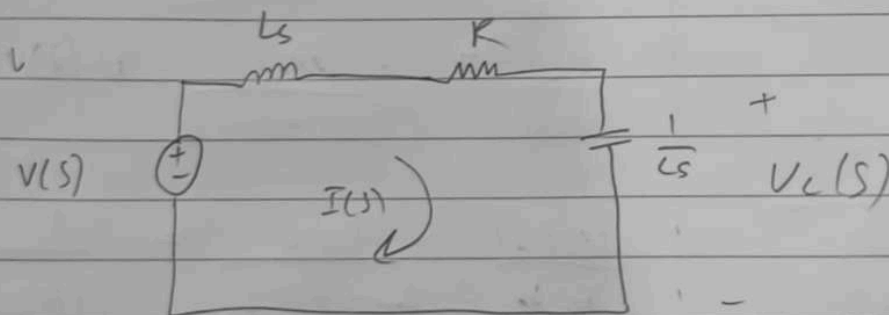
$$C \rightarrow 1/Cs$$

$\rightarrow$ Apply mesh/loop/Node Analysis to solve!

Question:



Find $\frac{I(s)}{V(s)} = ?$



$$Ls(I(s)) + R(I(s)) + \frac{1}{Cs}(I(s)) = V(s)$$



$$I(s) \left(Ls + R + \frac{1}{Cs} \right) = V(s)$$

$$\frac{I(s)}{V(s)} = \frac{1}{Ls + R + \frac{1}{Cs}}$$

$$\frac{I(s)}{V(s)} = \frac{Cs}{LCs^2 + RCs + 1}$$

$$= \frac{(1/L)s}{s^2 + (R/L)s + (1/LC)}$$

$$\frac{V_c(s)}{V(s)} = ?$$

$$V_c(s) = \frac{1}{Cs} (I(s))$$

$$= \frac{(1/Cs)(I(s))}{V(s)}$$

$$= \left(\frac{1}{Cs} \right) \left(\frac{I(s)}{V(s)} \right)$$

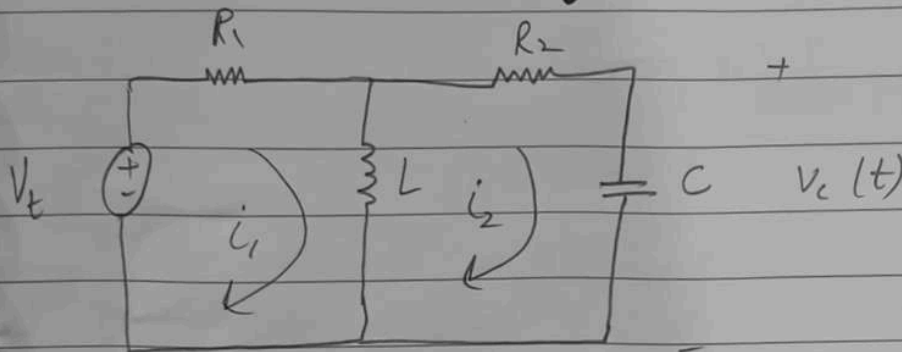
$$= \frac{\left(\frac{1}{Cs} \right) \left(\frac{s}{L} \right)}{s^2 + \left(\frac{R}{L} \right)s + \frac{1}{LC}}$$

$$\frac{V_c(s)}{V(s)} = \frac{1/LC}{s^2 + (R/L)s + 1/LC}$$

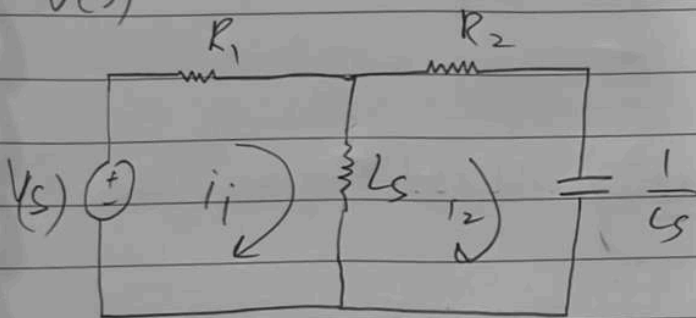
$$V_c(s) = \left(\frac{1/Cs}{1/Cs + R + Ls} \right) V(s)$$

$$\frac{V_c(s)}{V(s)} = \frac{1/Cs}{\frac{1}{Cs} + R + Ls} = \frac{1}{1 + RCs + LCs^2}$$

Day - 5



$$\frac{I_2(s)}{V(s)} = ?$$



Loop 1:

$$R_1 I_1(s) + L_s (I_1(s) - I_2(s)) = V(s)$$

$$(R_1 + L_s) I_1(s) - (L_s) I_2(s) = V(s) \quad \text{--- (1)}$$

Loop 2:

$$L_s (I_2(s) - I_1(s)) + R_2 I_2(s) + \frac{1}{C_s} (I_2(s)) = 0$$

$$-(L_s) I_1(s) + \left(R_2 + \frac{1}{C_s} + L_s \right) I_2(s) = 0 \quad \text{--- (2)}$$



From (2)

$$I_1(s) = \frac{(R_2 + 1/Cs + Ls)}{Ls} I_2(s) \quad \text{--- (3)}$$

Put (3) in (1)

$$(R_1 + Ls) \left(\frac{R_2 + 1/Cs + Ls}{Ls} \right) I_2(s) - (Ls) I_2(s) = V(s)$$

$$I_2(s) \left[\frac{(R_1 + Ls)(R_2 + 1/Cs + Ls)}{Ls} - Ls \right] = V(s)$$

$$I_2(s) \left[\frac{(R_1 + Ls)(R_2 + 1/Cs + Ls) - (Ls)^2}{Ls} \right] = V(s)$$

$$\frac{I_2(s)}{V(s)} = \frac{Ls}{(R_1 + Ls)(R_2 + \frac{1}{Cs} + Ls) - (Ls)^2}$$

$$\frac{V_c(s)}{Vs} = ?$$

$$\begin{aligned} \frac{V_c(s)}{Vs} &= \frac{(1/Cs) I_2(s)}{Vs} = \left(\frac{1}{Cs} \right) \left(\frac{I_2(s)}{V(s)} \right) \\ &= \frac{1}{Cs} \left[\frac{Ls}{(R_1 + Ls) \left(R_2 + \frac{1}{Cs} + Ls \right) - (Ls)^2} \right] \end{aligned}$$

Equation making technique

Taking example of previous question circuit

Loop 1:

$$\left(\begin{array}{l} \text{Sum of all impedances} \\ \text{in the primary loop} \end{array} \right) I_1 - \left(\begin{array}{l} \text{Sum of impedances} \\ \text{shared with the 2nd loop} \end{array} \right) I_2 = V_1$$

Loop 2:

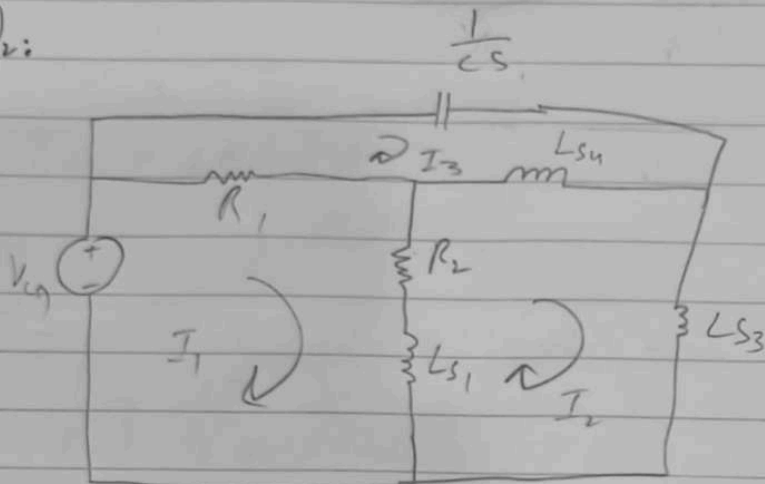
$$\left(\begin{array}{l} \text{Sum of all the impedances} \\ \text{in loop 2} \end{array} \right) I_2 - \left(\begin{array}{l} \text{Sum of all the impedances} \\ \text{shared with loop 1} \end{array} \right) I_1 = 0$$

Example:

$$(R_1 + L_s) I_1 - (L_s) I_2 = V_1 \quad \text{--- (1)}$$

$$-(L_s) I_1 + \left(R_2 + \frac{1}{C_s} + L_s \right) I_2 = 0 \quad \text{--- (2)}$$

Q2:



$$(R_1 + R_2 + L_{S1}) I_1 - (R_2 + L_{S1}) I_2 + R_1 I_3 = V(s) \quad (1)$$

$$-(L_{S1} + R_2) I_2 + (R_2 + L_{S1} + L_{S3} + L_{S4}) I_2 - L_{S4} I_3 = 0 \quad (2)$$

$$-I_1(R_1) - (L_{S4}) I_2 + (R_1 + L_{S4} + \frac{1}{C_s}) I_3 = 0 \quad (3)$$

Cramer's rule of solving equation.

$$(a_{11})x_1 + (a_{12})x_2 = b_1$$

$$(a_{21})x_1 + (a_{22})x_2 = b_2$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\Delta A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

$$u_1 = \begin{array}{|cc|} \hline b_1 & a_{12} \\ \hline b_2 & a_{22} \\ \hline \end{array}, \quad u_2 = \begin{array}{|cc|} \hline a_{11} & b_1 \\ \hline a_{21} & b_2 \\ \hline \end{array}$$

Δ_A Δ_A

$$\begin{aligned} a_{11}u_1 + a_{12}u_2 + a_{13}u_3 &= b_1 \\ a_{21}u_1 + a_{22}u_2 + a_{23}u_3 &= b_2 \\ a_{31}u_1 + a_{32}u_2 + a_{33}u_3 &= b_3 \end{aligned}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

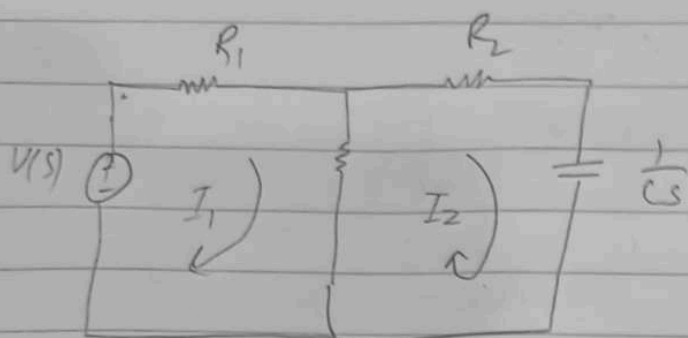
$$\Delta = \begin{array}{|ccc|} \hline a_{11} & a_{12} & a_{13} \\ \hline a_{21} & a_{22} & a_{23} \\ \hline a_{31} & a_{32} & a_{33} \\ \hline \end{array}$$

$$u_1 = \begin{array}{|ccc|} \hline b_1 & a_{12} & a_{13} \\ \hline b_2 & a_{22} & a_{23} \\ \hline b_3 & a_{32} & a_{33} \\ \hline \end{array}, \quad u_2 = \begin{array}{|ccc|} \hline a_{11} & b_1 & a_{13} \\ \hline a_{21} & b_2 & a_{23} \\ \hline a_{31} & b_3 & a_{33} \\ \hline \end{array}$$

Δ Δ

$$u_3 = \begin{array}{|ccc|} \hline a_{11} & a_{12} & b_1 \\ \hline a_{21} & a_{22} & b_2 \\ \hline a_{31} & a_{32} & b_3 \\ \hline \end{array}$$

Δ



$$\begin{aligned} (R_1 + L_s) I_1 - (L_s) I_2 &= V(s) \\ - (L_s) I_1 - (R_2 + L_s + \frac{1}{Cs}) I_2 &= 0 \end{aligned}$$

$$\frac{I_2(s)}{V(s)} = ?$$

$$\begin{bmatrix} R_1 + L_s & -L_s \\ -L_s & R_2 + L_s + \frac{1}{Cs} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \end{bmatrix}$$

$$I_2(s) = \frac{\begin{vmatrix} R_1 + L_s & V(s) \\ -L_s & 0 \end{vmatrix}}{\Delta} \quad \text{--- (1)}$$

$$\Delta = \begin{vmatrix} R_1 + L_s & -L_s \\ -L_s & R_2 + L_s + \frac{1}{Cs} \end{vmatrix}$$

$$= (R_1 + L_s) \left(R_2 + L_s + \frac{1}{Cs} \right) - (L_s)^2$$

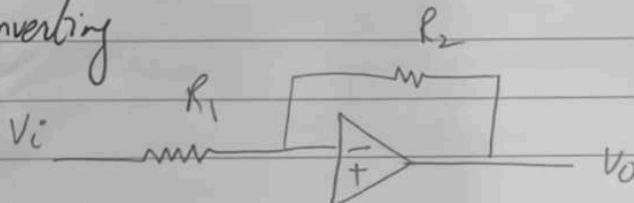
put s in (1)

$$I_2(s) = \frac{(Ls)(V_s)}{(R_1 + Ls)(R_2 + Ls + \frac{1}{Cs}) - (Ls)^2}$$

$$\frac{I_2(s)}{V_s} = \frac{Ls}{(R_1 + Ls)(R_2 + Ls + \frac{1}{Cs}) - (Ls)^2}$$

Transfer function of op-amp circuit

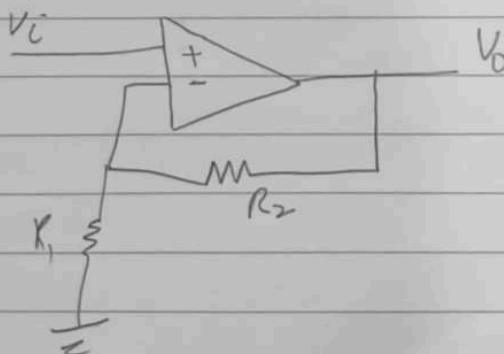
inverting



$$V_o = - \left(\frac{R_2}{R_1} \right) V_i$$

$$\boxed{\frac{V_o}{V_i} = - \frac{R_2}{R_1}}$$

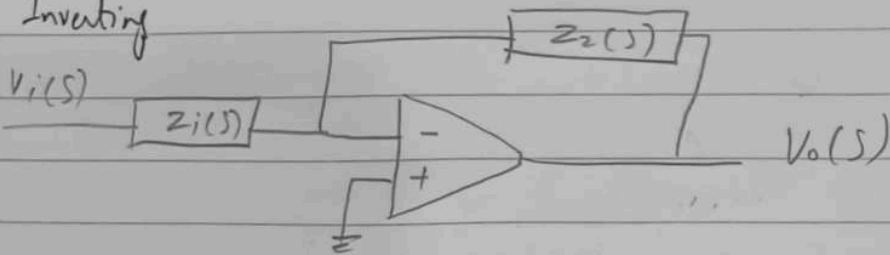
non-inverting



$$V_o = \left(1 + \frac{R_2}{R_1} \right) V_i$$

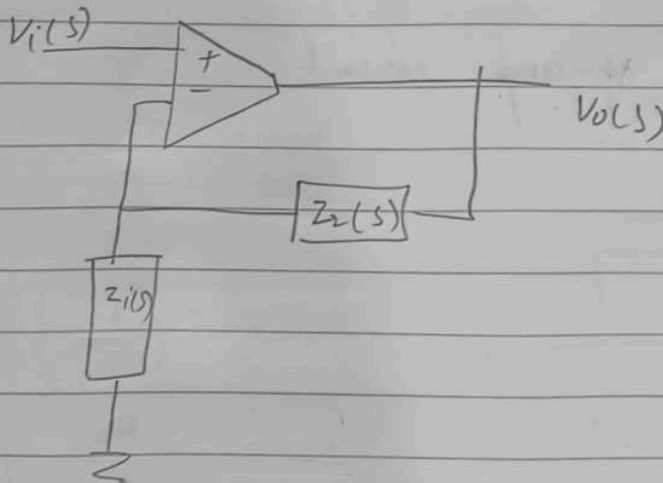
$$\boxed{\frac{V_o}{V_i} = 1 + \frac{R_2}{R_1}}$$

Inverting



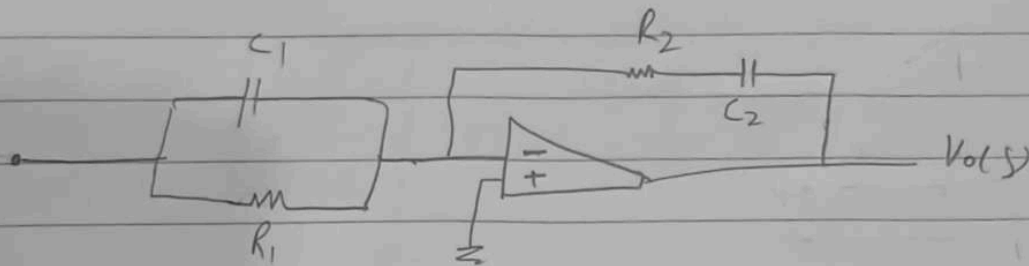
$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

Non-inverting



$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{Z_2(s)}{Z_1(s)}$$

Question 1:

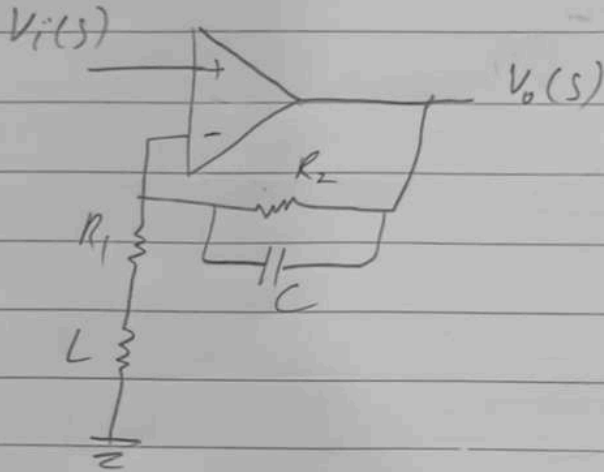


$$\frac{V_o(s)}{V_i(s)} = - \frac{Z_2}{Z_1} = - \frac{R_2 + \frac{1}{C_2 s}}{\left(R_1 \parallel \frac{1}{C_1 s} \right)}$$

$$= \frac{-R_2 + \frac{1}{C_2 s}}{\frac{R_1 \left(\frac{1}{C_1 s} \right)}{R_1 + \frac{1}{C_1 s}}}$$

$$= - \left(\frac{(R_2 + \frac{1}{C_2 s}) (R_1 + \frac{1}{C_1 s})}{R_1 \left(\frac{1}{C_1 s} \right)} \right)$$

(2)



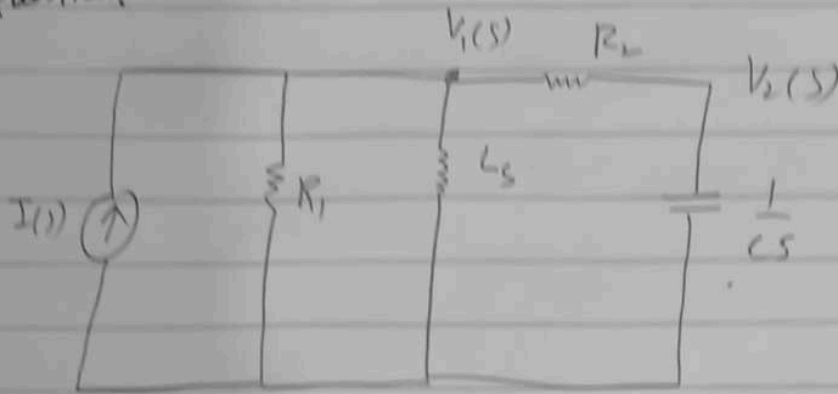
$$\frac{V_o}{V_i} = 1 + \frac{Z_2}{Z_1}$$

$$= 1 + \frac{R_2 \parallel \frac{1}{Cs}}{R_1 + Ls}$$

$$= 1 + \frac{(R_2) \left(\frac{1}{Cs} \right)}{(R_2) + \left(\frac{1}{Cs} \right)} \cdot \frac{1}{R_1 + Ls}$$

$$\frac{V_o}{V_i} = 1 + \frac{R_2 \left(\frac{1}{Cs} \right)}{\left(R_2 + \frac{1}{Cs} \right) (R_1 + Ls)}$$

Question:



$$\frac{V_2(s)}{I(s)} = ?$$

$$(1) \quad \frac{V_1(s) - 0}{R_1} + \frac{V_1 - 0}{Ls} + \frac{V_1 - V_2}{R_2} = I(s)$$

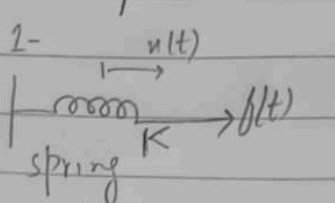
$$\left(\frac{1}{R_1} + \frac{1}{Ls} + \frac{1}{R_2} \right) V_1 - \left(\frac{1}{R_2} \right) V_2 = I(s) \quad \text{--- (1)}$$

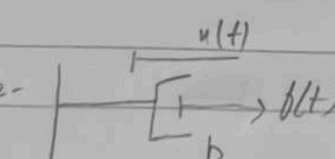
$$(2) \quad \frac{V_2 - V_1}{R_2} + \frac{V_2 - 0}{1/cs} = 0$$

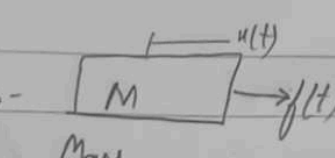
$$\left(-\frac{1}{R_2} \right) V_1 + \left(\frac{1}{R_2} + cs \right) V_2 = 0 \quad \text{--- (2)}$$

Day - 6

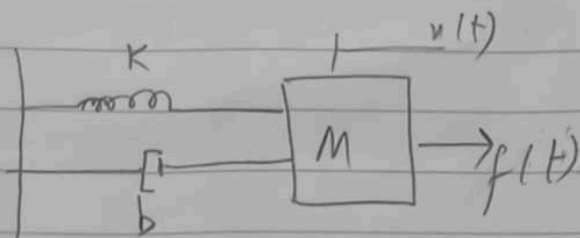
Mathematical Modeling of Translational mechanical system.

Components	Response	Impedance	Transfer function
1-  spring	$f(t) = K u(t)$	$\frac{F(s)}{X(s)} = K$	$F(s) = (K) X(s)$

2-  viscous damper	$f(t) = b \dot{u}(t)$	$\frac{F(s)}{X(s)} = bs$	$F(s) = (bs) X(s)$
--	-----------------------	--------------------------	--------------------

3-  Mass	$f(t) = M \ddot{u}(t)$	$\frac{F(s)}{X(s)} = Ms^2$	$F(s) = (Ms^2) X(s)$
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Example 1:



$$M\ddot{u}(t) + b\dot{u}(t) + Ku(t) = f(t)$$

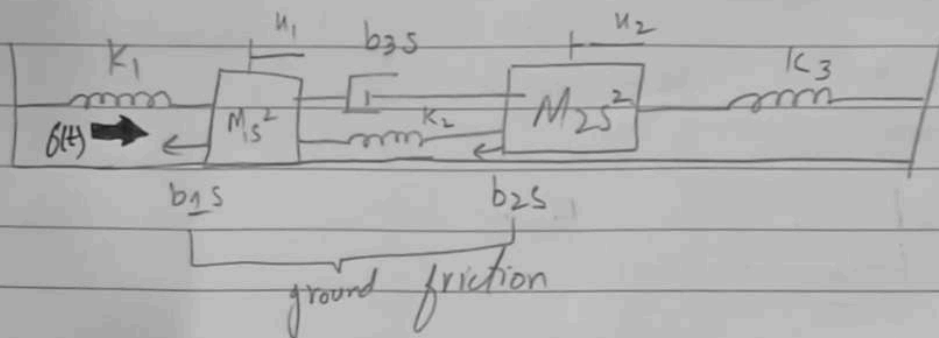
Laplace on b/s

$$ms^2 X(s) + bs X(s) + k X(s) = F(s)$$

$$(ms^2 + bs + k)X(s) = F(s)$$

$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$

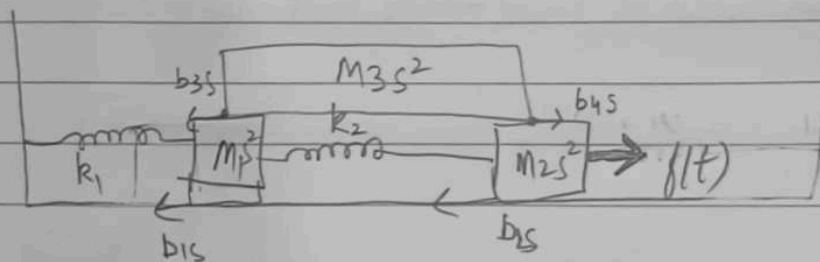
Example 2:



$$1) (M_1 s^2 + K_1 + b_1 s + b_3 s + K_2) X_1(s) - (b_3 s + K_2) X_2(s) = F(s) \quad \text{--- (1)}$$

$$2) (M_2 s^2 + k_3 + k_2 + b_3 s) X_2(s) - (b_3 s + k_2) X_1(s) = F(s) \\ -(b_3 s + k_2) X_1(s) + M_2 s^2 + k_3 + k_2 + b_3 s + b_2 s = 0 \quad \text{--- (2)}$$

Example 3



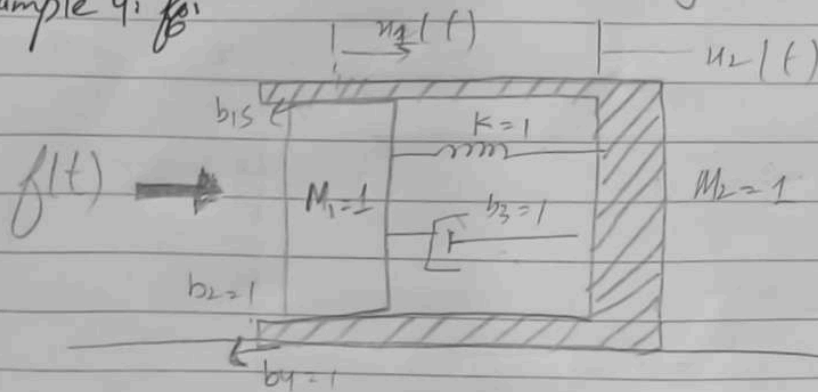
$$(1) \rightarrow (M_1 s^2 + k_1 + k_2 + b_3 + b_1) X_1(s) - (k_2) X_2(s) - (b_3 s) X_3(s) = 0$$

$$(2) \rightarrow -(k_2) X_1(s) + (M_2 s^2 + b_2 s + k_2 + b_4 s) X_2(s) - (b_4 s) X_3(s) = F(s)$$

$$(3) \rightarrow -(b_3 s) X_1(s) - (b_4 s) X_2(s) + (M_3 s^2 + b_3 s + b_4 s) X_3(s) = 0$$

(solve simultaneously)

Example 4:



$$(s^2 + 1 + s + s) X_1(s) - (1s + 1 + s + s) X_2(s) = F(s)$$

$$(s^2 + 3s + 1) X_1(s) - (3s + 1) X_2(s) = F(s) \quad (1)$$

$$-(s + s + s + 1) X_1(s) + (s^2 + s + s + s + 1) X_2(s) = 0$$

$$-(3s + 1) X_1(s) + (s^2 + 4s + 1) X_2(s) = 0 \quad (2)$$

$$\begin{bmatrix} s^2 + 3s + 1 & -3s + 1 \\ -3s + 1 & s^2 + 4s + 1 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$$\frac{X_1}{F} = ?$$

$$\Delta = \begin{vmatrix} F(s) & -3s+1 \\ 0 & s^2+4s+1 \end{vmatrix}$$

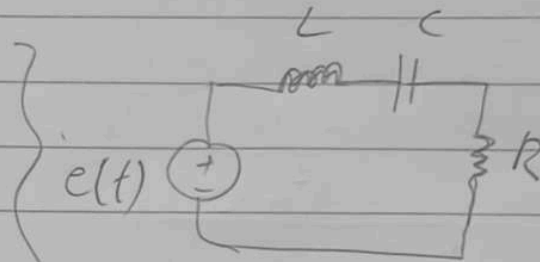
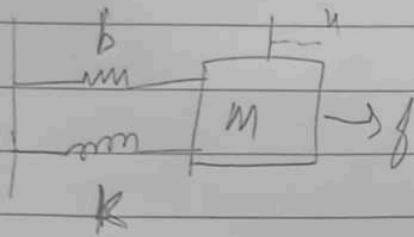
$$\begin{aligned} \Delta &= ((s^2+4s+1)(s^2+3s+1)) - ((-3s+1)(-3s+1)) \\ &= (s^4+3s^3+s^2+4s^3+12s^2+4s+s^2+3s+1) - (-9s^2-3s-3s+1) \\ &= (s^4+7s^3+14s^2+3s+1) - (-9s^2-6s+1) \\ &= (s^4+7s^3+14s^2+3s+1) + 9s^2+6s-1 \\ &= s^4+7s^3+23s^2+9s \end{aligned}$$

$$\Delta = s^4+7s^3+23s^2+9s$$

$$X_1 = \frac{F(s^2+4s+1)}{s^4+7s^3+23s^2+9s}$$

$$\frac{X_1}{F} = \frac{s^2+4s+1}{s^4+7s^3+23s^2+9s}$$

Electrical-to-Mechanical system correlation.



$$(Ms^2 + bs + k)X = F$$

$$(Ms + b + \frac{k}{s})sX = F$$

$$(Ms + b + \frac{k}{s}) = F$$

$$(Ls + R + \frac{1}{Cs})I(s) = E(s)$$

$$(Ls + R + \frac{1}{Cs})I(s) = E(s)$$

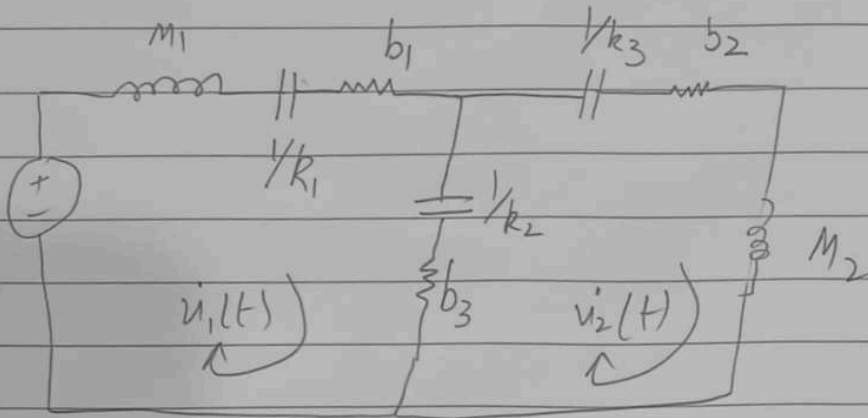
(Mass) $M \iff L$ (Inductor)

(damper) $b \iff R$ (Resistor)

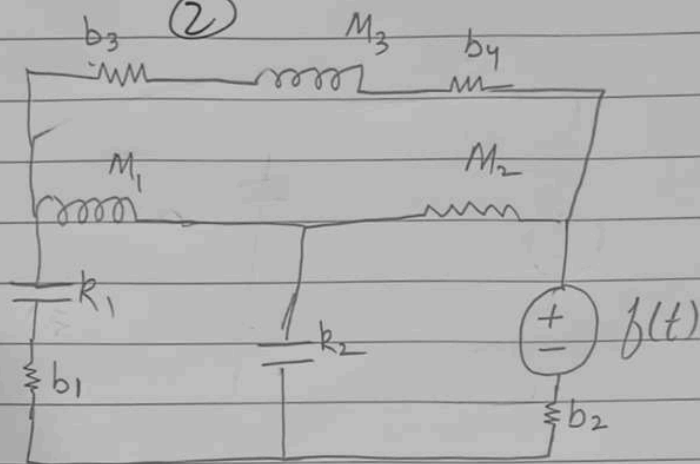
(Spring) $K \iff C$ (Capacitor)
 (velocity) $\dot{x}(t) \iff i(t)$ (current)
 (correlation)

(Force) $f(t) \iff e(t)$ (voltage) [Inputs]

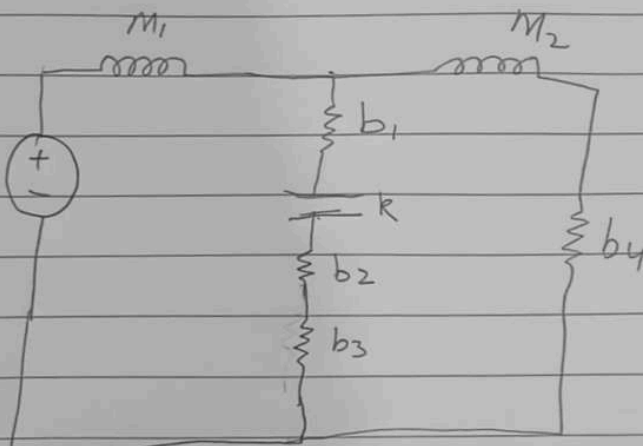
①



②

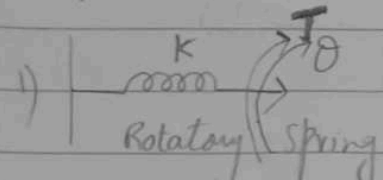
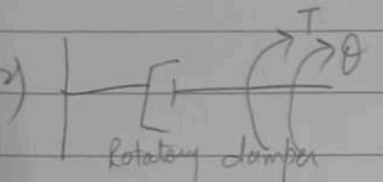
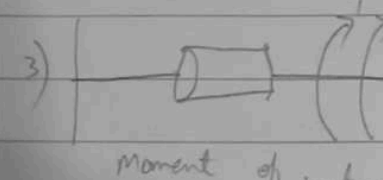


③

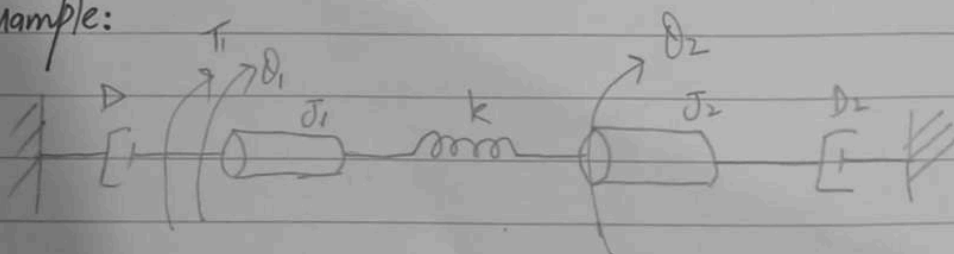


Day - 7

Modeling of a Rotational Mechanical System

Component	Response	Impedance $\left(\frac{T(s)}{\theta(s)} \right)$
1)  Rotatory Spring	$T(t) = K \theta(t)$	$\frac{T(s)}{\theta(s)} = K$
2)  Rotatory damper	$T(t) = D \dot{\theta}(t)$	$\frac{T(s)}{\theta(s)} = Ds$
3)  moment of inertia	$T(t) = J \ddot{\theta}(t)$	$\frac{T(s)}{\theta(s)} = Js^2$

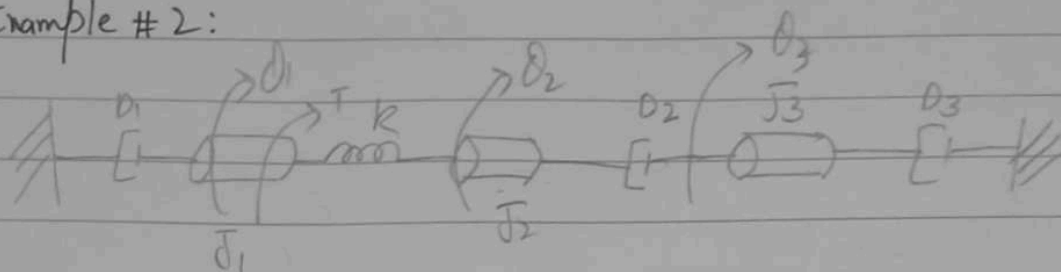
Example:



$$(J_1 s^2 + Ds + R) \theta_1 - k \theta_2 = T_1 \quad \text{--- (1)}$$

$$-k \theta_1 + (J_2 s^2 + Ds + R) \theta_2 = 0 \quad \text{--- (2)}$$

Example #2:

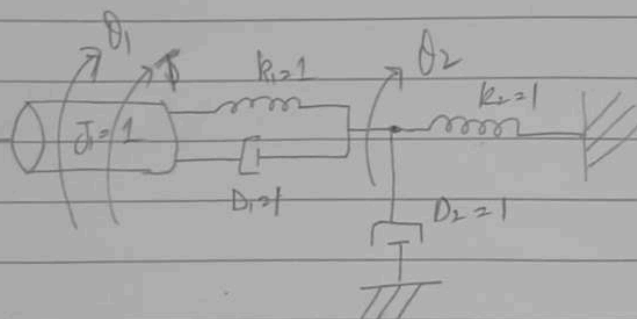


$$(J_1 s^2 + D_1 s + k) \theta_1 - k \theta_2 = F \quad \text{--- (1)}$$

$$-k \theta_1 + (J_2 s^2 + D_2 s + k) \theta_2 - (D_2 s) \theta_3 = 0 \quad \text{--- (2)}$$

$$0 - (D_2 s) \theta_2 + (J_3 s^2 + D_2 s + D_3 s) \theta_3 = 0$$

Example #3



$$(J_1 s^2 + D_1 s + k_1) \theta_1 - (D_1 s + k_1) \theta_2 = T \quad \text{--- (1)}$$

$$-(D_1 s + k_1) \theta_1 + (D_2 s + k_2 + k_1 D_1 s) \theta_2 = 0 \quad \text{--- (2)}$$

$$(s^2 + s + 1) \theta_1 - (s + 1) \theta_2 = T \quad \text{--- (1)}$$

$$-(s + 1) \theta_1 + (2s + 2) \theta_2 = 0 \quad \text{--- (2)}$$

$$\begin{bmatrix} s^2 + s + 1 & -s - 1 \\ -s - 1 & 2s + 2 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} T \\ 0 \end{bmatrix}$$

$$\begin{array}{r|l} s^2+s+1 & T \\ -s-1 & 0 \end{array}$$

Δ

$$\Delta_2 \left| \begin{array}{r|l} s^2+s+1 & -(s+1) \\ -(s+1) & 2(s+1) \end{array} \right|$$

$$\begin{aligned} &= 2(s+1)(s^2+s+1) - (s+1)^2 \\ &= (s+1)(2s^2+2s+2-s-1) \\ &= 2(s+1)(2s^2+s+1) \end{aligned}$$

$$\Delta_2 \left| \begin{array}{r|l} s^2+s+1 & T \\ -s+1 & 0 \end{array} \right|$$

Δ

$$= \frac{T(s+1)}{(s+1)(2s^2+s+1)}$$

$$\Delta_2 = \frac{T}{2s^2+s+1}$$

$$\boxed{\frac{\Delta_2}{T} = \frac{1}{2s^2+s+1}}$$

$$(Js^2 + Ds + k) \left(\frac{N_1}{N_2} \right) \theta_1 = \left(\frac{N_2}{N_1} \right) T_1$$

$$(Js^2 + Ds + k) \left(\frac{N_1}{N_2} \right)^2 \theta_1 = T_1$$

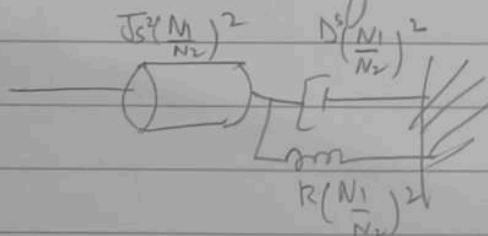
$$\frac{\theta_1}{T_1} = \frac{1}{\left(\frac{N_1}{N_2} \right)^2 (Js^2 + Ds + k)}$$

$$\frac{\theta_1}{T_1} = \frac{1}{J \left(\frac{N_1}{N_2} \right)^2 s^2 + D \left(\frac{N_1}{N_2} \right)^2 s + k \left(\frac{N_1}{N_2} \right)^2}$$

Method #2

∴ Reflect impedances on the other side using

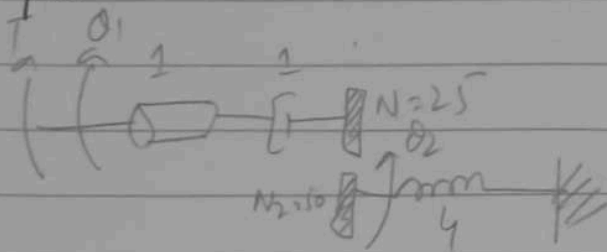
(original impedences) $\left(\frac{\text{"N" of destination shaft}}{\text{"N" of source shaft}} \right)^2$



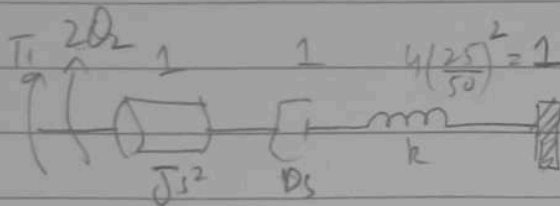
$$\left(Js^2 \left(\frac{N_1}{N_2} \right)^2 + Ds \left(\frac{N_1}{N_2} \right)^2 + k \left(\frac{N_1}{N_2} \right)^2 \right) \theta_1 = T_1$$

$$\frac{\theta_1}{T_1} = \frac{1}{Js^2 \left(\frac{N_1}{N_2} \right)^2 + Ds \left(\frac{N_1}{N_2} \right)^2 + k \left(\frac{N_1}{N_2} \right)^2}$$

Example:



$$\frac{\theta_2}{T} = ?$$



(2nd moving to primary)

$$\theta_2 = \left(\frac{N_1}{N_2} \right) \theta_1$$

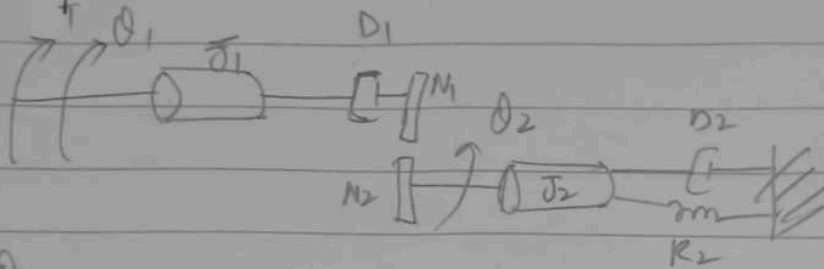
$$\theta_2 = \frac{\theta_1}{2} \Rightarrow \theta_1 = 2\theta_2$$

$$(Js^2 + Ds + k) 2\theta_2 = T_1$$

$$2(s^2 + s + 1) \theta_2 = T_1$$

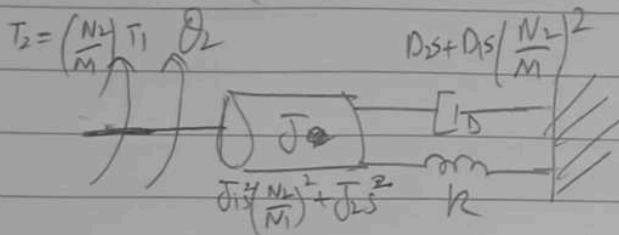
$$\frac{\theta_2}{T} = \frac{1}{2(s^2 + s + 1)}$$

Example:



$$\frac{\theta_2}{T}$$

$$D_1 \left(\frac{N_2}{N_1} \right)^2$$



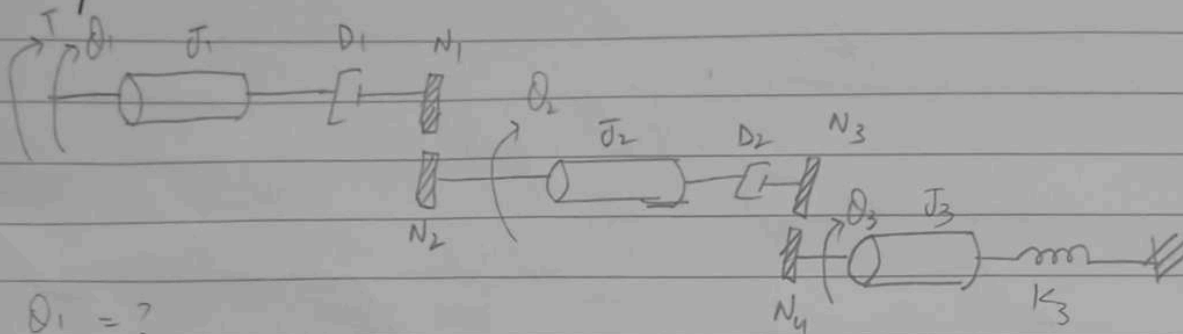
$$\left(\frac{N_2}{N_1} \right)^2 + J_2 s^2 + \left(D_{2S} + D_{1S} \left(\frac{N_2}{N_1} \right)^2 + K \right) \theta_2 = \left(\frac{N_2}{N_1} \right) T$$

$$\frac{\theta_2}{T} = \frac{\left(\frac{N_2}{N_1} \right)}{J_2 \left(\frac{N_2}{N_1} \right)^2 + J_2 s^2 + D_{2S} + D_{1S} \left(\frac{N_2}{N_1} \right)^2 + K}$$

$$(J_T s^2 + D_{TS} + K_T) \theta_u = T_u \rightarrow \text{general equation}$$

Day - 8

Examples:-



$$\frac{\theta_1}{T} = ?$$

$$J_T = J_1 + J_2 \left(\frac{N_1}{N_2} \right)^2 + J_3 \left(\left(\frac{N_1}{N_2} \right) \left(\frac{N_3}{N_4} \right) \right)^2$$

$$D_T = D_1 + D_2 \left(\frac{N_1}{N_2} \right)^2$$

$$K = k_3 \left(\left(\frac{N_1}{N_2} \right)^2 \left(\frac{N_3}{N_4} \right)^2 \right)$$

$$\frac{\theta_1}{T} = \frac{1}{\left(J_1 + J_2 \left(\frac{N_1}{N_2} \right)^2 + J_3 \left(\frac{N_1}{N_2} \right)^2 \left(\frac{N_3}{N_4} \right)^2 \right) s^2 + \left(D_1 + D_2 \left(\frac{N_1}{N_2} \right)^2 \right) s + k_3 \left(\left(\frac{N_1}{N_2} \right)^2 \left(\frac{N_3}{N_4} \right)^2 \right)}$$

$$\frac{\theta_1}{\theta_2} = \frac{N_2}{N_1}$$

$$\frac{\theta_1}{\theta_3} = \frac{\theta_1}{\theta_2} \times \frac{\theta_2}{\theta_3}$$

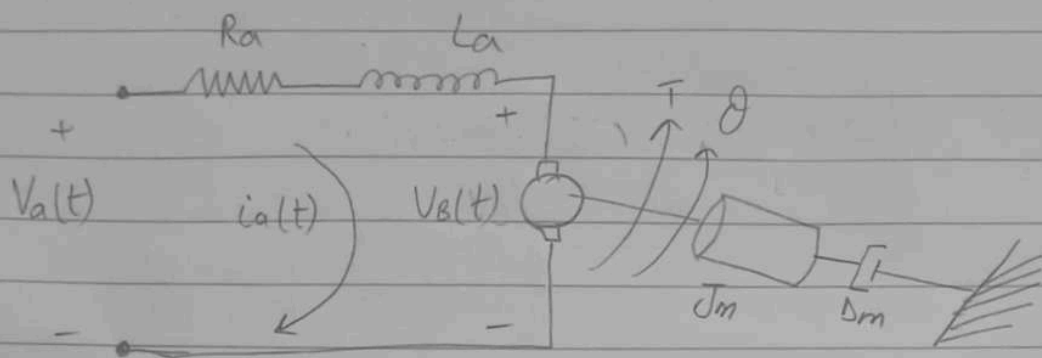
$$= \left(\frac{N_2}{N_1} \right) \left(\frac{N_3}{N_4} \right)$$

$$\frac{\theta_2}{\theta_3} = \frac{N_3}{N_4}$$

$$\boxed{\frac{\theta_1}{\theta_3} = \frac{N_3}{N_1}}$$

Mathematical Model of Electro-mechanical system.

DC motor



model of simple DC motor



motor
 $\Rightarrow$ Torque is directly proportion to current
 motor
 $\Rightarrow$ speed is depend to voltage

$$EMF = V_B(t) = K_B \dot{\theta}(t) \Rightarrow V_B(s) = K_B s \theta(s) \quad (1)$$

$$\text{circuit: } R_a I_a(s) + s L_a I_a(s) + V_B(s) = V_a(s) \quad (2)$$

$$\text{Torque: } T(t) = K_t i_a(t) \Rightarrow T(s) = K_T I_a(s) \quad (3)$$

$$\text{Rotational: } (J_m s^2 + D_m s) \theta(s) = T(s) \quad (4)$$

mechanical system

put (1) & (3) in eq (2)

$$(R_a + s L_a) I_a(s) + V_B(s) = V_a(s)$$

$$(R_a + s L_a) \left(\frac{T(s)}{K_T} \right) + K_B s \theta(s) = V_a(s) \quad (5)$$

put eq (4) in eq (5)

$$\frac{(R_a + sL_a)(J_m s^2 + D_m s)}{k_T} \theta(s) + k_B s \theta(s) = V_a(s)$$

$$\left[\frac{(R_a + sL_a)(J_m s^2 + D_m s)}{k_T} + k_B s \right] \theta(s) = V_a(s)$$

$$\frac{\theta(s)}{V_a(s)} = \frac{k_T}{(R_a + sL_a)(J_m s^2 + D_m s) + k_B k_T s}$$

in practical model $L_a \ll R_a$ (So " L_a " can be ignored)

$$\frac{\theta(s)}{V_a(s)} = \frac{k_T}{R_a (J_m s^2 + D_m s) + k_B k_T s}$$

$$\frac{\theta(s)}{V_a(s)} = \frac{k_T}{s (R_a J_m s + R_a D_m + k_B k_T)}$$

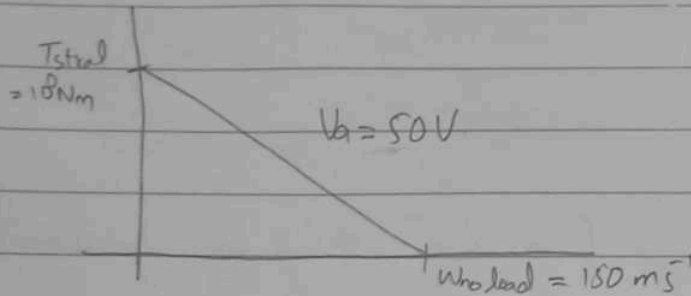
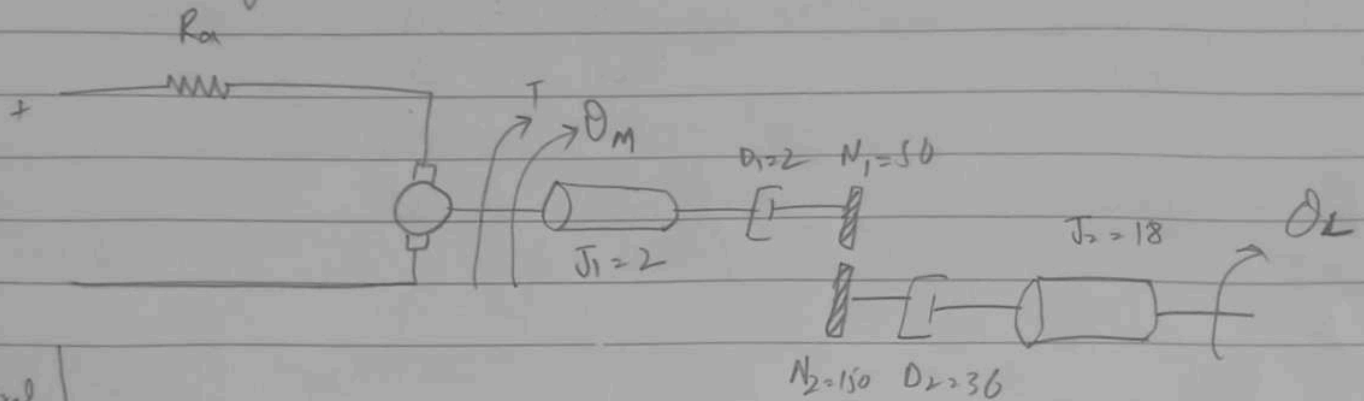
$$\frac{\theta(s)}{V_a(s)} = \frac{k_T / R_a J_m}{s \left(s + \frac{D_m}{J_m} + \frac{k_B k_T}{R_a J_m} \right)}$$

$$\boxed{\frac{\theta(s)}{V_a(s)} = \frac{G}{s(s + \alpha)}}$$

$$\therefore G = \frac{k_T}{R_a J_m}$$

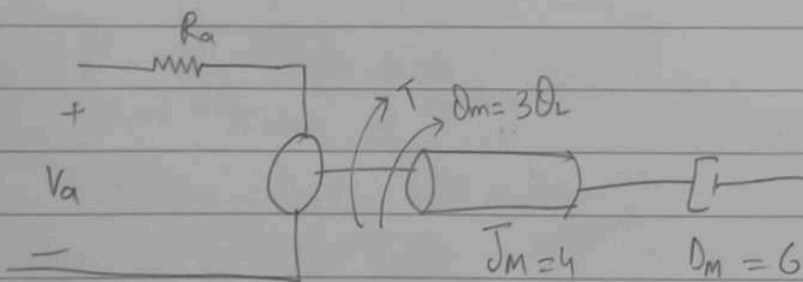
$$\therefore \alpha = \frac{D_m}{J_m} + \frac{k_B k_T}{R_a J_m}$$

⇒ If inductor is so small there is only one pole at origin.



$$K_T = \frac{T_{stall}}{I_a} = 2$$

$$K_B = \frac{V_a}{\omega_{no load}} = 0.33$$



$$J_m = 2 + \left(\frac{50}{150} \right)^2 (18) = 2 + 2 = 4$$

$$D_m = 2 + \left(\frac{50}{150} \right) (36) = 2 + 4 = 6$$

$$\frac{D_m}{D_L} = \frac{150}{50}$$

$$\boxed{\begin{aligned} D_m &= 30 \\ D_L &= \frac{D_m}{3} \end{aligned}}$$

$$G_1 = \frac{k_r}{R_a J_m} = \frac{2}{4} = 0.5$$

$$\alpha = \frac{D_m}{J_m} + \frac{k_r k_B}{R_a J_m} = \frac{6}{4} + \frac{7(0.33)}{4 \times 2} = \frac{5}{3}$$

$$\frac{D_m}{V_a(s)} = \frac{0.5}{s(s+5/3)}$$

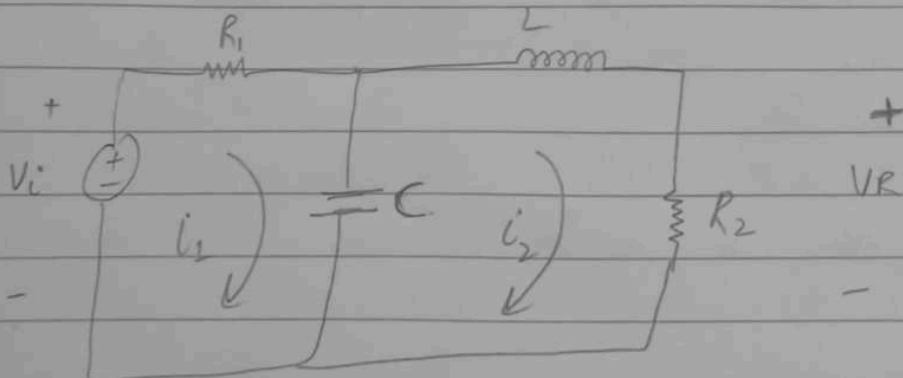
$$\therefore D_m = 3D_L$$

$$\frac{3D_L}{V_a(s)} = \frac{0.5}{s(s+5/3)}$$

$$\frac{D_L}{V_a(s)} = \frac{1/6}{s(s+5/3)}$$

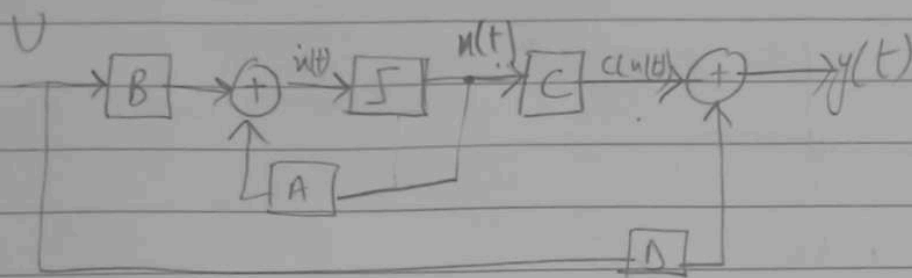
Day - 9 (a)

State Space Representation:-



Transfer function has lack in information

"State space model is in time domain"



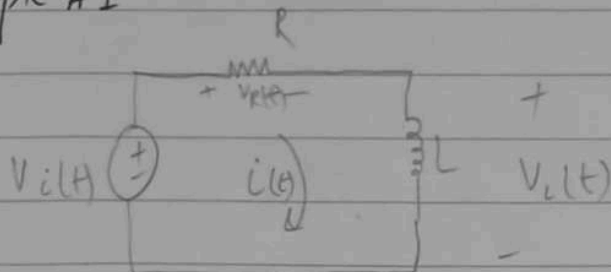
"If the system is dynamic there is a differential equation."

Example:

Rules:

- 1) The no. of state variable and hence, the state equation is equal to number of Reactive elements in the circuit (i.e. L & C)
- 2) All state equation & output equation must be written in terms of all state variables & input variable.
- 3) For capacitor $\rightarrow V_c$ (Voltage across capacitor) is taken as state variable.
For Inductor $\rightarrow i_L$ (Current across inductor) is taken as state variable.

Example #1



$$\begin{aligned} \text{① } V_i &= V_R + V_L \\ \text{② } V_L &= L \frac{di}{dt} \\ \text{③ } i &= C \frac{dv}{dt} \end{aligned}$$

$$\Rightarrow \begin{aligned} u(t) &= i(t) \\ y(t) &= V_R(t) \end{aligned}$$

$$Ri(t) + L \dot{i}(t) = V_i(t)$$

$$\dot{i}(t) = \begin{bmatrix} -R \\ L \end{bmatrix} i(t) + \begin{bmatrix} 1 \\ L \end{bmatrix} V_i(t)$$

$$\dot{x}(t) = A x(t) + B u(t)$$

$$\text{output} \Rightarrow y = V_R(t) \Rightarrow Ri(t) \Rightarrow \text{ohm law}$$

$$A = \begin{bmatrix} -R \\ L \end{bmatrix}, B = \begin{bmatrix} 1 \\ L \end{bmatrix}, C = [R], D = [0]$$

$$\text{output} \Rightarrow y = L \frac{di}{dt}$$

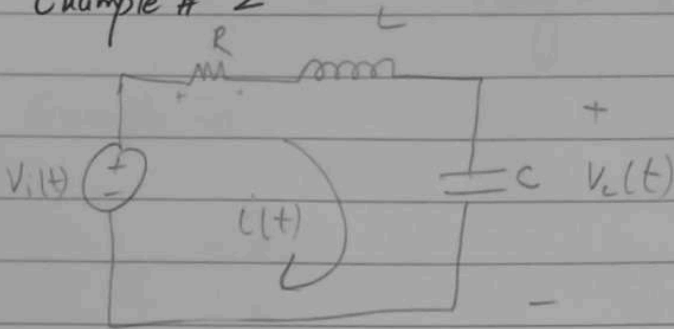
$$= L \left[\begin{bmatrix} -R \\ L \end{bmatrix} i(t) + \begin{bmatrix} 1 \\ L \end{bmatrix} V_i(t) \right]$$

$$y = -Ri(t) + V_i(t)$$

$$y = Cx(t) + Du(t)$$

$$C = [-R], D = [1]$$

Example # 2



"L & C" $\Rightarrow$ $x(t) = i(t)$, $V_c(t)$
 $u(t) = V_i(t)$
 $y(t) = V_R(t)$

$$V_{in}(t) = V_R(t) + V_L(t) + V_C(t) \quad (\text{VDR})$$

$$V_{in}(t) = R i(t) + L \dot{i}(t) + V_C(t)$$

$$\dot{i}(t) = -\frac{R}{L} i(t) - \frac{1}{L} V_C(t) + \frac{1}{L} V_i(t) \quad \text{--- (1)}$$

$$i(t) = C \frac{dV_C}{dt}$$

$$\dot{i}(t) = C \dot{V}_C(t)$$

$$\dot{V}_C(t) = \frac{1}{C} \dot{i}(t)$$

$$V_C(t) = \frac{1}{C} i(t) + 0 V_C(t) + (0) V_C(t) \quad \text{--- (2)}$$

$$\begin{bmatrix} \dot{i}(t) \\ \dot{V}_C(t) \end{bmatrix} = \begin{bmatrix} -R/L & -1/L \\ 1/C & 0 \end{bmatrix} \begin{bmatrix} i(t) \\ V_C(t) \end{bmatrix} + \begin{bmatrix} 1/L \\ 0 \end{bmatrix} V_i(t)$$



$$y = v_R(t)$$

$$y = R i(t)$$

$$y = R i(t) + 0 v_c(t) + 0 v_i(t)$$

$$y = \begin{bmatrix} R & 0 \end{bmatrix} \begin{bmatrix} i(t) \\ v_c(t) \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} v_i(t)$$

$$\text{if } y = \begin{bmatrix} v_R \\ v_c \end{bmatrix}$$

$$v_R = i(t)R + 0v_c + 0v_i$$

$$v_c = v_i(t) - v_R - v_c$$

$$v_c(t) = -R i(t) - v_c + v_i(t)$$

$$y(t) = \begin{bmatrix} v_R \\ v_c \end{bmatrix} = \begin{bmatrix} R & 0 \\ -R & -1 \end{bmatrix} \begin{bmatrix} i(t) \\ v_c(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} v_i$$

Day - 9 (b)

$$V_c(t) = R(i_1 - C\ddot{V}_c(t)) \quad (d) \text{ P-grad}$$

$$V_c(t) = Ri_1 - RC\ddot{V}_c(t)$$

$$\ddot{V}_c(t) = \frac{1}{C}i_1 - \frac{1}{RC}V_c(t) \quad \text{--- (B)}$$

$$\begin{bmatrix} \dot{i}_1 \\ \dot{V}_c \end{bmatrix} = \begin{bmatrix} 0 & -1/L \\ 1/C & -1/RC \end{bmatrix} \begin{bmatrix} i_1(t) \\ V_c(t) \end{bmatrix} + \begin{bmatrix} 1/L \\ 0 \end{bmatrix} V_c^o(t)$$

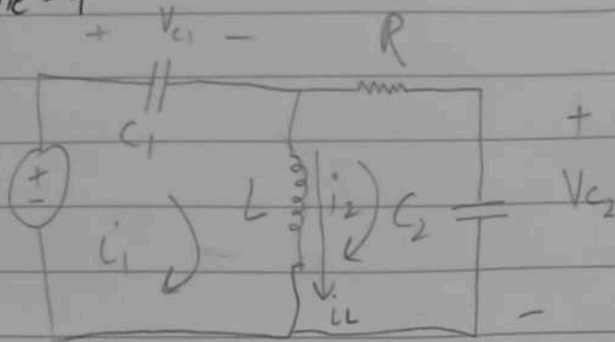
$$y(t) = i_1(t)$$

$$y(t) = i_1(t) - L_2(t) \cdot X$$

$$y(t) = \frac{V_c(t)}{R} \quad \checkmark$$

$$y = \begin{bmatrix} 0 & 1/R \end{bmatrix} \begin{bmatrix} i_1(t) \\ V_c(t) \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} V_c^o(t)$$

Example - 4



$$x(t) = \begin{bmatrix} i_1(t) \\ V_{c1}(t) \\ V_{c2}(t) \end{bmatrix}$$

∴ $i_L = i_1(t)$

$$L \dot{i}_L(t) + V_{c1} = V_{in} \quad \text{--- (1)}$$

$$\dot{i}_L(t) = -\frac{1}{L} V_{c1} - 0 i_L + \frac{1}{L} V_{in} + 0 V_{c2} \quad \text{--- (A)}$$

$$i_2 = C_2 \dot{V}_{c2} \quad \text{--- (3)}$$

$$R i_2 + V_{c2} - L \dot{i}_L = 0 \quad \text{--- (4)}$$

put (3) in (4)

$$R C_2 \dot{V}_{c2} + V_{c2} = L \dot{i}_L$$

$$R C_2 \dot{V}_{c2}(t) + V_{c2} = V_{in} - V_{c1}$$

$$\dot{V}_{c2}(t) = \frac{1}{R C_2} V_{in} - \frac{1}{R C_2} V_{c1} - \frac{1}{R C_2} V_{c2} \quad \text{--- (B)}$$

$$i_1 = C_1 \dot{V}_{c1} \quad \text{--- (5)}$$

taking eq (1)

$$i_L = i_1 - i_2$$

$$i_1 = i_L + C_2 \dot{V}_{c2} \quad \text{from eq (B)}$$

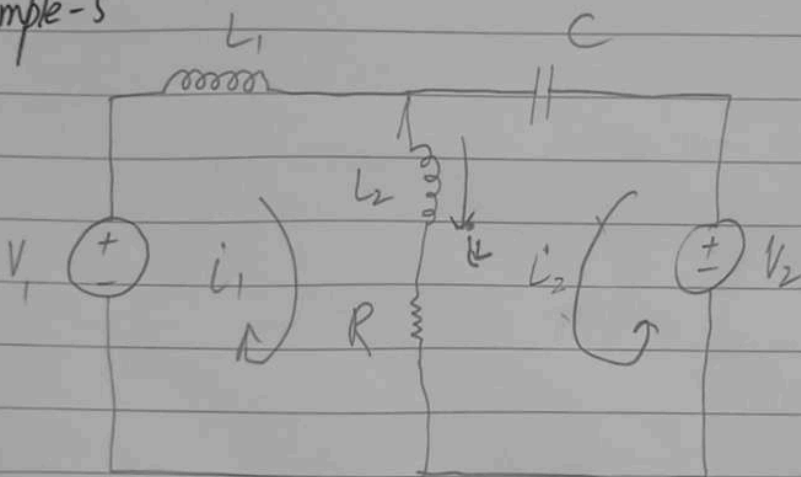
$$i_L = i_L - \frac{1}{R} V_{c1} - \frac{1}{R} V_{c2} + \frac{1}{R} V_i$$

put this in eq (1)

$$\dot{V}_{c1} = \frac{1}{C_1} i_L - \frac{1}{RC_1} V_{c1} - \frac{1}{RC_1} V_{c2} + \frac{1}{RC_1} V_i \quad \text{--- (C)}$$

$$\begin{bmatrix} \dot{i}_L \\ \dot{V}_{c1} \\ \dot{V}_{c2} \end{bmatrix} = \begin{bmatrix} 0 & -1/R & 0 \\ 1/C_1 & -1/RC_1 & -1/RC_1 \\ 0 & -1/RC_1 & -1/RC_2 \end{bmatrix} \begin{bmatrix} i_L \\ V_{c1} \\ V_{c2} \end{bmatrix} + \begin{bmatrix} +1/R \\ +1/RC_1 \\ +1/RC_2 \end{bmatrix} V_{in}$$

Example-5



$$u = \begin{bmatrix} i_1 \\ V_C \\ i_L \end{bmatrix}$$

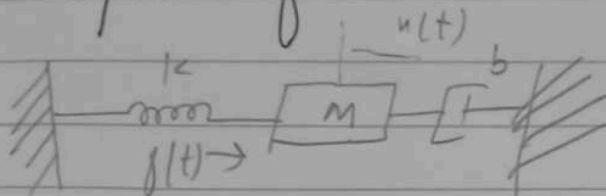
$$y = V_R(t)$$

$$u(t) = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$i_L = i_1(t) + i_2(t) \quad \text{--- (1)}$$

~~Study~~
state

Space of Mechanical System



Example #1

$$(Ms^2 + bs + k)X(s) = F(s)$$

$$Ms^2X(s) + bs(X(s)) + kX(s) = F(s)$$

Inverse Laplace

$$M\ddot{u}(t) + b\dot{u}(t) + ku(t) = f(t) \quad \text{--- (1)}$$

$$\because Z_1 = u(t), Z_2 = \dot{u}(t) \Rightarrow \dot{Z}_2 = \ddot{u}(t), \dot{Z}_1 = Z_2$$

leave the highest order term of differential eqn untouched

Substitution all this in eq (1)

$$M\dot{Z}_2(t) + bZ_2 + kZ_1 = f(t)$$

$$\dot{Z}_2(t) = \frac{f(t)}{M} - \frac{bZ_2}{M} - \frac{k}{M}Z_1$$

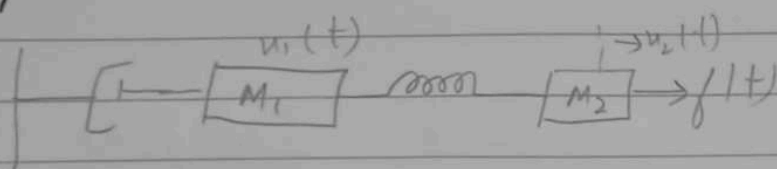
$$\dot{Z}_1 = Z_2$$

$$\begin{bmatrix} \dot{Z}_1 \\ \dot{Z}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k/M & -b/M \end{bmatrix} \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/M \end{bmatrix} f(t)$$

$$u(t) = z_1$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} f(t)$$

Example # 2



$$M_1 \ddot{u}_1(t) + b \dot{u}_1(t) + k u_1 - k u_2 = 0 \quad \text{--- (1)}$$

$$-(k x_1) + (M_2 \ddot{u}_2(t) + R u_2) = f(t)$$

$$M_2 \ddot{u}_2(t) - k u_1 - k u_2 = f(t) \quad \text{--- (2)}$$

$$z_1 = u_1(t)$$

$$z_2 = \dot{u}_1(t) = \dot{z}_1$$

$$\dot{z}_2 = \ddot{u}_1(t)$$

$$z_3 = u_2(t)$$

$$z_4 = \dot{u}_2(t) = \dot{z}_3$$

$$\dot{z}_4 = \ddot{u}_2(t)$$

Substitute this in eq (1)

$$M_1 \dot{z}_2(t) + b z_2 + k z_1 - k z_3 = 0$$

$$\dot{z}_2(t) = \frac{b}{M_1} z_2 + \frac{k}{M_1} z_1 - \frac{k}{M_1} z_3 = 0 \quad \text{--- (A)}$$

$$M_2 \dot{z}_4(t) - k z_1(t) + k z_3 = f(t)$$

$$\dot{z}_4(t) = \frac{k}{M_2} z_1(t) - \frac{k}{M_2} z_3 + \frac{1}{M_2} f(t) \quad \text{--- (B)}$$

$$\dot{z}_1 = z_2 \quad \text{--- (C)}$$

$$\dot{z}_3 = z_4 \quad \text{--- (D)}$$

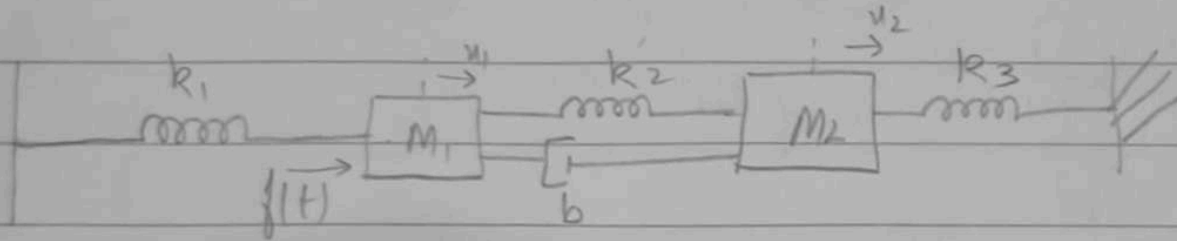
$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \\ \dot{z}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -k/m_1 & -b/m_1 & k/m_1 & 0 \\ 0 & 0 & 0 & 1 \\ k/m_2 & 0 & -k/m_2 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/m_2 \end{bmatrix} f(t)$$

$$y = x_2(t)$$

$$y = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} f(t)$$

Day-10 (3 Page →)

Question:



$$m_1 \ddot{u}_1 + b \dot{u}_1 + k_1 u_1 + k_2 u_1 - k_2 u_2 - b \dot{u}_2 = f(t)$$

$$M_2 \ddot{u}_2 + b \dot{u}_2 + k_2 (u_2 - u_1) + k_3 u_2 - k_2 u_1 - b \dot{u}_1 = 0$$

$$z_1 = u_1(t)$$

$$z_2 = \dot{u}_1(t) = \dot{z}_1$$

$$\dot{z}_2 = \ddot{u}_1(t)$$

$$z_3 = u_2(t)$$

$$z_4 = \dot{u}_2(t) = \dot{z}_3$$

$$\dot{z}_4 = \ddot{u}_2(t)$$

$$M_1 \dot{z}_2 + b z_2 + (k_1 + k_2) z_1 - b z_4 - k_2 z_3 = f$$

$$-b z_2 - k_2 z_1 + M_2 \dot{z}_4 + b z_4 + (k_2 + k_3) z_3 = 0$$

Day - 10

Transfer Function to State Space conversion:-

Q:
$$\frac{C(s)}{R(s)} = \frac{b_0}{a_0 s^3 + a_1 s^2 + a_2 s + a_3}$$
 $\left(\frac{\text{output}}{\text{Input}} \right)$

$$C(s)(a_0 s^3 + a_1 s^2 + a_2 s + a_3) = b_0 R(s)$$

$$(a_0 \ddot{\ddot{c}}(t) + a_1 \ddot{c}(t) + a_2 \dot{c}(t) + a_3 c(t)) = b_0 r(t)$$

$$z_1 = c(t)$$

$$z_2 = \dot{c}(t) = \dot{z}_1$$

$$z_3 = \ddot{c}(t) = \dot{z}_2$$

$$\dot{z}_3 = \ddot{\ddot{c}}(t)$$

$$a_0 \dot{z}_3 + a_1 z_3 + a_2 z_2 + a_3 z_1 = b_0 r$$

$$\dot{z}_1 = z_2 \quad \text{--- (A)}$$

$$\dot{z}_2 = z_3 \quad \text{--- (B)}$$

$$\dot{z}_3 = -\frac{a_3}{a_0} z_1 + \frac{a_2}{a_0} z_2 - \frac{a_1}{a_0} z_3 + \frac{b_0}{a_0} r$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{a_3}{a_0} & -\frac{a_2}{a_0} & -\frac{a_1}{a_0} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{b_0}{a_0} \end{bmatrix} r$$

output $\Rightarrow y = c(t) = z_1$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} r$$

Question:

$$\frac{C(s)}{R(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$

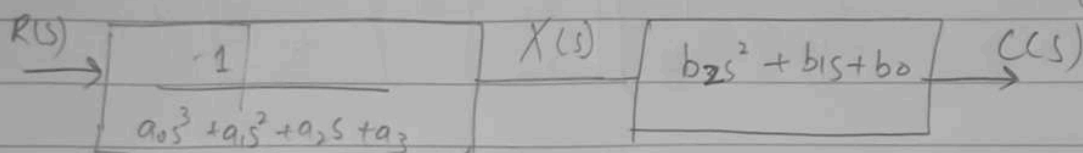
(Sign (-ve) put in List 1 Diagonally & put in the box
 & Power (highest) sh in 1st row & Power $\frac{1}{2}$ sh in 2nd row
 - & divide

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

Question:

$$\frac{C(s)}{R(s)} = \frac{b_2 s^2 + b_1 s + b_0}{a_0 s^3 + a_1 s^2 + a_2 s + a_3}$$

We divide this in state (denominator first)



$$\frac{X(s)}{R(s)} = \frac{1}{a_0 s^3 + a_1 s^2 + a_2 s + a_3}$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{a_3}{a_0} & -\frac{a_2}{a_0} & -\frac{a_1}{a_0} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{a_0} \end{bmatrix} \quad \sim \text{(short method)}$$

not recommended
↑ in this question

$$(a_0 s^3 + a_1 s^2 + a_2 s + a_3) X(s) = R(s)$$

$$a_0 \ddot{u}(t) + a_1 \dot{u}(t) + a_2 u(t) + a_3 u(t) = r(t)$$

$$u = z_1$$

$$\dot{u} = \dot{z}_2 = \dot{z}_1$$

$$\ddot{u} = \dot{z}_3 = \dot{z}_2$$

$$\ddot{\dot{u}} = \dot{z}_3$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{a_3}{a_0} & -\frac{a_2}{a_0} & -\frac{a_1}{a_0} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/a_0 \end{bmatrix} r(t)$$

$$C(s) = b_2 s^2 + b_1 s + b_0$$

$$X(s)$$

$$c(t) = b_2 \ddot{u}(t) + b_1 \dot{u}(t) + b_0 u(t)$$

$$c(t) = b_2 z_3 + b_1 z_2 + b_0 z_1 \Rightarrow \text{output eq}$$

$$y(t) = \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} r$$

Question.

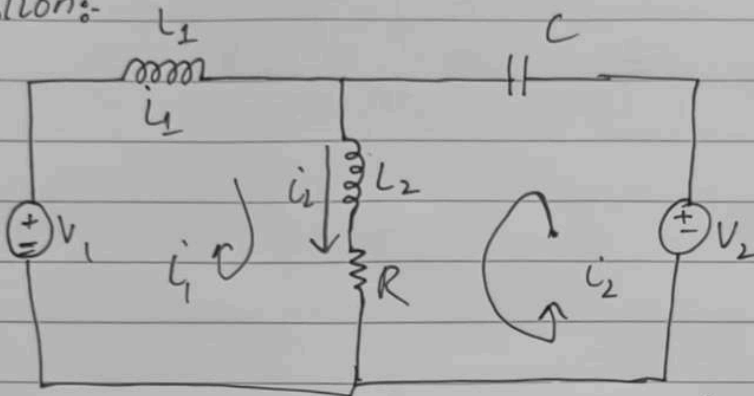
$$\frac{C(s)}{R(s)} = \frac{2s+1}{s^2+7s+9}$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -9 & -7 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} r$$

Day - 11

Question:-



(Previous Example)

Example # 5 on day 9(b)

$$x = \begin{bmatrix} V_C \\ i_1 \\ i_R \end{bmatrix}, \quad y = V_R$$

$$i_R = i_1 + i_2 \quad \text{--- (1)}$$

$$L_1 \dot{i}_1 + L_2 \dot{i}_R + R i_R = V_1 \quad \text{--- (2)}$$

$$\begin{aligned} V_C + L_2 \dot{i}_R + R i_R &= V_2 \\ \Rightarrow \ddot{i}_R &= -\frac{1}{L_2} V_C - \frac{R}{L_2} \dot{i}_R + \frac{1}{L_2} V_2 \quad \text{--- (3)} \end{aligned}$$

$$i_2 = C \dot{V}_C$$

$$i_R - i_1 = i_2 \quad \text{from eq (1)}$$

$$i_R - \dot{V}_C = C \dot{V}_C$$

$$\Rightarrow \ddot{V}_C + \frac{1}{L_2} \dot{V}_C - \frac{1}{L_2} \dot{i}_1 \quad \text{--- (4)}$$



put eq (3) in (2)

$$L_1 \dot{i}_1 + L_2 \left(-\frac{1}{L_2} V_c - \frac{R}{L_2} \dot{i}_R + \frac{1}{L_2} V_2 \right) + R \dot{i}_R = V_1$$

$$L_1 \dot{i}_1 - V_c - R \dot{i}_R + V_2 + R \dot{i}_R = V_1$$

$$\Rightarrow \dot{i}_1 = V_1 \left(\frac{1}{L_1} \right) + V_c \left(\frac{1}{L_1} \right) - V_2 \left(\frac{1}{L_1} \right) \quad \text{--- (5)}$$

$$\begin{bmatrix} \dot{V}_c \\ \dot{i}_1 \\ \dot{i}_R \end{bmatrix} = \begin{bmatrix} 0 & -1/C & 0 \\ 1/L_1 & 0 & 0 \\ -1/L_2 & 0 & -R/L_2 \end{bmatrix} \begin{bmatrix} V_c \\ i_1 \\ i_R \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1/L_1 & -1/L_1 \\ 1/L_2 & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$y = \begin{bmatrix} 0 & 0 & R \end{bmatrix} \begin{bmatrix} V_c \\ i_1 \\ i_R \end{bmatrix} + \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$



State Space to Transfer Function:

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

taking

^ Laplace transform:

$$sX(s) = AX(s) + BU(s) \quad \text{--- (1)}$$

$$Y(s) = CX(s) + DU(s) \quad \text{--- (2)}$$

from (1)

$$sX(s) - AX(s) = BU(s)$$

$$X(s)(sI - A) = BU(s)$$

↳ its incorrect cause "s" is variable
and A is matrix

$$X(s)(sI - A) = BU(s)$$

↳ identity matrix

$$\cancel{X(s)} = \frac{BU(s)}{sI}$$

$$X(s) = (sI - A)^{-1} BU(s) \quad \text{--- (3)}$$

put (3) in eq (2)

$$Y(s) = CX(s) + DU(s)$$

$$Y(s) = C(sI - A)^{-1}B U(s) + D U(s)$$

$$Y(s) = [C(sI - A)^{-1}B + D] U(s)$$

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

Question:

$$\dot{n} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} n + \begin{bmatrix} .2 \\ 0 \end{bmatrix} U$$

$$y = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} n$$

Solution:-

$$sI - A = s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s+4 & 1.5 \\ -4 & s \end{bmatrix}$$

$$|sI - A| = s(s+4) - 1.5(-4)$$

$$= s^2 + 4s + 6$$



$$(SI - A)^{-1} = \frac{1}{s^2 + 4s + 6} \begin{bmatrix} s & -1.5 \\ 4 & s+4 \end{bmatrix}$$

As him may 'B' ko multiply krna hai always.

$$(SI - A)^{-1} B = \frac{1}{s^2 + 4s + 6} \begin{bmatrix} s & -1.5 \\ 4 & s+4 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

Post multiply

$$(SI - A)^{-1} B = \frac{1}{s^2 + 4s + 6} \begin{bmatrix} 2s \\ 8 \end{bmatrix}$$

$$C(SI - A)^{-1} B = \frac{1}{s^2 + 4s + 6} \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} \begin{bmatrix} 2s \\ 8 \end{bmatrix}$$

pre multiply

$$C(SI - A)^{-1} B = \frac{1}{s^2 + 4s + 6} [3s + 5]$$

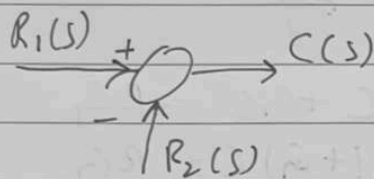
$$C(SI - A)^{-1} B + D = \frac{3s + 5}{s^2 + 4s + 6} = \frac{Y(s)}{U(s)}$$

Day-12

Ch# 5

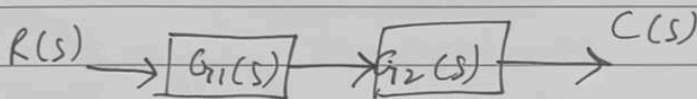
Block diagram ~~detect~~ reduction:

Example # 1



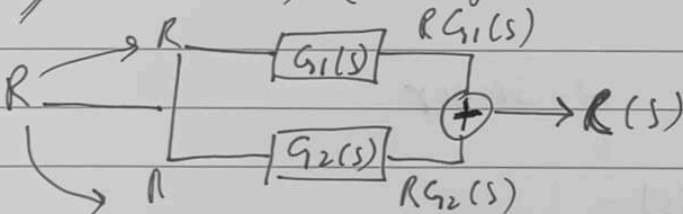
$$C(s) = R_1(s) - R_2(s)$$

2) Series (Multiply known by)



$$C(s) = (G_1(s) G_2(s)) R(s)$$

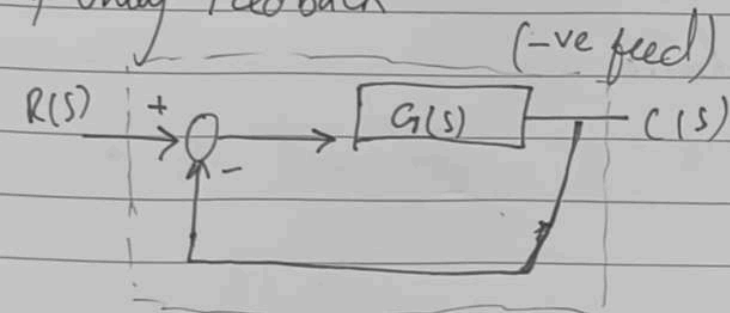
3) Parallel $\Rightarrow$ (Ending and starting point are same) Adding



$$C(s) = RG_1 + RG_2$$

$$= R(G_1 + G_2)$$

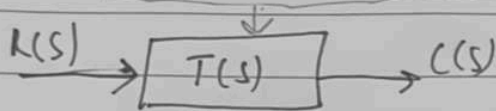
4) Unity Feedback



Note:-

(In matlab if there is nothing in feedback block then is gain of

$$1 \Rightarrow \boxed{1}$$



$$E(s) = R(s) - C(s) \quad \text{--- (1)}$$

$$C(s) = G(s)E(s) \quad \text{--- (2) put in (1)}$$

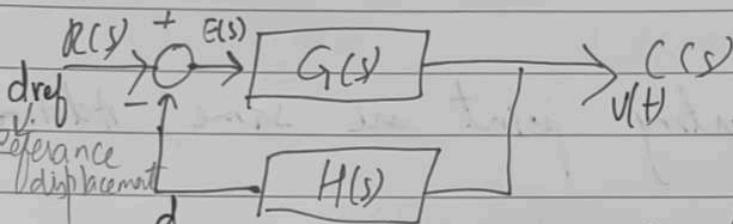
$$C = G(R - C) \Rightarrow C = RG - GC$$

$$\left. \begin{aligned} C + GC &= RG \\ (1+G)C &= RG \\ T_{cl} = \frac{C(s)}{R(s)} &= \frac{G(s)}{1+G(s)} \end{aligned} \right\}$$

5) Non-unity Feedback

-ve feed

("In practical system always use negative feedback to stable the system")



transfer function of sensor

$$\because V(s) \xrightarrow{\text{Integrator}} d(s) \quad \therefore H(s) = \frac{1}{s}$$

$$E(s) = R(s) - C(s)H(s)$$

$$C(s) = -E(s)G(s)$$

put (1) into (2)

$$C(s) = (R(s) - C(s)H(s))G(s)$$

$$C(s) = G(s)R(s) - G(s)C(s)H(s)$$

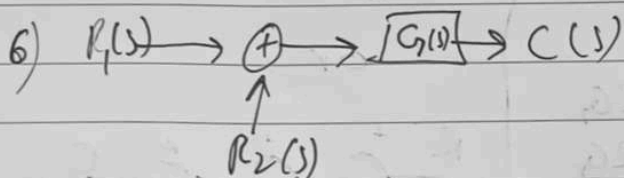
$$C(s) + G(s)C(s)H(s) = G(s)R(s)$$

$$C(s)(1 + G(s)H(s)) = G(s)R(s)$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

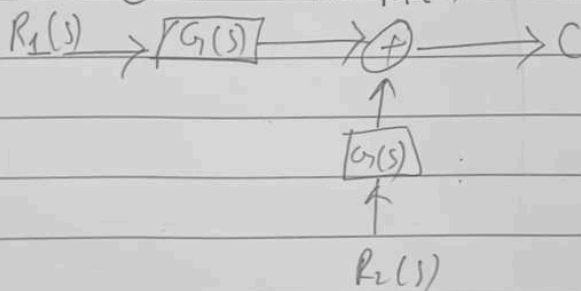
+ve feedback

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)}$$



$$C(s) = G(s)(R_1(s) + R_2(s))$$

$$C(s) = G(s)R_1(s) + G(s)R_2(s)$$



$\Rightarrow$ Use in moving the blocks.

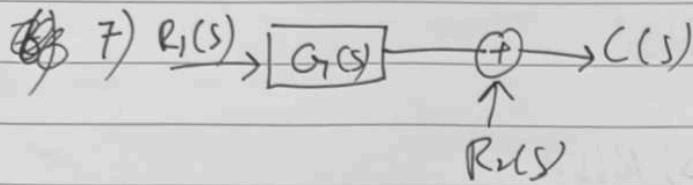
$\because$ output may block ho to or hum nay usay input may push karna hay wo same push ho ga and voice versa

Remainder
(10th 2nd batch of OS)

Mon Tue Wed Thu Fri Sat Sun



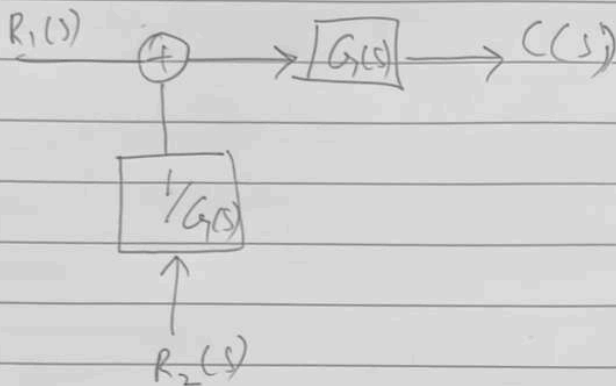
Date: / /



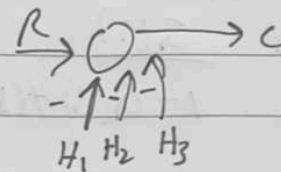
$$C(s) = R_1 G + R_2$$

$$C(s) = G(s) R_1(s) + \frac{1}{G(s)} R_2(s)$$

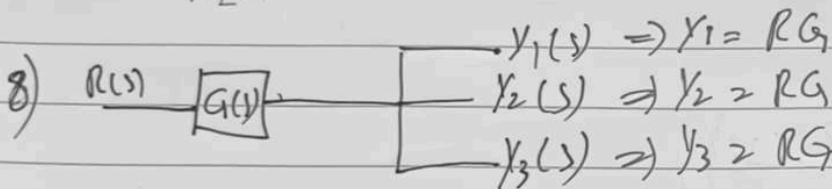
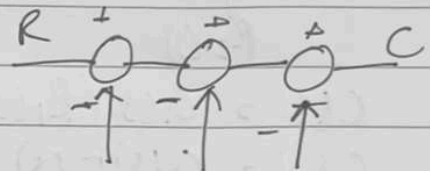
∴ Agar kisi block the ~~neighbouring~~ input may shift hona hai to us ka reciprocal ata hai.



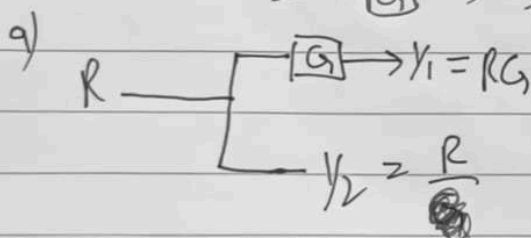
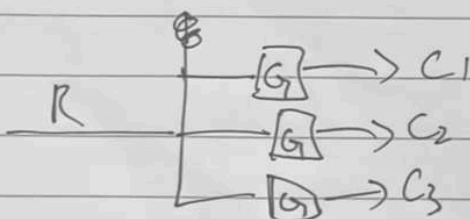
10-



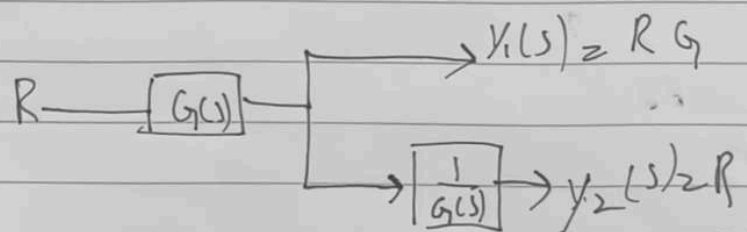
OR



OR



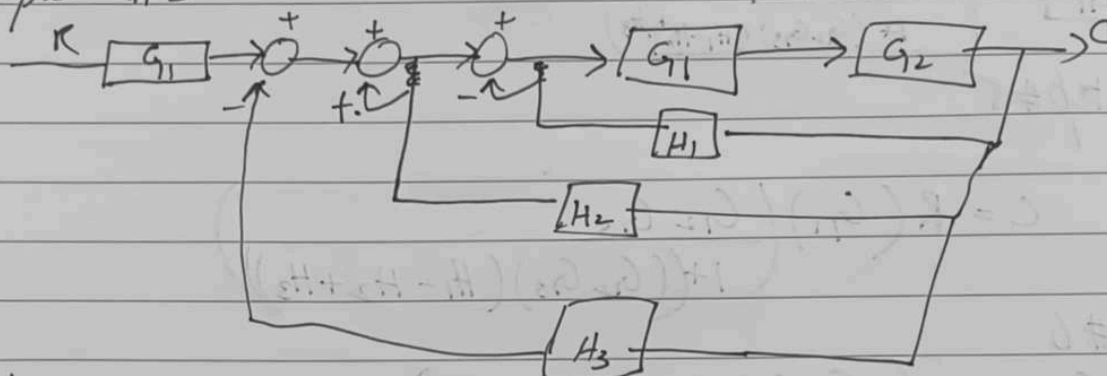
or $\Rightarrow$



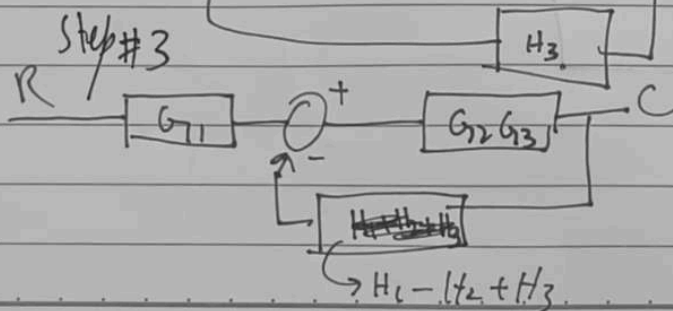
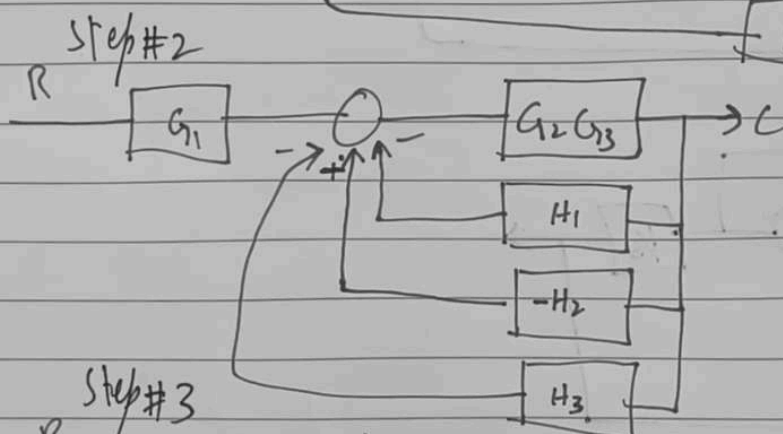
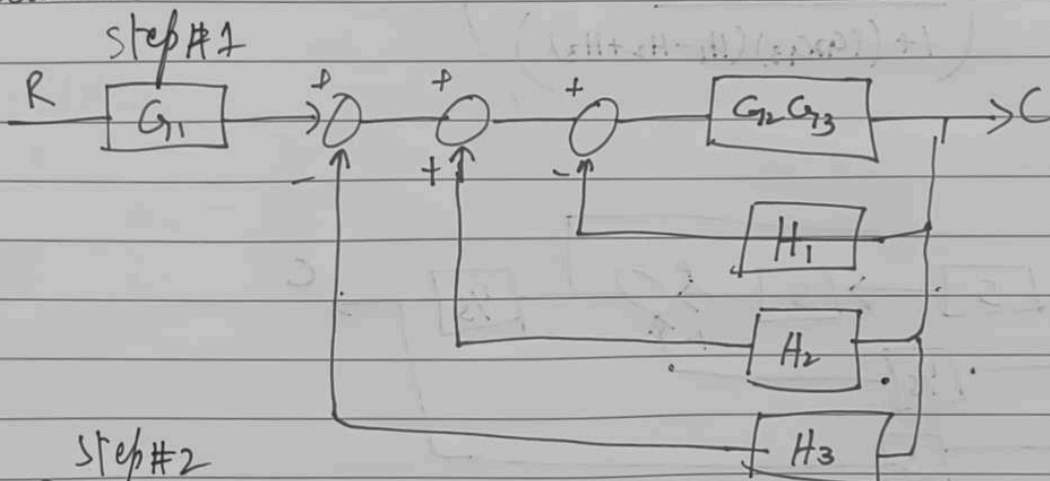


Day - 13

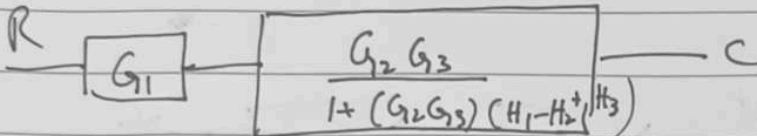
Example #1



Solution:



step #4



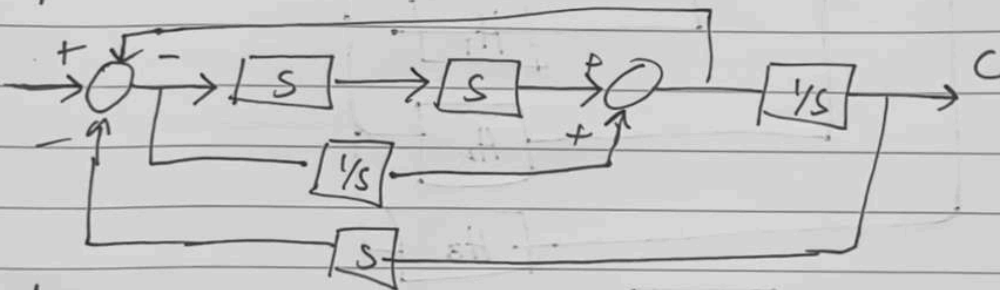
step #5

~~$$C = R(G_1) \left(\frac{G_2 G_3}{1 + ((G_2 G_3)(H_1 - H_2 + H_3))} \right)$$~~

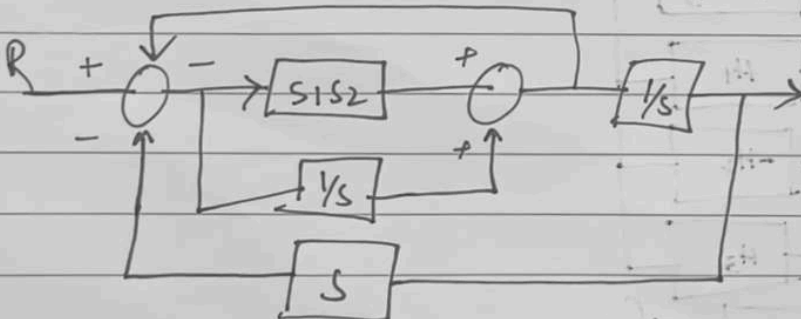
step #6

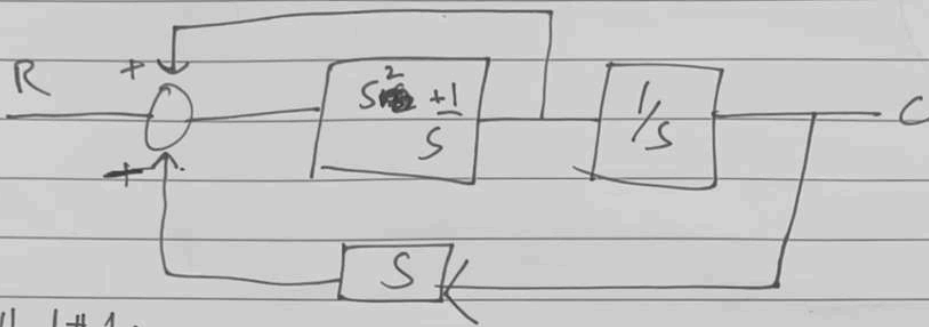
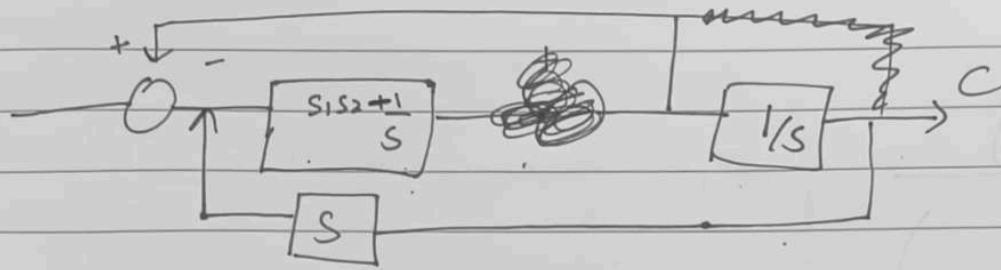
$$\frac{C}{R} = \left(\frac{G_1 G_2 G_3}{1 + (G_2 G_3)(H_1 - H_2 + H_3)} \right)$$

Example #2

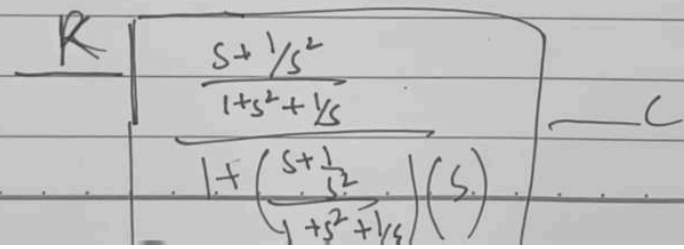
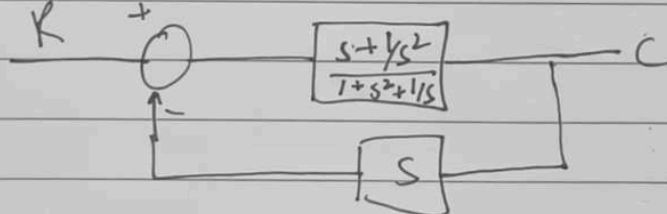
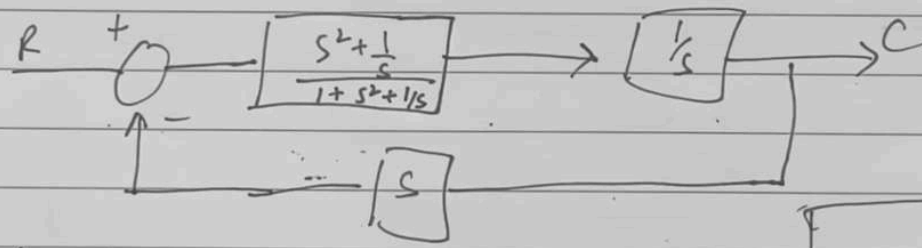


Solution:





Method #1:



$$R \rightarrow \frac{\frac{S^3 + 1}{S^2}}{\frac{S + S^3 + 1}{S}} = 1 + \left(\frac{S^2 + 1}{S} \right) \frac{1}{1 + S^2 + \frac{1}{S}}$$



$$1 + \left(\frac{\frac{s^3+1}{s(s+s^3+1)}}{\frac{s}{s+s^3+1}} \right)$$

$$2 \frac{s^3+1}{s^2+s^4+s}$$

$$1 + \frac{s^3+1}{s^3+s+1}$$

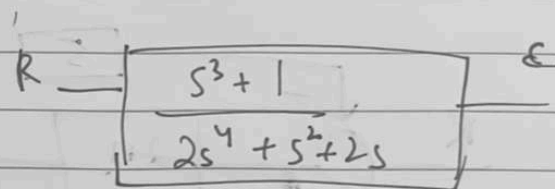
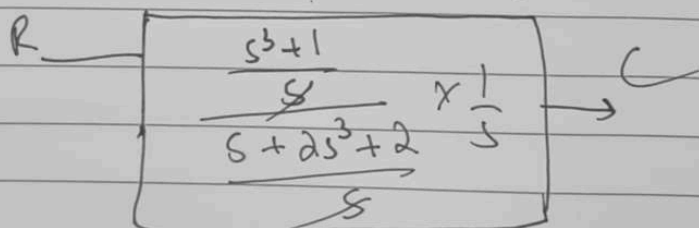
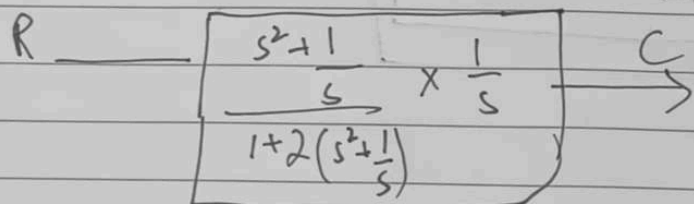
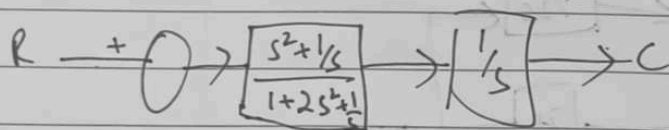
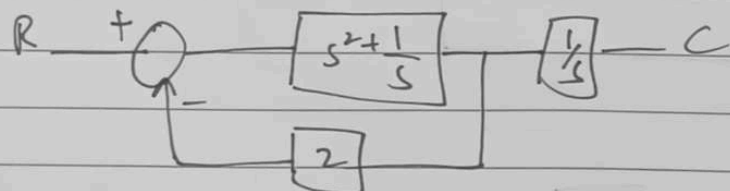
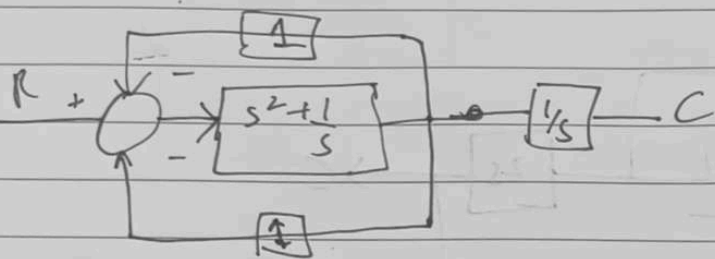
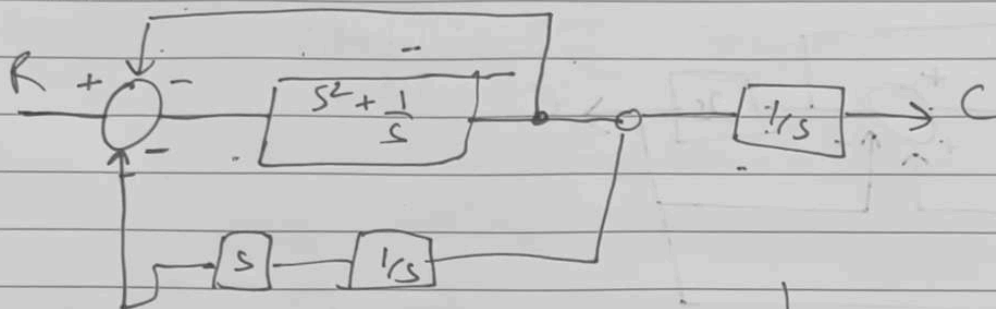
$$2 \frac{s^3+1}{s^2+s^4+s}$$

$$\frac{s^3+s+1 + s^3+1}{s^3+s+1}$$

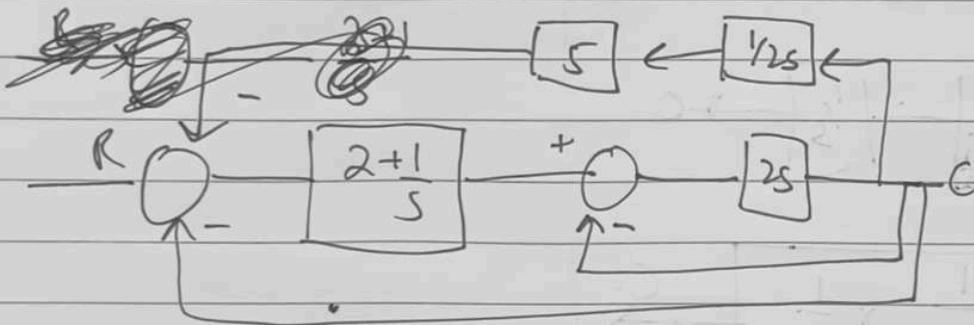
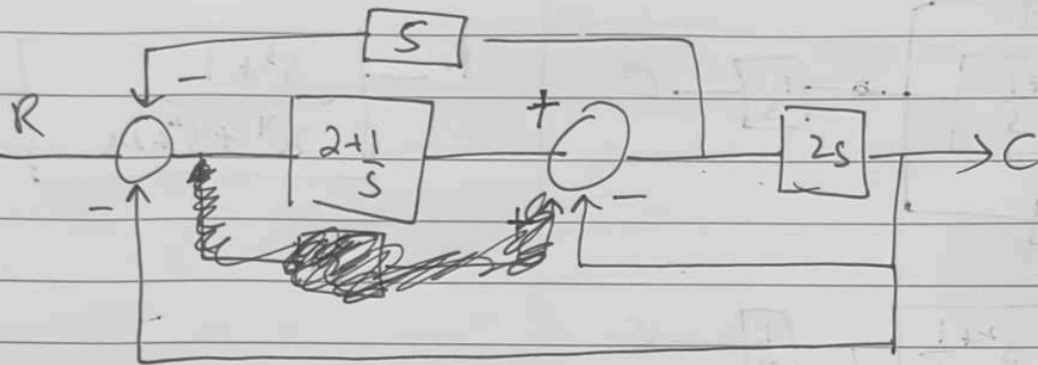
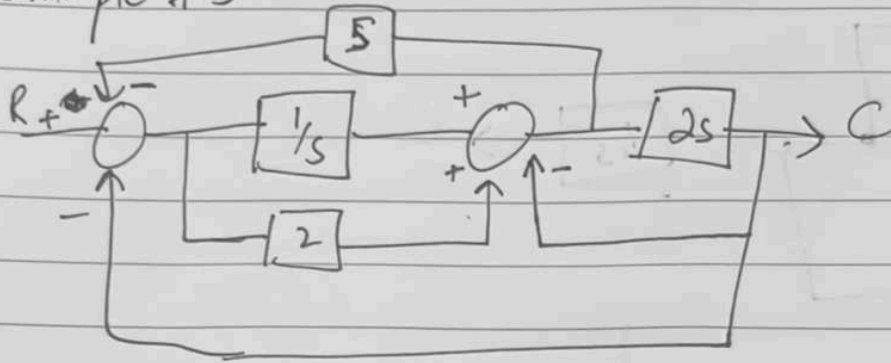
$$2 \frac{s^3+1}{2s^4+s^2+2s}$$



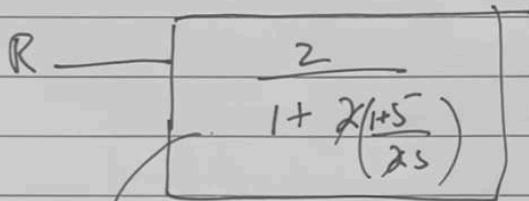
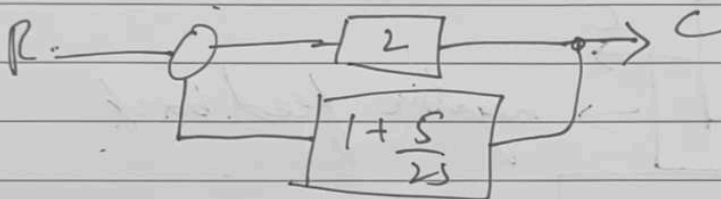
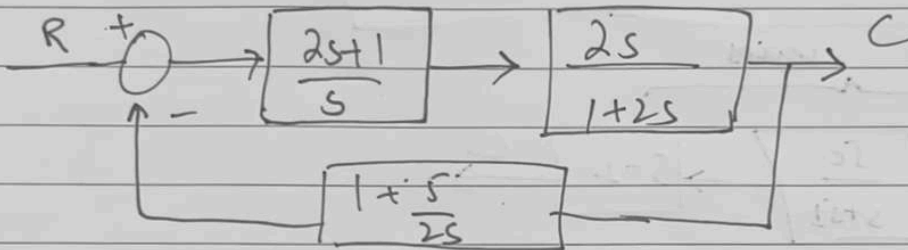
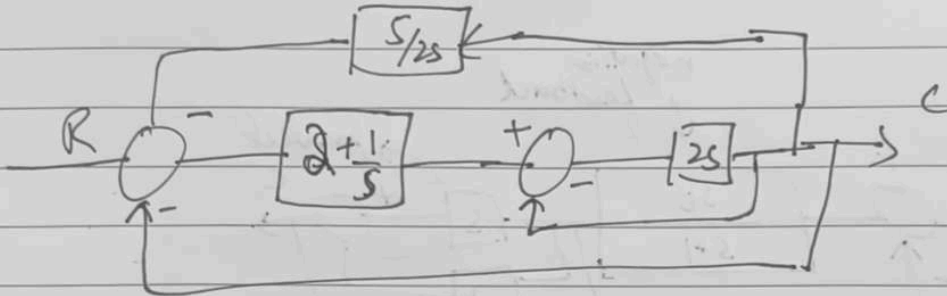
Method 2:



Example #3

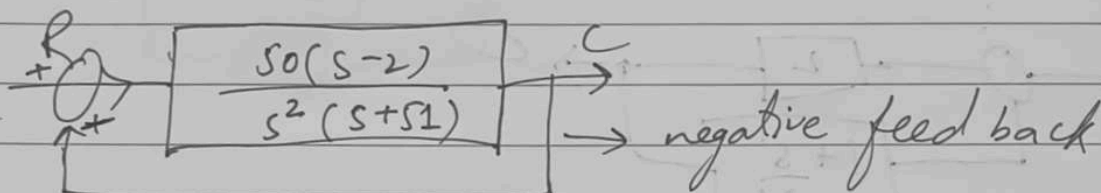
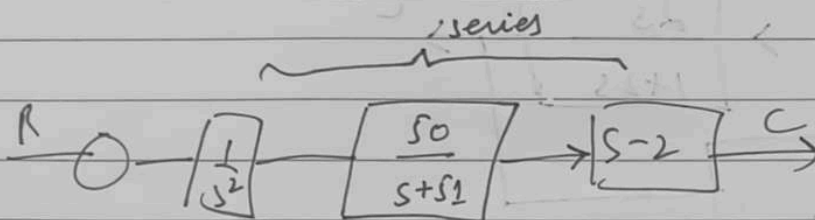
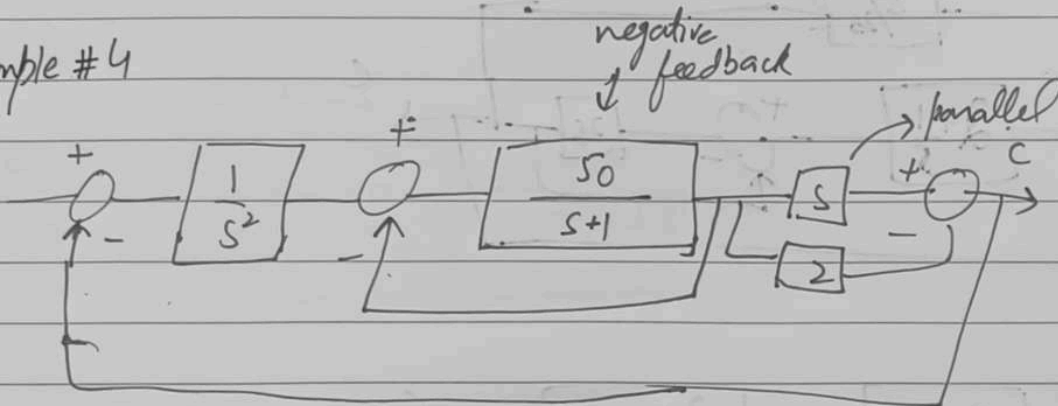


~~Handwritten scribbles~~



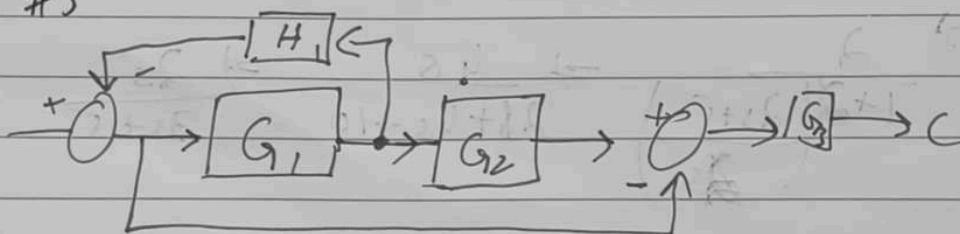
$$\frac{2}{1 + \frac{2(s+5)}{2s}} \Rightarrow \frac{4s}{2s + 4s + 10} \Rightarrow \frac{2s}{3s + 5}$$

Example #4



$$\frac{C}{R} = \frac{50(s-2)}{s^3 + 51s^2 + 50s - 100}$$

Example #5

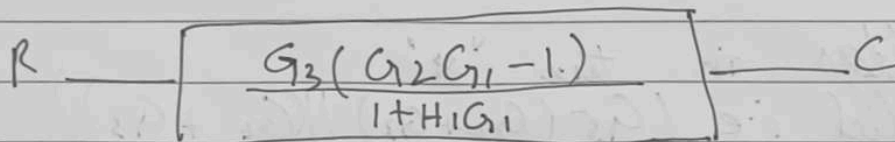
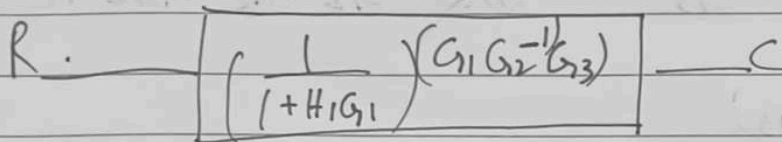
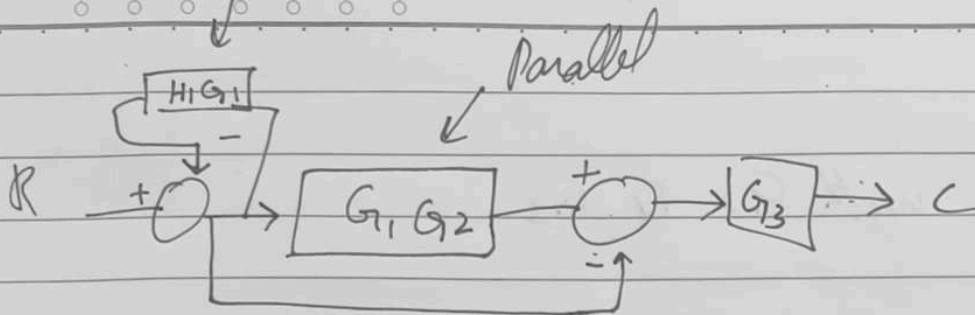


Mon Tue Wed Thu Fri Sat Sun

non-volatility feedback

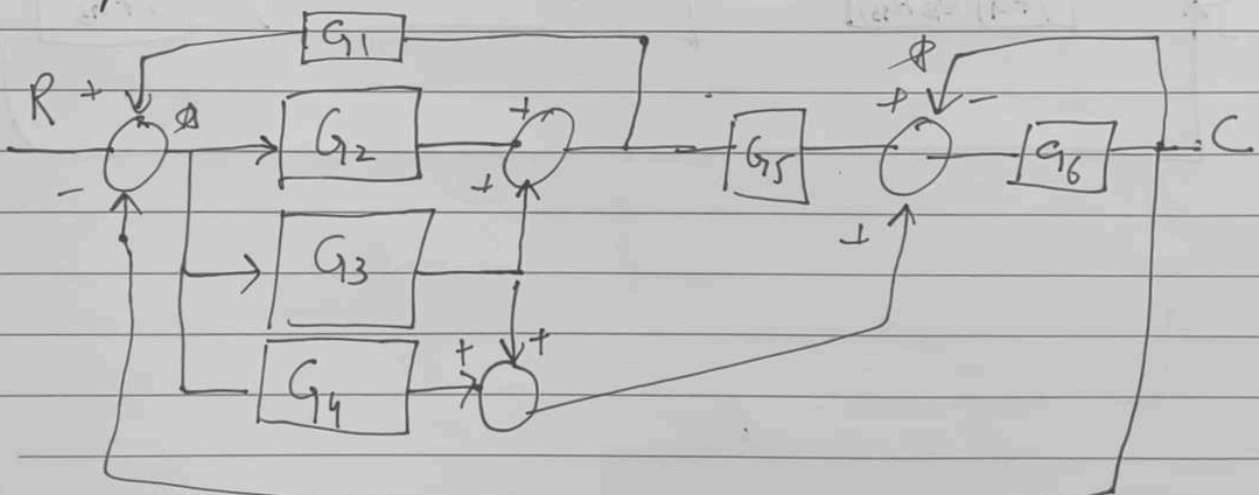


Date: / /



$$\frac{C}{R} = \frac{G_3 (G_2 G_1 - 1)}{1 + H_1 G_1}$$

Example # 6



* G_3 is parallel with G_2 & G_3
so it, become

$$\rightarrow G_2 + G_3$$

$$\rightarrow G_4 + G_3$$

* $G_2 + G_3$ move to G_5 and $G_2 + G_3$ move to G_1

$$\rightarrow G_5 (G_2 + G_3)$$

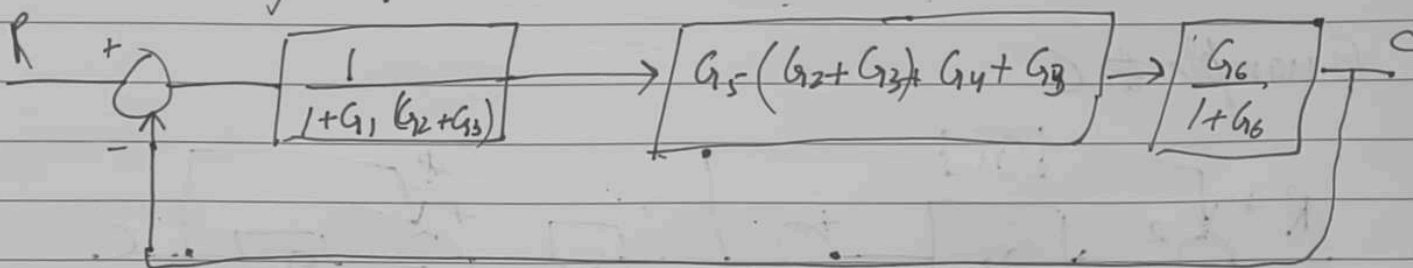
$$\rightarrow G_1 (G_2 + G_3)$$

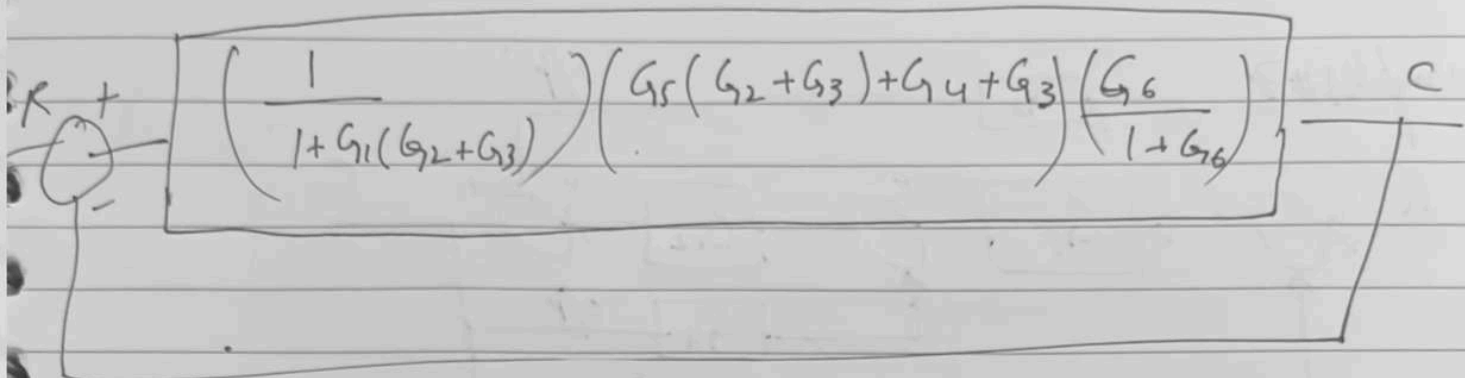
* Break the adders in two '*'

* solve the parallel i.e. $(G_5 (G_2 + G_3) \parallel (G_4 + G_3))$

* unity feedback of G_6

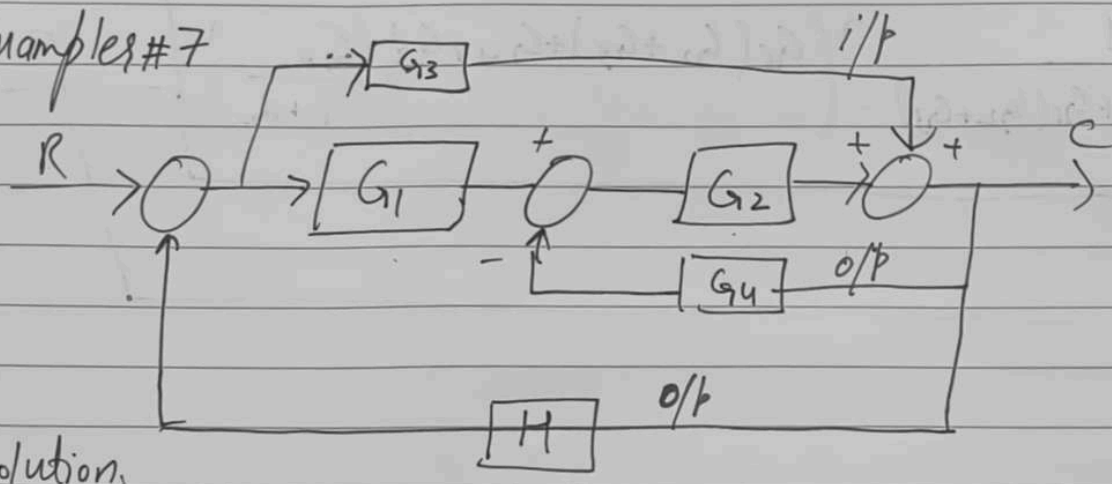
* non unity feed back



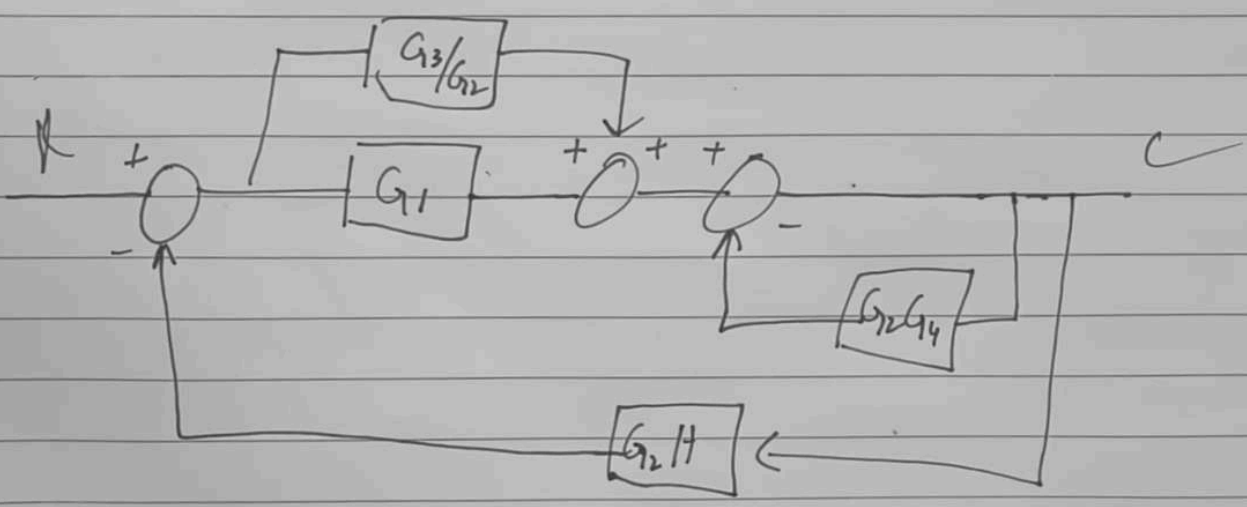
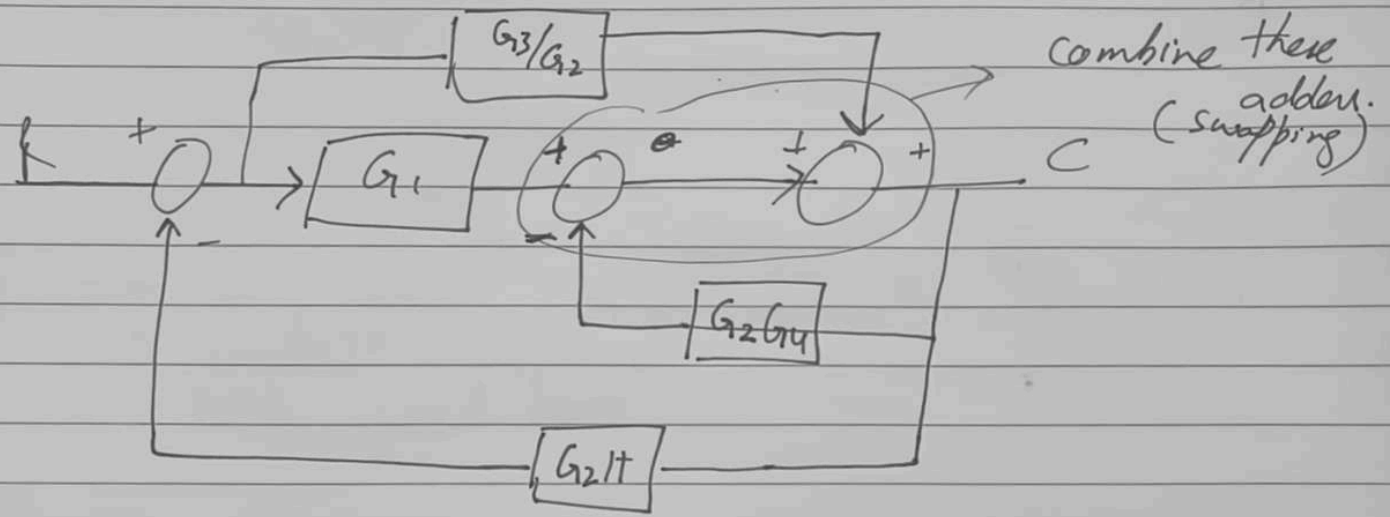


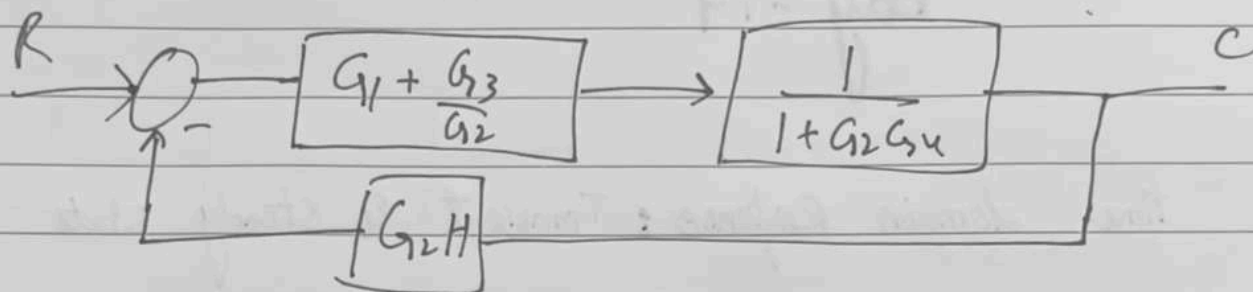
$$\frac{C}{R} =$$

Examples #7



solution,





$$\frac{C}{R} = \left(G_1 + \frac{G_3}{G_2} \right) \left(\frac{1}{1 + G_2 G_4} \right)$$

$$1 + (G_2 H) \left(G_1 + \frac{G_3}{G_2} \right) \left(\frac{1}{1 + G_2 G_4} \right)$$

Day - 14

Chapter #4

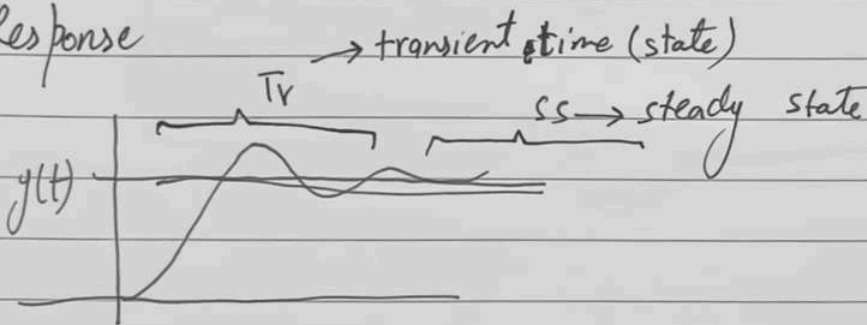
Time domain Response : Transient & steady state

* Derive the mathematical model
* Once the model is derived, we would now look at various method to analyze the performance

To find how our system is doing we use Test signal that are given below

- 1- Impulse input (sudden but momentarily)
- 2- Step signal input (sudden but constant)
- 3- Ramp signal input (linearly increasing)

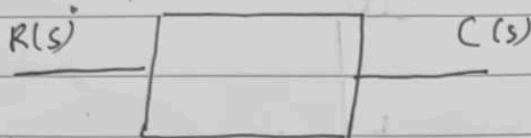
Response



$$y(t) = y_{Tr} + y_{ss} \rightarrow e_{ss} \rightarrow \text{steady state error} \rightarrow \text{performance parameter}$$

Performance parameters {
 $t_{rise} \rightarrow$ rise time
 $t_{peak} \rightarrow$ peak time
 $t_{settle} \rightarrow$ settling time
 $OS \rightarrow$ over shoot

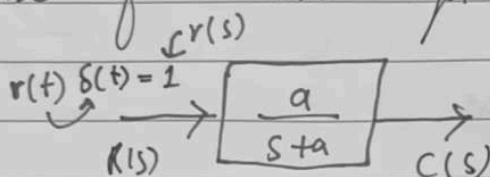
First order system



zero $\rightarrow$ roots of numerator
pole $\rightarrow$ roots of denominator

$$\frac{C(s)}{R(s)} = \frac{a}{s+a}$$

Response of 1st order system at unit impulse



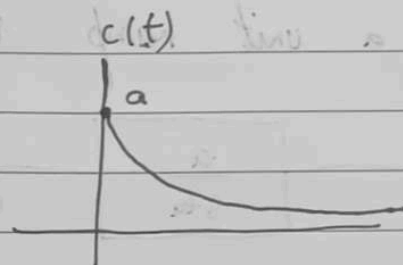
$$\frac{C(s)}{R(s)} = T(s)$$

$$C(s) = R(s) T(s)$$

$$C(s) = 1 \left(\frac{a}{s+a} \right)$$

$$C(s) = \frac{a}{s+a}$$

$$c(t) = a e^{-at}$$

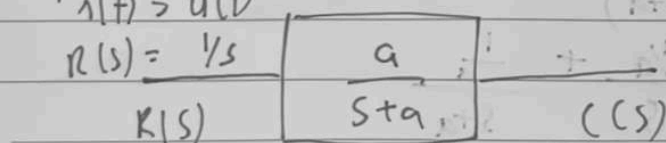


"a" $\rightarrow$ system natural
decaying frequency
 $a \uparrow \rightarrow$ decaying is fast
and vice versa

Response at unit step input

$$r(t) = u(t)$$

$$R(s) = 1/s$$





$$\dot{C}(s) = T(s)$$

$$R(s)$$

$$C(s) = R(s)T(s)$$

$$C(s) = \frac{1}{s} \left(\frac{a}{s+a} \right)$$

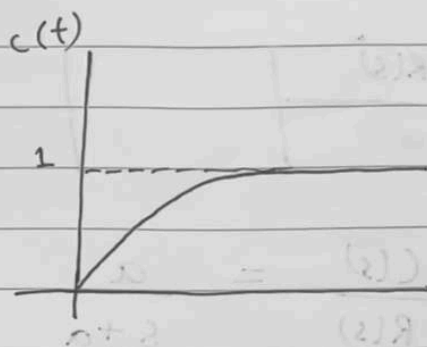
$$C(s) = \frac{a}{s(s+a)}$$

$$C(s) = \frac{1}{s} \left(\frac{1}{s} - \frac{1}{s+a} \right)$$

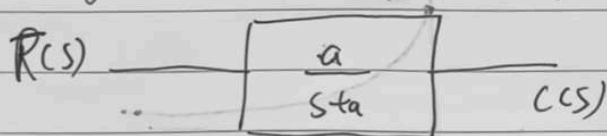
$$c(t) = 1 - e^{-at}$$

forced
Response

natural system response



Response of a unit ramp response:-



$$r(t) = t$$

$$r(s) = \frac{1}{s^2}$$

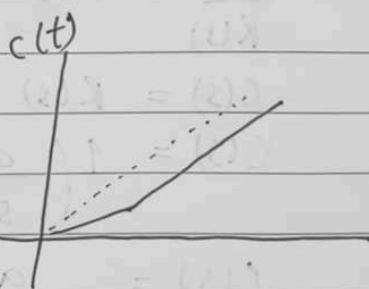
$$C(s) = R(s)T(s)$$

$$C(s) = \frac{1}{s^2} \left(\frac{a}{s+a} \right)$$

$$= \frac{a}{s^2(s+a)}$$

$$C(s) = \frac{1}{s^2} - \frac{1/a}{s} + \frac{1/a}{s+a}$$

$$c(t) = t - 1 + e^{-at}$$

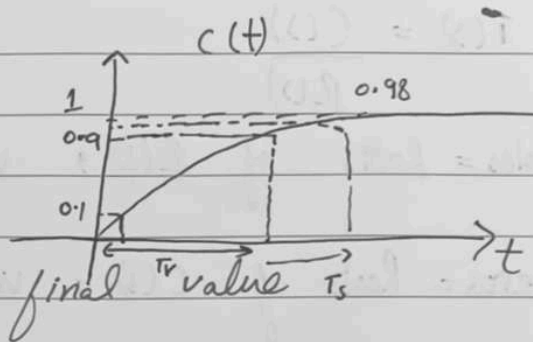


Analysis of step Response

Rise time:

Time taken by response to get from 10% to 90% of

$$T_r = \underline{2.2}$$



settling time: T_s

Time taken by response to reach the final value (or 98% of final value)

$$\overline{I_s} = \underline{4}$$

Time domain response analysis

$r(t)$	$c(t)$
Ramp = t $\downarrow d/dt$	$t - \frac{1}{a} + \frac{1}{a} e^{-at}$ $\downarrow d/dt$
Step = $1 = u(t)$ $\downarrow d/dt$	$1 - e^{-at}$ $\downarrow d/dt$
impuls = $\delta(t)$	$a e^{-at}$ $\downarrow d/dt$

If we derivate the ramp we get step and then again we derivate the step it come out with impulse response and vice versa.



$$T(s) = \frac{C(s)}{R(s)}$$

denominator
↑ polynomial

poles = Roots of $R(s)$, values of "s" that make $R(s) = 0$

zeros = Roots of $C(s)$, values of "s" that make $C(s) = 0$
↓
numerator polynomial

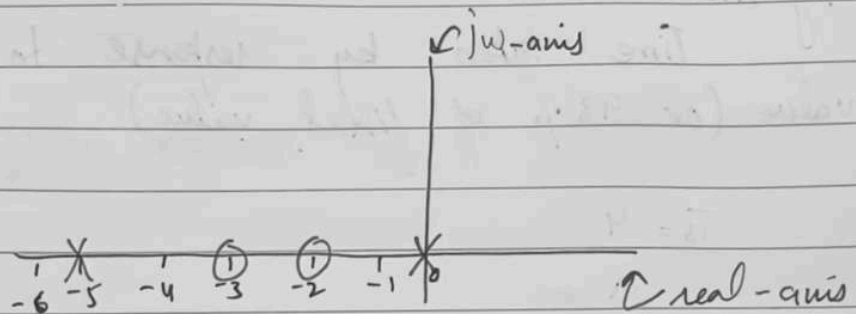
$$T(s) = \frac{(s+2)(s+3)}{s(s+5)}$$

poles = $s = 0, -5$
zeros = $s = -2, -3$

↓

(O) represent

→ (X)



Day - 15

Second order system:-

$$H(s) = \frac{b}{s^2 + as + b}$$

2nd order system $\rightarrow$ have 2 poles. that are

- * Real and Distinct poles. ~~✗~~
- * Real and equal poles. ~~✗~~
- * complex conjugate poles. ~~✗~~
- * Imaginary poles. ~~✗~~

eq

$$H(s) = \frac{b}{s^2 + as + b}$$

ω_n = natural frequency ; $\omega_n = \sqrt{b}$

ξ = damping ratio ; $\xi = \frac{\text{exponential decay frequency}}{\text{natural frequency}} = \frac{a/2}{\omega_n} = \frac{a/2}{\sqrt{b}}$

ω_d = damping natural frequency = $\omega_n \sqrt{1 - \xi^2}$

$$H(s) = \frac{\omega_n^2}{s^2 + (2\xi\omega_n)s + \omega_n^2}$$

eg $\frac{4}{s^2 + 4s + 4}$

$\hookrightarrow \omega_n = 2$

$\xi = 1$

Generalized poles of a 2nd order system.

$$H(s) = \frac{\omega_n^2}{s^2 + (2\xi\omega_n)s + \omega_n^2}$$

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$s = \frac{-2\xi\omega_n \pm \sqrt{(2\xi\omega_n)^2 - 4(1)(\omega_n^2)}}{2}$$

$$s = \frac{-2\xi\omega_n \pm \sqrt{4\xi^2\omega_n^2 - 4\omega_n^2}}{2}$$

$$s = \frac{-2\xi\omega_n \pm \sqrt{4\omega_n^2(\xi^2 - 1)}}{2}$$

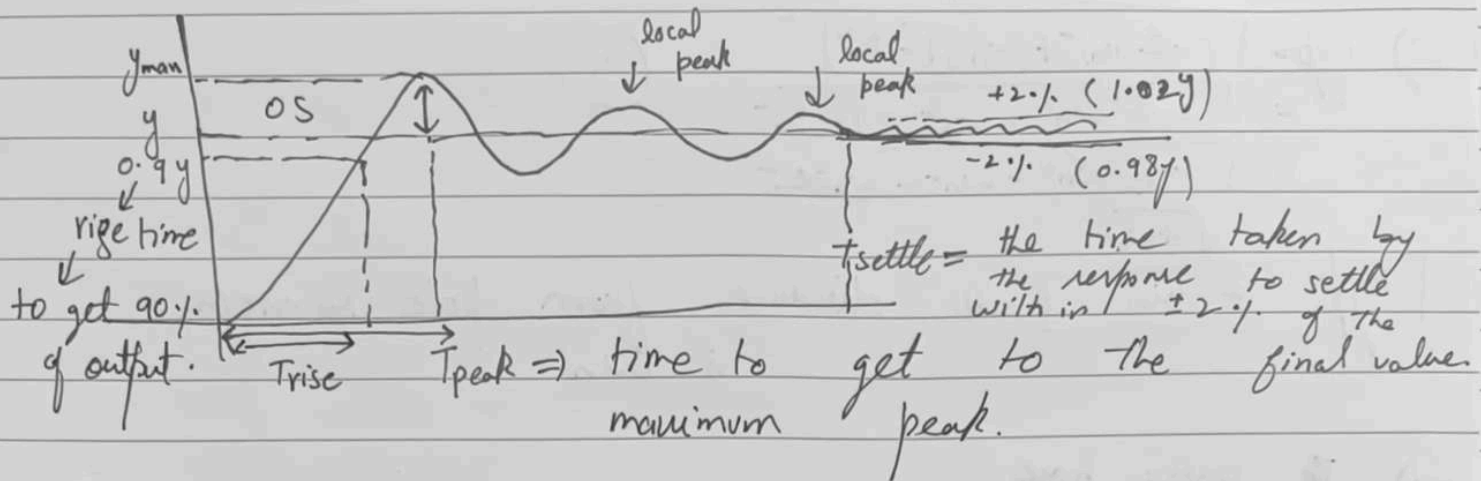
$$s = -2\xi\omega_n \pm 2\omega_n\sqrt{\xi^2 - 1}$$

$$s = -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$$

$$s = \underbrace{-\xi\omega_n}_{\text{real}} \pm \underbrace{j\omega_n\sqrt{1-\xi^2}}_{\text{imaginary}}$$

$\therefore$ damp natural is the imaginary part.

$$s = -\xi\omega_n \pm j\omega_d$$



⇒ Rise time, T_r (1)

$$T_r = \frac{1.8}{\omega_n}$$

⇒ Settling time, T_s (4)

$$T_s = \frac{4}{\zeta \omega_n}$$

⇒ Peak time, T_p (2)

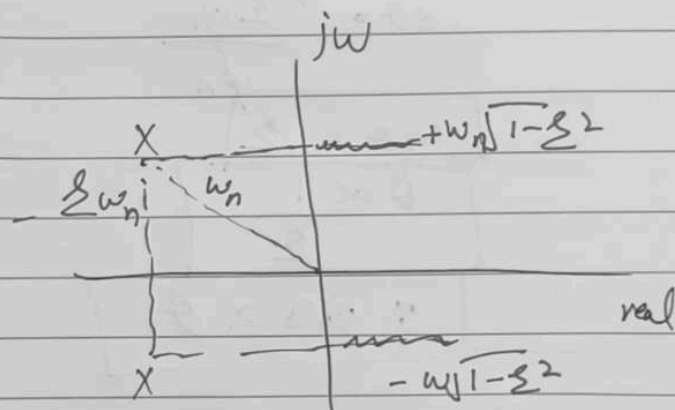
$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

⇒ Overshoot, % OS (3)

$$\% OS = \frac{y_{max} - y_{min}}{y_{final}} \times 100\%$$

$$\% OS = \left(e^{\frac{-\pi \zeta}{\sqrt{1-\zeta^2}}} \right) \times 100\%$$

$$\% OS \propto \frac{1}{\zeta}$$





$$\Rightarrow \text{hyp} = \sqrt{(-\xi \omega_n)^2 + (\omega_n \sqrt{1-\xi^2})^2} \quad (5)$$

$$= \sqrt{-\xi^2 \omega_n^2 + \omega_n^2 (1-\xi^2)}$$

$$= \sqrt{-\xi^2 \omega_n^2 + \omega_n^2 - \omega_n^2 \xi^2}$$

$$= \sqrt{\omega_n^2}$$

$\boxed{\text{hyp} = \omega_n} \Rightarrow$ distance from pole to origin is ω_n

$$\Rightarrow \cos \theta = \frac{\text{base}}{\text{hyp}}$$

$$\cos \theta = \frac{\xi \omega_n}{\omega_n}$$

$$\boxed{\cos \theta = \xi}$$

$$\boxed{\theta \propto \frac{1}{\xi}}$$

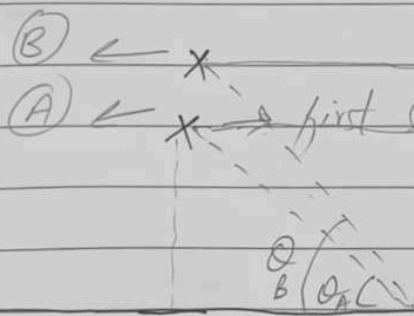
$$\boxed{\therefore \text{P.S} \propto \theta}$$

$\Rightarrow$ distance of pole from origin is " ω_n "

Day - 16

General roots of 2nd order system:-

$$s = -\underbrace{\xi \omega_n}_{\text{Real part}} \pm j \underbrace{\omega_n \sqrt{1-\xi^2}}_{\text{imaginary part}}$$



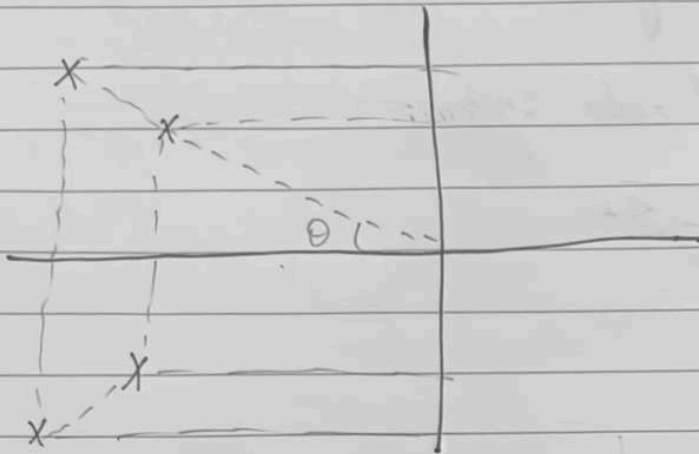
first system here then after some
can changing it move to B
if system gone from A to B

- T_s is same
- T_P is decrease
- ξ is decrease
- %OS is increase
- ω_n is increase
- T_r is decrease

If the real part is change and imaginary
is same

- T_s is decrease
- T_P is same
- ξ is increase
- %OS is decrease
- ω_n is increase
- T_r is decrease

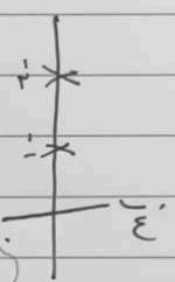
∴ for good response
change the real part
of the system.



T_s is decrease
 T_P is decrease
 Σ is same
 $\%OS$ is same
 W_n is increase
 TR is decrease

Pole location and their Input and response

1) Real & Distinct

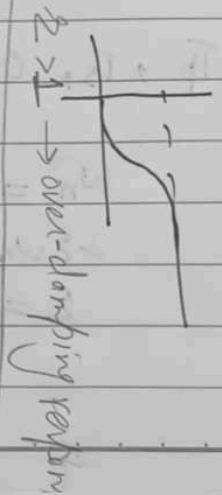


$$H(s) = \frac{2}{s^2 + 3s + 2}$$

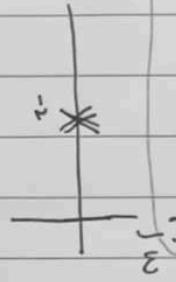
$$\omega_n = \sqrt{2}$$

$$2\zeta\omega_n = 3$$

$$\zeta = 1.06$$



2) Real & Equal



$$H(s) = \frac{4}{s^2 + 4s + 4}$$

$$\omega_n = 2$$

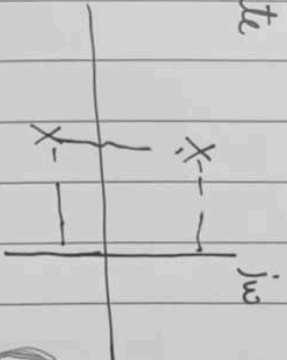
$$2\zeta\omega_n = 4$$

$$\zeta = 1$$

$\zeta = 1 \rightarrow$ critically-damped response



3) complex conjugate



(95% of system lie in this)

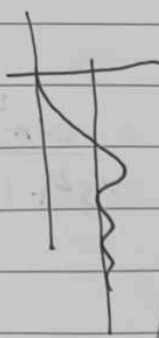
$$H(s) = \frac{9}{s^2 + 2s + 9}$$

$$\omega_n = 3$$

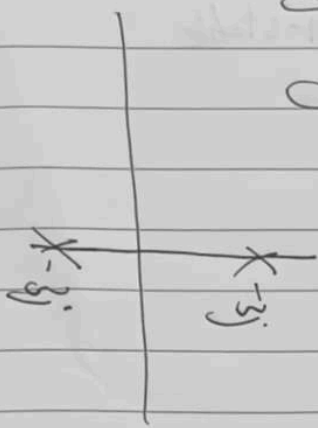
$$2\zeta\omega_n = 2$$

$$\zeta = \frac{1}{3} = 0.33$$

$0 < \zeta < 1 \rightarrow$ Under-damped response



4) Pure imaginary



$$H(s) = \frac{9}{s^2 + 9}$$

$$\omega_n = 3$$

$$2\zeta\omega_n = 0$$

$$\zeta = 0$$

$\zeta = 0 \rightarrow$ Un-damped response



5)

$$H(s) = \frac{2}{s^2 - 3s + 2}$$

$\zeta < 0 \rightarrow$ unstable response

★ Imp Note (Rule of thumb)

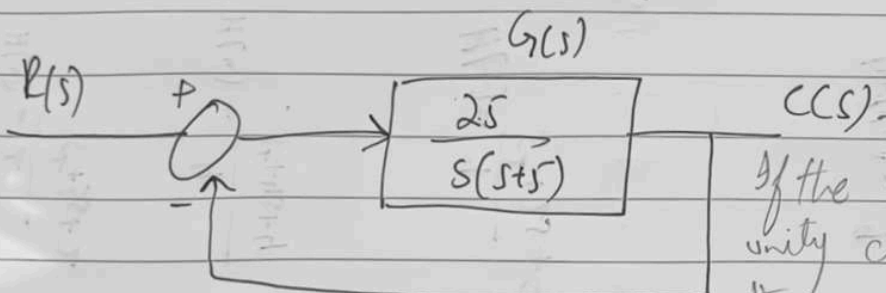
★ When the poles enter into the right half plane the system will be unstable

Stable

left

Un-stable
Right

Q1



a) $T(s) = \frac{2s}{s^2 + 5s + 2s}$ — rule of thumb →

If the system is negative unity close loop system then write the complete function and write the numerator value

→ closed loop transfer function, overall transfer function is same as single block T.F

b) $T_p, T_s, OS\%$

$$\omega_n = 5 \Rightarrow \frac{\omega_n^2}{s^2 + (2\zeta\omega_n)s + \omega_n^2}$$

$$2\zeta\omega_n = 5$$

$$\zeta = 0.5$$

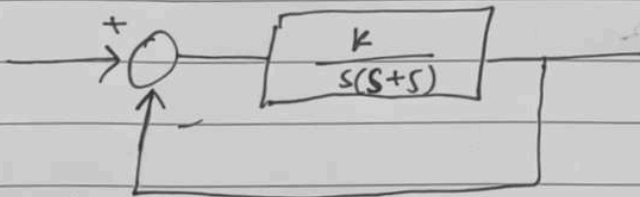
$$T_P = \frac{4}{\sum w_n} = \frac{4}{(0.5)(5)} = 1.6 \text{ sec}$$

$$T_s = \frac{\pi}{\omega_n \sqrt{1 - \xi^2}} = \frac{\pi}{5 \sqrt{0.75}} = 0.725 \text{ sec}$$

$$\begin{aligned} \%OS &= \left(e^{-\frac{\pi \xi}{\sqrt{1 - \xi^2}}} \right) \times 100\% \\ &= \left(e^{-\frac{0.5\pi}{\sqrt{0.75}}} \right) \times 100\% \end{aligned}$$

$$\%OS \approx 16.3\%$$

Q#2



- 1) Find overall T_s
- 2) Design K , if $\%OS = 10\%$

$$a) \quad T(s) = \frac{G(s)}{1 + G(s)} = \frac{K}{s(s+5) + K}$$

$$b) \quad \%OS = 10\%$$



$$\left(e^{-\frac{\pi \Sigma}{\sqrt{1-\Sigma^2}}} \right) \times 100\% = 10\%$$

$$e^{-\frac{\pi \Sigma}{\sqrt{1-\Sigma^2}}} = 0.1$$

Taking natural log on b/s

$$\frac{-\pi \Sigma}{\sqrt{1-\Sigma^2}} = \ln(0.1)$$

$$\frac{-\pi \Sigma}{\sqrt{1-\Sigma^2}} = -2.30$$

$$\frac{\Sigma}{\sqrt{1-\Sigma^2}} = 0.7329$$

$$\frac{\Sigma^2}{1-\Sigma^2} = 0.537$$

$$\Sigma^2 = 0.5371 - 0.5371 \Sigma^2$$

$$1.5371 \Sigma^2 = 0.5371$$

$$\Sigma^2 = \frac{0.5371}{1.5371}$$

$$1.5371$$

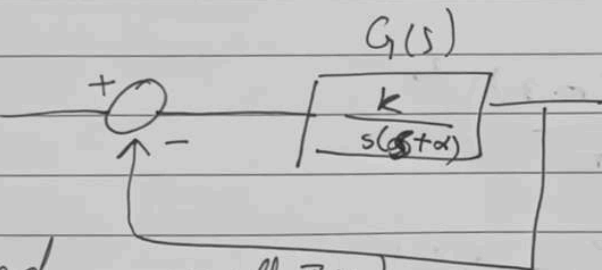
$$\Sigma^2 = 0.349$$

$$\Sigma = 0.5911$$

sq. root

$$\begin{aligned}
 2 \xi \omega_n &= 5 \\
 2(0.5911) \omega_n &= 5 \\
 \omega_n &= 4.229 \\
 (\omega_n)^2 &= k = 17.887 \\
 \boxed{k &= 17.9}
 \end{aligned}$$

Q#3



- a) Find overall $T(s)$
 b) Design k & α if $T.S = 0.2s$ & $O.S = 30\%$

$$a) T(s) = \frac{G(s)}{1 + G(s)} = \frac{k}{s^2 + \alpha s + k}$$

$$\begin{aligned}
 i) \quad k &= \omega_n^2 \\
 \alpha &= 2 \xi \omega_n
 \end{aligned}$$

$$\begin{aligned}
 T.S &= \frac{\sigma}{\xi \omega_n} \\
 0.2 &= \frac{\sigma}{\xi \omega_n}
 \end{aligned}$$

$$\boxed{\xi \omega_n = 20}$$

$$\alpha = 2 \xi \omega_n$$

$$\alpha = 2(20)$$

$$\boxed{\alpha = 40}$$

$$ii) O.S = 30\%$$

$$e^{\frac{-\pi \xi}{\sqrt{1-\xi^2}}} \times 100\% = 30\%$$

$$e^{\frac{-\pi \xi}{\sqrt{1-\xi^2}}} = 0.3$$

taking natural log on b/s

$$\frac{-\pi z}{\sqrt{1-z^2}} = \ln(0.3)$$

$$\frac{z}{\sqrt{1-z^2}} = \frac{\ln(0.3)}{-\pi}$$

$$\frac{z^2}{1-z^2} = \left(\frac{\ln(0.3)}{-\pi} \right)^2$$

$$z^2 = 0.1468 (1-z^2)$$

$$z^2 = 0.1468 - 0.1468 z^2$$

$$1.1468 z^2 = 0.1468$$

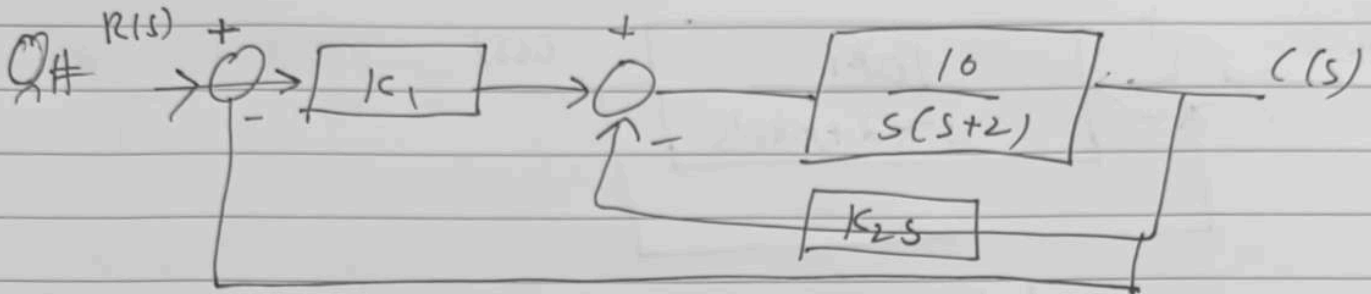
$$\sqrt{z^2} = \sqrt{\frac{0.1468}{1.1468}}$$

$$z = 0.3576$$

$$z \omega_n = 20$$

$$\omega_n = 56.022$$

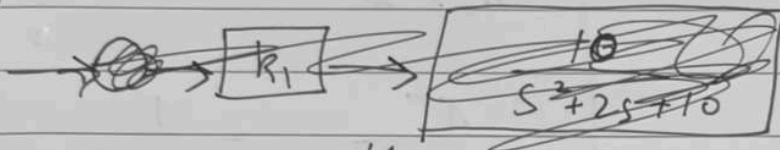
$$K = \omega_n^2 = 3138.4$$



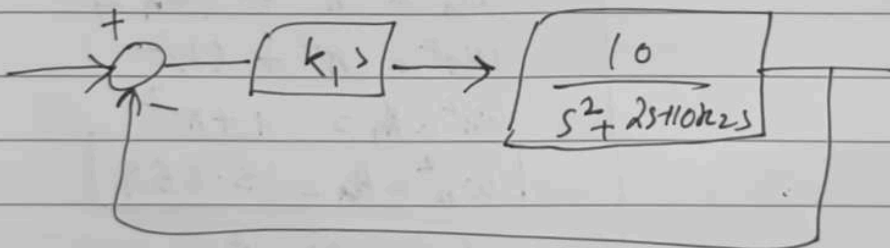
$T_p = 1.0 \text{ sec}$, $T_s = 2.0 \text{ sec}$

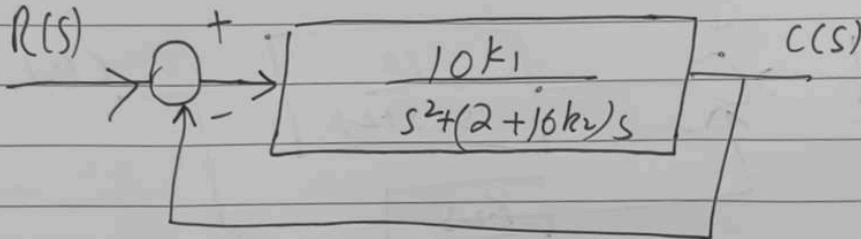
- Find $T(s)$
- Find K_1, K_2

a)



$$\begin{aligned}
 \text{a) } \frac{G}{1+G_H} &= \frac{10}{s(s+2)} \\
 &= \frac{10}{1 + \frac{10}{s(s+2)} \cdot K_2(s)} \\
 &= \frac{10}{\frac{s(s+2) + 10K_2s}{s(s+2)}} = \frac{10}{s^2 + 2s + 10K_2s}
 \end{aligned}$$





$$T(s) = \frac{10k_1}{s^2 + (2 + 10k_2)s + 10k_1}$$

$$\omega_n^2 = 10k_1 \quad \text{--- (A)}$$

$$2\zeta\omega_n = 2 + 10k_2 \quad \text{--- (B)}$$

$$T_s = 2$$

~~2~~

$$\frac{4}{\zeta\omega_n} = 2$$

$$\zeta\omega_n$$

$$\frac{4}{2} = \zeta\omega_n$$

$$\zeta\omega_n = 2$$

$$2(2) = 2 + 10k_2$$

$$4 - 2 = 10k_2$$

$$\frac{2}{10} = k_2$$

$$10$$

$$k_2 = 0.2$$

$$T_p = 1$$

$$\pi = 1$$

$$\omega_n \sqrt{1 - \zeta^2}$$

$$\frac{\pi^2}{\omega_n^2(1 - \zeta^2)} = 1$$

$$\frac{\pi^2}{\omega_n^2 - \omega_n^2\zeta^2} = 1$$

$$\frac{\pi^2}{\omega_n^2 - (\omega_n\zeta)^2} = 1$$

$$\omega_n^2 = (\omega_n\zeta)^2$$

$$\frac{\pi^2}{\omega_n^2} = \omega_n^2 - \omega_n^2\zeta^2$$

$$\omega_n^2 = \frac{\pi^2}{\omega_n^2} + (\omega_n\zeta)^2$$

$$\omega_n^2 = \frac{\pi^2}{\omega_n^2} + (2)^2$$

$$\omega_n^2 = \frac{\pi^2}{\omega_n^2} + 4 + \pi^2$$

$$\omega_n^2 = \frac{\pi^2}{\omega_n^2} + 13.869$$

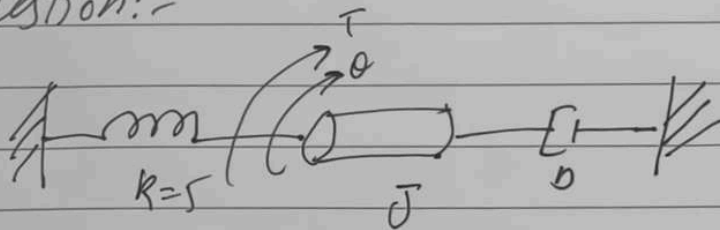
from eq (A)

$$13.869 = 10k_1$$

$$k_1 = 1.3869$$

$$k_1 = 1.39$$

Question:-



$$(Js^2 + Ds + k)\theta(s) = T(s)$$

$$\frac{\theta(s)}{T(s)} = \frac{1}{Js^2 + Ds + 5}$$

find J & D if
 $T_s = 2 \text{ sec.}$
 $\therefore OS = 20\%$

$$\frac{1}{Js^2 + Ds + 5} = \frac{1}{s^2 + (2\zeta\omega_n)s + \omega_n^2}$$

$$\frac{1/J}{s^2 + \frac{D}{J}s + \frac{5}{J}} = \frac{1}{s^2 + (2\zeta\omega_n)s + \omega_n^2}$$

$$2\zeta\omega_n = \frac{D}{J} \quad \text{--- (A)}$$

$$\omega_n^2 = \frac{5}{J} \quad \text{--- (B)}$$

$$\frac{4}{\sum W_n} = 2$$

$$2 = \sum W_n \quad \text{--- (C)}$$

$$2 \sum W_n = \frac{D}{J}$$

$$2(2) = \frac{D}{J}$$

$$\frac{D}{J} = 4 \quad \text{--- (1)}$$

$$D = 4J$$

$$\left(\frac{-\bar{n} \sum}{\sqrt{1-\sum^2}} \right) \times 100\% = 20\%$$

$$e^{\frac{-\bar{n} \sum}{\sqrt{1-\sum^2}}} = 0.2$$

$$\frac{-\bar{n} \sum}{\sqrt{1-\sum^2}} = \ln(0.2)$$

$$\frac{\sum}{\sqrt{1-\sum^2}} = \left(\frac{\ln(0.2)}{-\bar{n}} \right)$$

$$\frac{\sum^2}{1-\sum^2} = \left(\frac{\ln(0.2)}{-\bar{n}} \right)^2$$



$$\frac{\xi^2}{1-\xi^2} = 0.262$$

$$\xi^2 = \frac{0.262}{1+0.262}$$

$$\boxed{\xi = 0.456}$$

put in c

$$2 = (0.456) \omega_n$$

$$\omega_n = \frac{2}{0.456}$$

$$\boxed{\omega_n = 4.38}$$

$$\omega_n^2 = 5/\delta$$

$$19.236 = \frac{5}{\delta}$$

$$\delta = \frac{5}{19.236}$$

$$\boxed{\delta = 0.26}$$

$$D = 4\delta$$

$$D = 1.04$$

Day - 17

Stability:

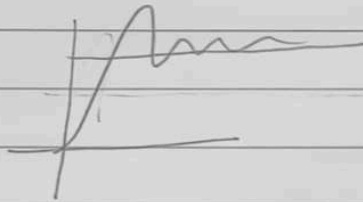
→ If the poles of the system one of them on the right half plane the system will be unstable.



→ If the poles are on jw axis the system will be a marginally system.

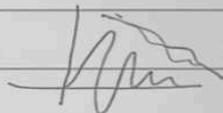


→ If the poles are on left hand then the system will be stable.



$$C(t) = C_{\text{natural}} + C_{\text{forced}}$$

For stability we check C_{natural}

if $C_{\text{natural}} \rightarrow 0$ at $t \rightarrow \infty$  stable

if $C_{\text{natural}} \rightarrow k$ at $t \rightarrow \infty$  marginally stable

if $C_{\text{natural}} \rightarrow \infty$ at $t \rightarrow \infty$  unstable

3) Descartes' Theorem

↳ It don't correct for all cases
it just give us the idea that: sign change.

$H(s) = \frac{1}{s^5 + 4s^4 + 3s^3 + 2s^2 + s + 1}$ can tell us location of RHP poles

$$+s^5 + 4s^4 + 3s^3 + 2s^2 + s + 1$$

x x x x x

Total poles = 5

stability depends on the location of poles.

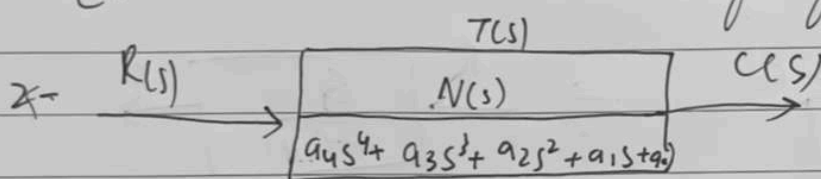
All poles should be in Left half plane

sign changes mean right half plane.

number of poles on LHP = 3
" " " " " RHP = 2

Routh Hurwitz criteria:- (Main topic of chapter)

1- evaluate the overall Transfer function.



3- Pick the denominator

~~(PTD)~~

4- Make Routh table

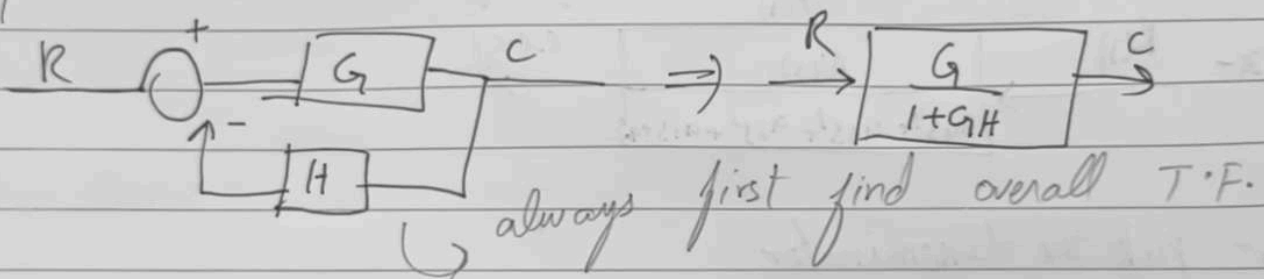
5- Look at the sign of the 1st entry of each row.

Routh table

→ Us ka co-efficient or phir us say 2 kam ka or. agr use hai to 0 must likhna

s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2	$b_1 = \frac{\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3}$ $b_2 = \frac{\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3}$ $b_3 = \frac{\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3}$		
s^1	$c_1 = \frac{\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1}$ $c_2 = \frac{\begin{vmatrix} a_3 & 0 \\ b_1 & b_3 \end{vmatrix}}{b_1}$ $c_3 = \frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1}$		
s^0	$d_1 = \frac{\begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}}{c_1}$ $d_2 = \frac{\begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}}{c_1}$ $d_3 = \frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1}$		

Example:



$$T(s) = \frac{1000}{s^3 + 10s^2 + 3s + 1030}$$

(P.T.V)

s^3	1	31	0	
s^2	10	1030	0	
	1	103	0	
s^1	-72	0	0	
s_0	103	0	0	

only the magnitude
(simplify by factor of 10)

$$- \begin{vmatrix} 1 & 31 \\ 1 & 103 \end{vmatrix} = -72$$

$$- \begin{vmatrix} 1 & 103 \\ -72 & 0 \end{vmatrix} = -72$$

Total poles = 3

RHP = 2

LHP = 1 $\Rightarrow$ 3-2

It's the unstable system.

2. $T(s) = \frac{200}{3s^7 + 9s^6 + 9s^5 + 4s^4 + 7s^3 + 8s^2 + 2s + 6}$

s_7	3	0	7	2	0	
s_6	9	4	8	6	0	
s_5	4.66	4.33	0	0	0	
s_4	-4.36	8	6	0	0	

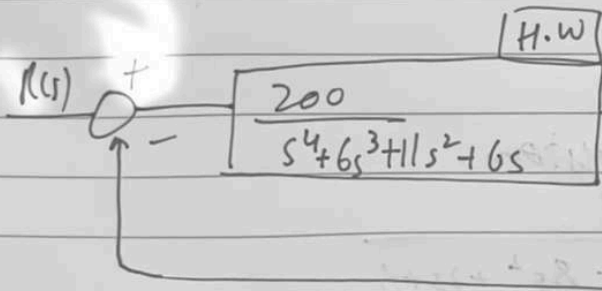
$$- \begin{vmatrix} 3 & 0 \\ 9 & 4 \end{vmatrix} = 12$$

$$- \begin{vmatrix} 9 & 4 \\ 4.66 & 4.33 \end{vmatrix} = 9$$

$$- \begin{vmatrix} 4.66 & 4.33 \\ -4.36 & 8 \end{vmatrix} = 4.66$$



s_3	12.90	6.41	0	0	0	4.66	4.33
						-4.36	8
						-4.36	
s_2	10.16	6	0	0	0	-4.36	8
						12.90	6.41
						12.90	
s_1	-1.20	0	0	0	0	10.16	6.41
						10.16	6
						10.16	
s_0	6	0	0	0	0		



- Find $T(s)$
- Check poles for stability

Solution:-

$$T(s) = \frac{200}{s^4 + 6s^3 + 11s^2 + 6s + 200}$$

s_4	1	11	200	
s_3	6 1	6 11	0	Simplify
	10	200	0	
s_2	1	20	0	→ Simplify
	-19	0	0	
s_1				
	20	0	0	
s_0				

Total poles = 4

signed change = 2

unsigned changed 2 $4 - 2 = 2$

unsigned changed = 2

Case 2 : Zero (only) in the 1st column

$$T(s) = \frac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3}$$

s^5	1	3	5	0
s^4	2	6	3	0
s^3	$\cancel{0} \epsilon$	$3 \cdot 5$	0	0
s^2	$\frac{6\epsilon - 7}{\epsilon}$	3	0	
s^1	$\frac{-3\epsilon^2 + 21\epsilon + 24.5}{6\epsilon - 7}$	0	0	
s^0	3	0	0	0

→ If zero appear we just replace 0 by " ϵ " and its value is very very very infinitesimally small.
→ The sign of the " ϵ " will be

$$-\left(\frac{3\epsilon - 3 \cdot 5}{6 - 7} \right) \Rightarrow -\left(\frac{3\epsilon - 21 - 24.5}{\epsilon} \right)$$

$$-3s^2 + 21s + 24.5$$

$\times$

$$6s - 7$$

$\times$

$$-3s^2 + 21s + 24.5$$

$$6s - 7$$

$\rightarrow 0.1$		$\rightarrow 0.1$
$s > 0$	$s < 0$	
+	+	s^5
+	+	s^4
+	-	s^3
-	+	s^2
+	+	s^1
+	+	s^0

2 sign changes

2 RHP

3 LHP

Case 3 $\Rightarrow$ Entire Row is zero.

$$T(s) = \frac{10}{s^5 + 7s^4 + 6s^3 + 42s^2 + 8s + 56}$$

always on even number.

s^5	1	6	8	0
cause $\rightarrow s^4$	7 1	42 6	56 8	0
effect $\rightarrow s^3$	4 1	12 30	0	0
	1	3	0	0
s^2	+3	8	0	0
s^1	$\frac{1}{3}$	0	0	0
s^0	8	0	0	0

simplify by "7"

simplify by "4"

(1) "cause" row had a order of s^4
we say that the system has 4-jw-poles.

Polynomial of s^4

$\therefore$ 1 pole is on LHP and 4 on jw

$$P(s) = s^4 + 6s^2 + 8$$

$$\frac{dP}{ds} = 4s^3 + 12s$$

Now remove the "0" and place the

$$T(s) =$$

$$20$$

$$s^8 + s^7 + 12s^6 + 22s^5 + 39s^4 + 59s^3 + 48s^2 + 38s + 20$$

s^8	1	12	39	48	20
s^7	1	22	59	38	0
s^6	-10 -1	-20 -2	+10 +1	20 2	0
s^5	20	166	340	2	0
s^4	+1	3	2	0	0
s^3	0 4	2 0	6 3	0	0
s^2	+1.5	2	0	0	0
s^1	1/3	0	0	0	0
s^0	0	0	0	0	0

$$P(s) = s^4 + 3s^2 + 2$$

$$\frac{dP}{ds} = 4s^3 + 6s$$

4 jw ~~axis~~ axis

2 LHP

2 RHP

(Unstable)

$$T(s) = \frac{1}{2s^5 + 3s^4 + 2s^3 + 3s^2 + 2s + 1}$$

s^5	2	2	2	0
s^4	3	3	1	0
s^3	$0E$	$\frac{4}{3}$	0	0
s^2	$4 - \frac{3}{E}$	1	0	0
s^1	$\frac{16 - 12E - 3E^2}{12 - 9E}$	0	0	0
s^0	+1	0	0	0

$$\frac{16 - 12E - 3E^2}{12 - 9E}$$

$$E = 0, 1$$

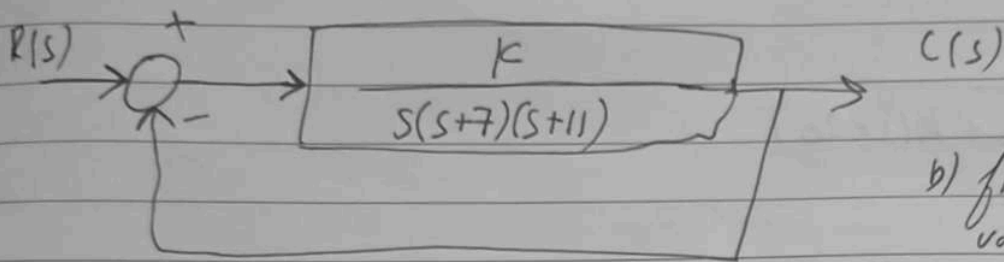
$E > 0$	$E < 0$	
+	+	s^5
+	+	s^4
+	-	s^3
-	+	s^2
+	+	s^1
+	+	s^0

3 LHP

2 RHP

(Unstable)

Stability Designing



b) find The range of value of "k" for which the system is stable.

$$\frac{K}{s^3 + 18s^2 + 77s + K}$$

s^3	1	77	0
s^2	18	K	0
s^1	$\frac{-K + 1386}{18}$	0	0
s^0	K	0	0

$\hookrightarrow \frac{1386 - K}{18}$

$$\frac{1386 - K}{18} > 0$$

$$K < 1386$$

$$K > 0$$

$$\boxed{0 < K < 1386}$$

$$s^2 + 3s + 2s + 6$$

$$s^2 + 5s + 6s$$

a) $T(s) = ?$

b2?

$$\frac{k(s+20)}{s^3 + 5s^2 + 6s + k(s+20)}$$

$$\frac{k(s+20)}{s^3 + 5s^2 + (6+k)s + 20}$$

s^3	1	$6+k$	0
s^2	5	$20k$	0
s^1	$6-2k$	0	0
s^0	$20k$	0	0

$$-(20-30-sk)$$

$$s$$

$$= -20 + 10 + sk$$

$$s$$

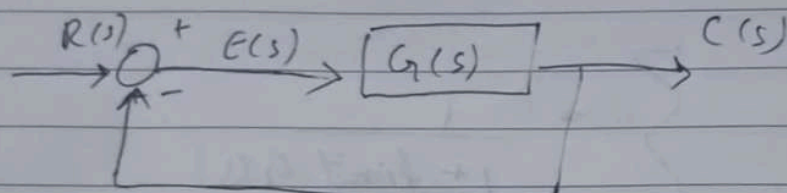
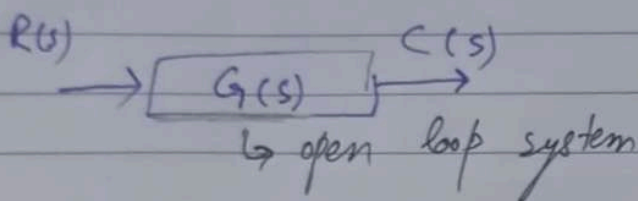
$$6-3k > 0 \Rightarrow k < 2$$

$$20k > 0 \Rightarrow k > 0$$

$$0 < k < 2$$

Chapter # 7

Steady State Error



$$E(s) = R(s) - C(s) \quad \text{--- (1)}$$

$$C(s) = E(s)G(s) \quad \text{--- (2)}$$

put (2) in eq (1)

$$E(s) = R(s) - E(s)G(s)$$

$$E(s)[1 + G(s)] = R(s)$$

$$E(s) = \frac{R(s)}{1 + G(s)}$$

final value theorem:

$$f(t), f(\infty) = \lim_{s \rightarrow 0} sF(s)$$

value of $f(t)$ as $t \rightarrow \infty$
final value, steady state value

→ steady state = $e(\infty)$
→ final value of error

$$e(t) \xrightarrow{L} E(s)$$

$$e_{ss} = e(\infty) \\ = \lim_{s \rightarrow 0} sE(s)$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1+G(s)}$$

a) Unit Step input

$$r(t) = u(t)$$

$$R(s) = \frac{1}{s}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{s(1/s)}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{1+G(s)}$$

$$= \frac{\lim_{s \rightarrow 0} 1}{\lim_{s \rightarrow 0} (1+G(s))}$$

$$= \frac{1}{1 + \lim_{s \rightarrow 0} G(s)}$$

$$\text{if } k_p = \lim_{s \rightarrow 0} G(s)$$

then,

$$e(\infty) = \frac{1}{1+k_p} \rightarrow \text{position}$$

→ for step input.

b) Unit Ramp input

$$r(t) = t u(t)$$

$$R(s) = \frac{1}{s^2}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^2)}{1 + G(s)}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{1}{s(1 + G(s))}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{1}{\lim_{s \rightarrow 0} (s(1 + G(s)))}$$

$$e(\infty) = \frac{1}{\lim_{s \rightarrow 0} (s(1 + G(s)))}$$

$$e(\infty) = \frac{1}{\lim_{s \rightarrow 0} (s + sG(s))}$$

$$e(\infty) = \frac{1}{\lim_{s \rightarrow 0} s + \lim_{s \rightarrow 0} sG(s)}$$

$$e(\infty) = \frac{1}{\lim_{s \rightarrow 0} s(G(s))}$$

$$\text{if } K_v = \lim_{s \rightarrow 0} s(G(s))$$

then,

$$e(\infty) = \frac{1}{K_v \rightarrow \text{velocity}}$$

for ramp input

c) For parabolic input

$$r(t) = \frac{1}{2} t^2 u(t)$$

$$R(s) = \frac{1}{s^3}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{s R(s)}{1+G(s)}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{s \left(\frac{1}{s^3} \right)}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2(1+G(s))}$$

$$= \frac{\lim_{s \rightarrow 0} (1)}{\lim_{s \rightarrow 0} (s^2 + s^2 G(s))}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s^2 + \lim_{s \rightarrow 0} s^2 G(s)}$$

$$e(\infty) = \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)}$$

$$\text{if } K_A = \lim_{s \rightarrow 0} s^2 G(s)$$

then,

$$e(\infty) = \frac{1}{K_A} \rightarrow \text{acceleration}$$

System Type

$\therefore$ order of system doesn't matter

1) Type 0

if $G(s)$ has No pole at the origin.

$$G(s) = \frac{1}{s+2}, \quad \frac{1}{s^3+s^2+3s+8}$$

2) Type I

if $G(s)$ has 1 pole at the origin.

$$G(s) = \frac{1}{s(s+2)}$$

3) Type II

if $G(s)$ has 2 poles at the origin

$$G(s) = \frac{1}{s^2(s+2)}$$

$v(t)$	error	Type 0	Type I	Type II
$\frac{1}{1+K_P}$	$K_P = \text{constant}$ $e(\infty) = \text{constant}$	$K_P = \infty$ $e(\infty) = 0$	$K_P = \infty$ $e(\infty) = 0$	
$\frac{1}{K_V}$	$K_V = 0$ $e(\infty) = \infty$	$K_V = \text{constant}$ $e(\infty) = \text{constant}$	$K_V = \infty$ $e(\infty) = 0$	

$\frac{1}{K_A}$	$K_A = 0$ $e(\infty) = \infty$	$K_A = \text{constant}$ $e(\infty) = \text{constant}$
-----------------	-----------------------------------	--

$K_P = \lim_{s \rightarrow 0} G(s)$	Type 0 = $\lim_{s \rightarrow 0} \frac{1}{s+2} = 0.5$	Type I = $\lim_{s \rightarrow 0} \frac{1}{s(s+2)} = \infty$	Type II = $\lim_{s \rightarrow 0} \frac{1}{s^2(s+2)} = \infty$
-------------------------------------	---	---	--

$K_V = \lim_{s \rightarrow 0} s G(s)$	Type 0 = $\lim_{s \rightarrow 0} s \left(\frac{1}{s+2} \right) = 0$	Type I = $\lim_{s \rightarrow 0} s \left(\frac{1}{s(s+2)} \right) = 0.5$	Type II = $\lim_{s \rightarrow 0} s \left(\frac{1}{s^2(s+2)} \right) = \infty$
---------------------------------------	--	---	---

$K_A = \lim_{s \rightarrow 0} s^2 G(s)$	Type 0 = $\lim_{s \rightarrow 0} s^2 \left(\frac{1}{s+2} \right) = 0$	Type I = $\lim_{s \rightarrow 0} s^2 \left(\frac{1}{s(s+2)} \right) = 0$	Type II = $\lim_{s \rightarrow 0} s^2 \left(\frac{1}{s^2(s+2)} \right) = 0.5$
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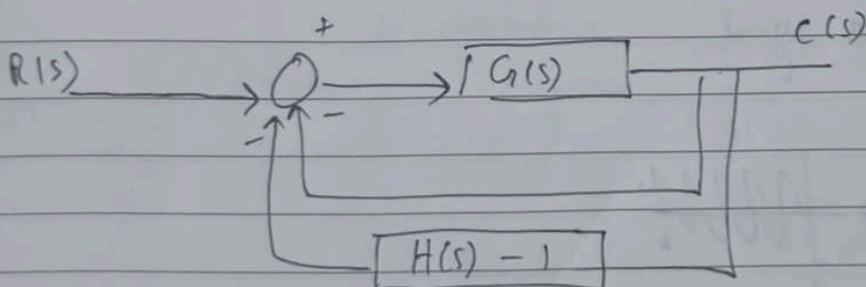
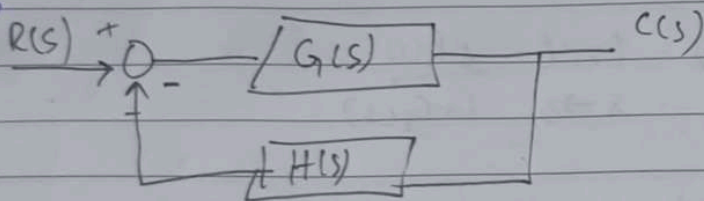
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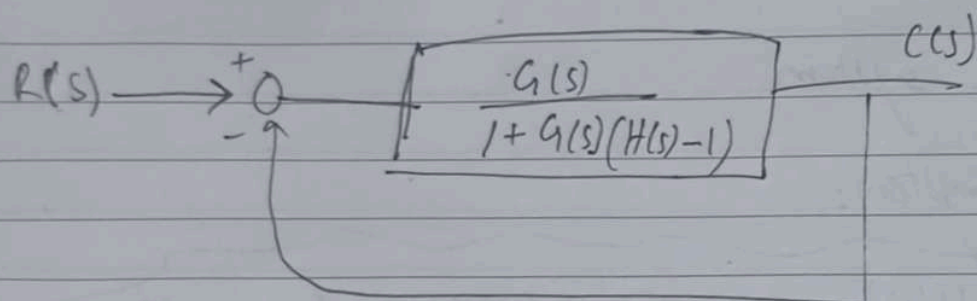
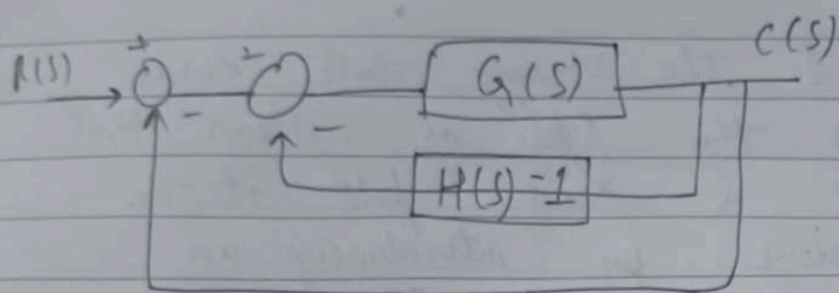
If we want to eliminate the steady state error we have to increase the type of system that is done by adding a new pole at the origin. Practically it's done by introducing an integral of error controller in series with $G(s)$

→ This slow the system

Non-unity feedback system:

$$e(\infty) = \lim_{s \rightarrow 0} \frac{s(R(s))}{1+G(s)} \rightarrow e(\infty) \text{ expression for unity Feedback system}$$

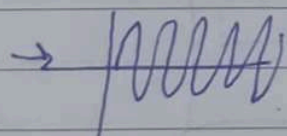




$$D(s) = \frac{G(s)}{1 + G(s)(H(s)-1)}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + D(s)} \Rightarrow \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)}$$

Differentiator & put the ^{system} but ~~it~~ it increase the frequency of error signal

→ 
↳ after passing to derivator

∴ steady state error $\rightarrow$ open loop T.F

∴ Routh table $\rightarrow$ close loop T.F
(stability)

Error via state space

~~$\dot{x} = Ax + Bu$~~

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

$$G(s) = C(sI - A)^{-1}B + D \Rightarrow \frac{C(s)}{R(s)} = G(s)$$

$$E(s) = R(s) - C(s)$$

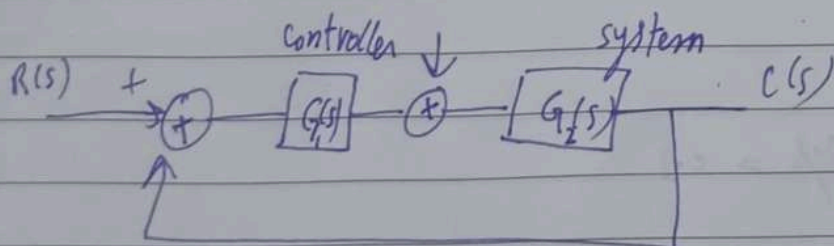
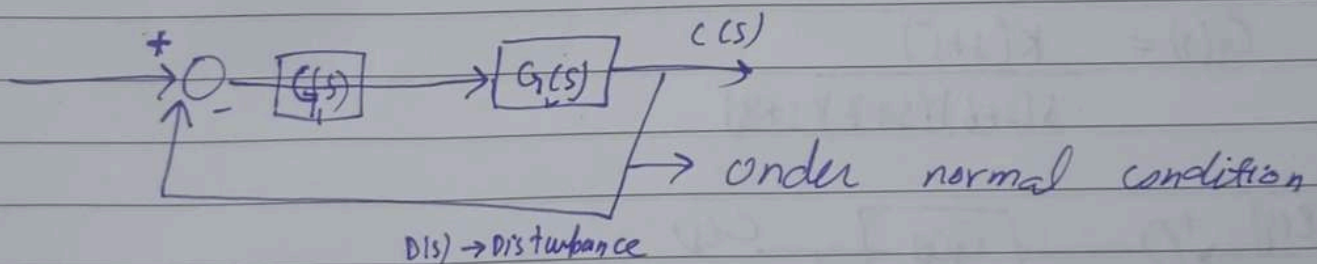
$$= R(s) - R(s)G(s)$$

$$= R(s) [1 - G(s)]$$

$$e(\infty) = \lim_{s \rightarrow 0} sE(s)$$

$$e(\infty) = \lim_{s \rightarrow 0} sR(s) [1 - G(s)]$$

ess under external disturbance:-



$$e(\infty) = \frac{1}{1 + G_1 G_2} R(s) - \frac{G_2}{1 + G_1 G_2} D(s)$$

error is now controlled by two terms = error due to actual input $R(s)$ error due to disturbance i/p

$$e(\infty) = \lim_{s \rightarrow 0} s E(s)$$

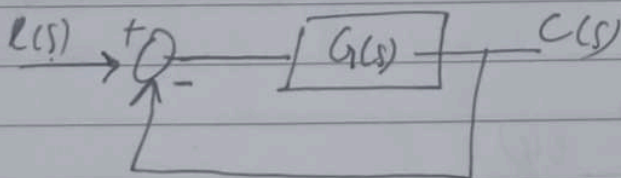
$$e(\infty) = \lim_{s \rightarrow 0} \frac{s}{1+G_1 G_2} R(s) - \lim_{s \rightarrow 0} \frac{s G_2}{1+G_1 G_2} D(s)$$

$$e_R(\infty) = \lim_{s \rightarrow 0} \frac{s}{1+G_1 G_2} R(s)$$

$$e_D(\infty) = \lim_{s \rightarrow 0} \frac{s G_2}{1+G_1 G_2} D(s)$$

Question:

$$G(s) = \frac{K(s+5)}{s(s+6)(s+7)(s+8)}$$



Find K so that $e_{ss} = 10\% = 0.1$

Solution:

input will be ramp input (Refer to table)

$$R(s) = \frac{1}{s^2} \quad \text{for constant } e_{ss} \text{ of Type I systems}$$

$$K_V = \lim_{s \rightarrow 0} s G(s)$$

$$= \lim_{s \rightarrow 0} s \left| \frac{k(s+5)}{s(s+6)(s+7)(s+8)} \right|$$

$$K_V = \frac{5k}{6(7)(8)} = \frac{5k}{336}$$

$$e(\infty) = \frac{1}{K_V} = \frac{336}{5k}$$

$$0.1 = \frac{336}{5k}$$

$$336 \cdot 0.1 = 5k$$

$$\boxed{k = 672}$$

$$F(s) = \frac{s+1}{s(s+2)}$$

Evaluate $\nearrow$ $s = -3+j4$

$$F(-3+j4) = \frac{-3+j4+1}{(-3+j4)(-3+j4+2)}$$

$$= \frac{-2+j4}{(-3+j4)(-1+j4)}$$

$$= \frac{\angle_1 \angle \phi_1}{\angle_2 \angle \phi_2 \cdot \angle_3 \angle \phi_3}$$

$$= \left(\frac{\angle_1}{\angle_2 \angle_3} \right) \angle \phi_1 - \phi_2 - \phi_3$$

$$-2+j4$$

$$L = \sqrt{(-2)^2 + 4^2}$$

$$\phi_1 = \tan^{-1}\left(\frac{4}{-2}\right)$$

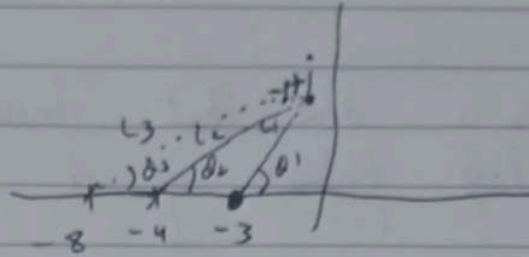
$$G(s) = \frac{s+3}{(s+4)(s+8)}$$

Evaluate at $s = -1+j1$

$$G(s) = \frac{-1+j1+3}{(-1+j1+4)(-1+j1+8)}$$

$$= \frac{2+j}{(3+j)(7+j)}$$

$$\frac{L_1 \angle \theta_1}{L_2 \angle \theta_2 \cdot L_3 \angle \theta_3}$$



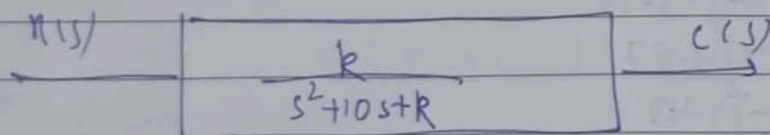
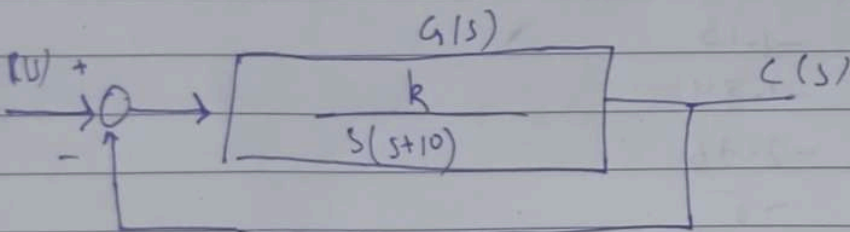
so, we conclude that

$$= m \angle \alpha$$

$$\therefore m = \frac{L_1}{L_2 L_3}$$

$$\alpha = \theta_1 - \theta_2 - \theta_3$$

Root locus:-

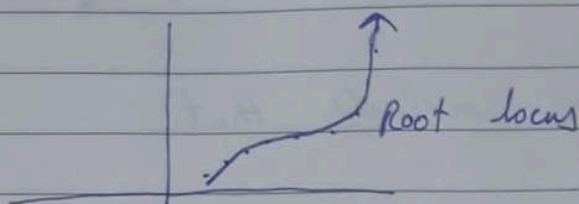


open-loop poles = 0, -10

closed loop poles = depends on the value of k

Root locus:

Is a line that passes through all possible closed loop poles of a system.



$\Rightarrow$ set of closed loop poles computed for different values of K is called root locus.

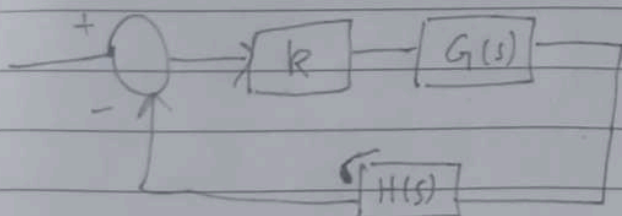
K	Pole 1	Pole 2
0	-10	-10
5	-9.47	-0.53
10	-8.87	-1.13
15	-8.16	-1.84
20	-7.24	-2.76
25	-5	-5
30	$-5 + j2.24$	$-5 - j2.24$
35	$-5 + j3.16$	$-5 - j3.16$
40	$-5 + j3.87$	$-5 - j3.87$
45	$-5 + j4.47$	$-5 - j4.47$
50	$-5 + j5$	$-5 - j5$

zeros are attracting
poles are repelling each other

To sketch the root locus we only need the open loop poles.

$$G(s) = \frac{K}{s(s+10)}$$

open loop poles = 0, -10



open loop function = $K G(s) H(s)$

$$\text{close loop function} = \frac{K G(s)}{1 + K G(s) H(s)}$$

$$\text{close loop poles} \Rightarrow 1 + K G(s) H(s) = 0$$

$$K G(s) H(s) = -1 \rightarrow \text{tell us whether the point is lying on the R.L.}$$

Phasor form:

$$K G(s) H(s) = +1 \angle (2n+1)180^\circ, n = 0, \pm 1, \pm 2, \dots$$

odd multiple of π

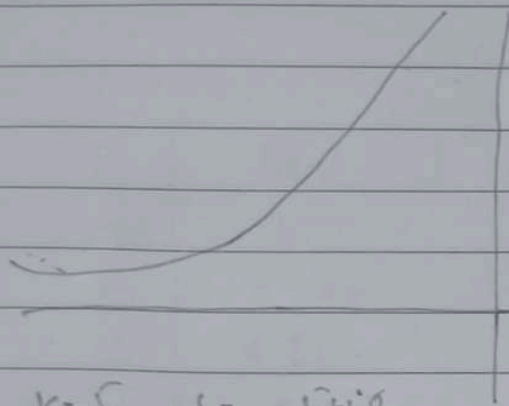
also we write as

$$\text{magnitude (m)} = |K G(s) H(s)| = +1$$

$$\text{Angle } (\theta) = \angle K G(s) H(s) = (2n+1)\pi$$

$$K = \frac{1}{|G(s)H(s)|} = \frac{1}{|G(s)| |H(s)|}$$

* If we want to see whether a point $s = a + jb$ lies on the root locus for a certain value of "K" we use the relation.



$$K=5, s = -5 + j8$$

$$\begin{aligned} K G(s) H(s) &= \frac{K}{s(s+10)} \\ &\text{Values from table} \\ &= \frac{20}{(-7.24)(-7.24+10)} \\ &= -1 \end{aligned}$$

Design Problem:-

$$s = -5$$

$$R = ?$$

$$K H(s) G(s) = \frac{K}{s(s+10)}$$

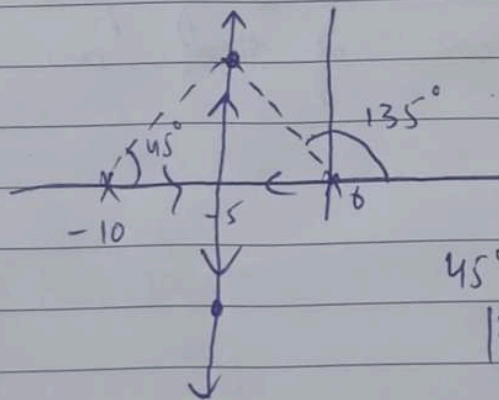
$$-1 = \frac{K}{-5(-5+10)}$$

$$-1 = \frac{K}{-25}$$

$$K = 25$$

Sketching the root locus:-

→ Root locus can be obtained by sweeping through every point in s-plane to locate the points for which angle add upto an odd multiple of 180°



$$45^\circ + 135^\circ = 180^\circ$$

Point on root locus.

Rules of sketching

1- Number of branches:-

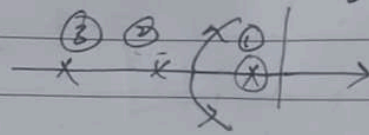
of Branches of RL = # of open loop poles

2- Symmetry:-

~~The~~ ~~RL~~ R.L. is symmetrical about real-axis (n-axis)

3- Real axis segment:-

The R.L. always originates at the left of odd-numbered pole or zero



4- Starting & ending or near

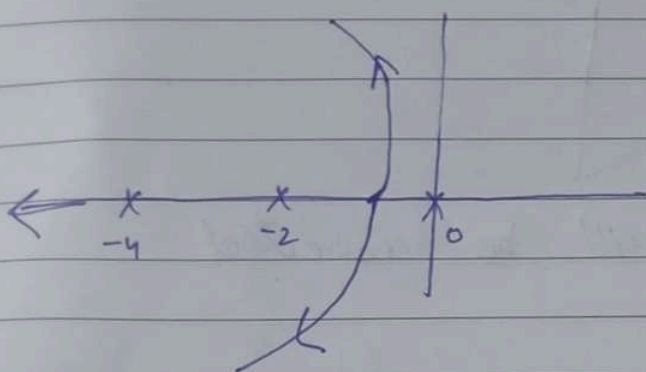
R.L. starts at open loop Poles & ends at or near open loop zeros

5- Behaviour at infinity

If there are no zero, the Root locus would end at infinity, or if the number of poles and zeros are unequal some of them are also goes to infinity.

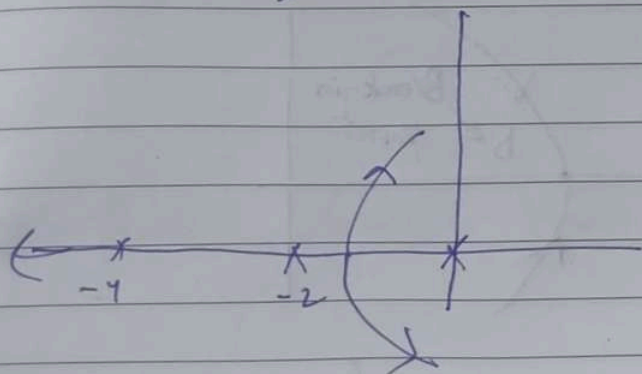
Behavior at Infinity:

$$G(s) = \frac{2}{s(s+2)(s+4)}$$



Apparently both root are correct cause the are satisfying all the rules.

Now we have to find asymptotes
 → starting point of asymptote
 → Angle inclination of asymptote



1) Starting point on real axis:

$$\sigma_a = \frac{\sum \text{poles} - \sum \text{zeros}}{\# \text{ of poles} - \# \text{ of zeros}}$$

$$\Rightarrow \frac{(0 + (-2) + (-4)) - (0)}{3 - 0} = -2$$

2) Angle of inclination

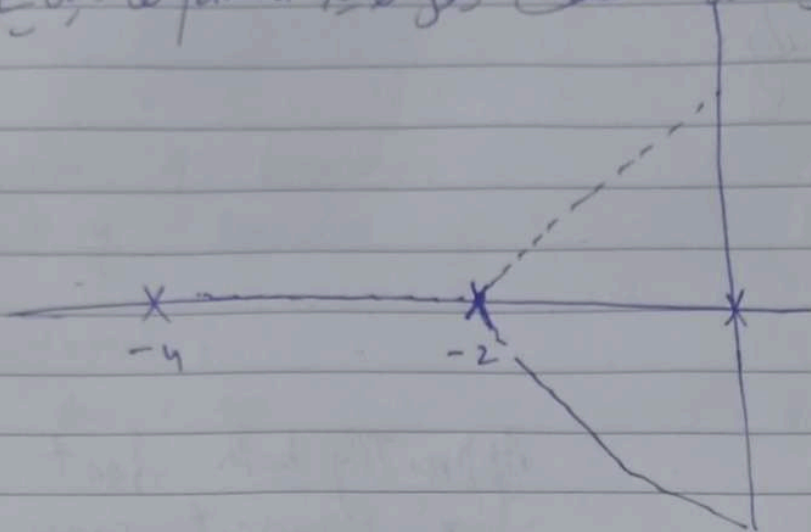
$$\theta_a = \frac{(2n+1)\pi}{\# \text{ of poles} - \# \text{ of zeros}}$$

We have to remain in the angle of $0-2\pi$

$$\Rightarrow \frac{(2n+1)\pi}{3-0} = \frac{(2n+1)\pi}{3} = \frac{\pi}{3}, \pi, \frac{5\pi}{3}, \dots$$

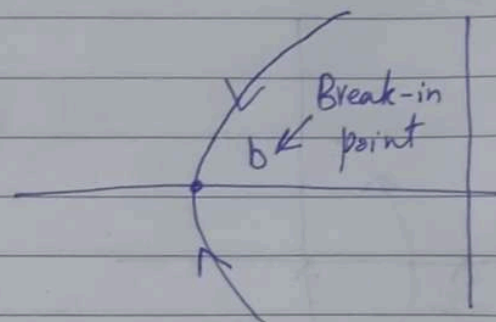
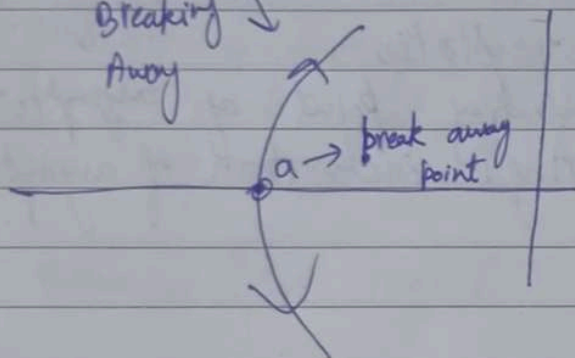
$n = 0, 1, 2, 3, 5, 7, \dots$

جب $\frac{N}{D}$ formula N د D سے زیادہ $\sqrt[n]{n}$ zeros, poles ہوں گے



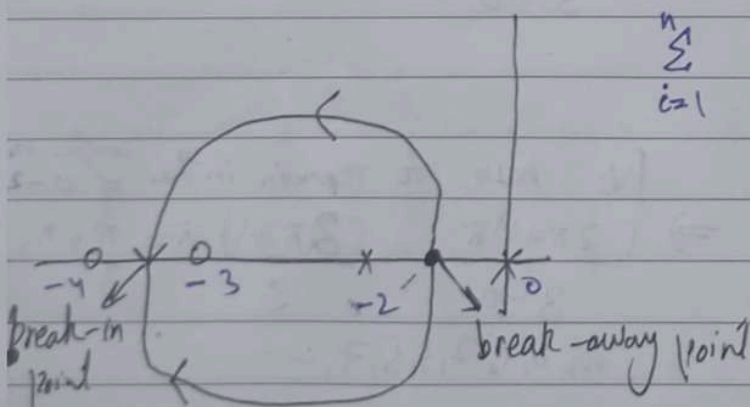
Now the second one will be corrected

RL is
Breaking
Away



Both point always on real axis

Real-axis Break away & Break-in points:-



$$\sum_{i=1}^n \frac{1}{s - p_i} = \sum_{j=1}^n \frac{1}{s - z_j}$$

$\downarrow$ Pole $\downarrow$ zero

$$\frac{1}{s-(0)} + \frac{1}{s-(-2)} = \frac{1}{s-(-3)} + \frac{1}{s-(-4)}$$

$$\frac{1}{s} - \frac{1}{s+2} = \frac{1}{s+3} + \frac{1}{s+4}$$

$$5s^2 + 24s + 24 = 0$$

$$s = -1.4, -3.4 \rightarrow \text{Break in}$$

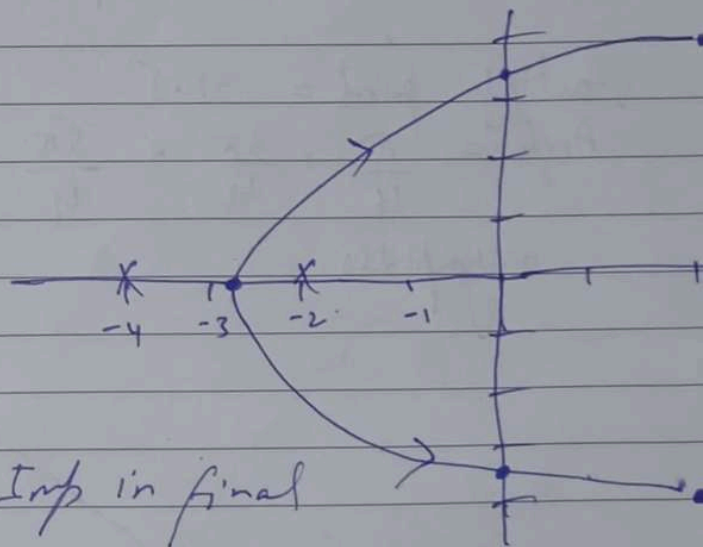
↓

Breakaway

$$G(s) = \frac{k(s^2 - 4s + 20)}{(s+2)(s+4)}$$

$$\text{poles} = -2, +4$$

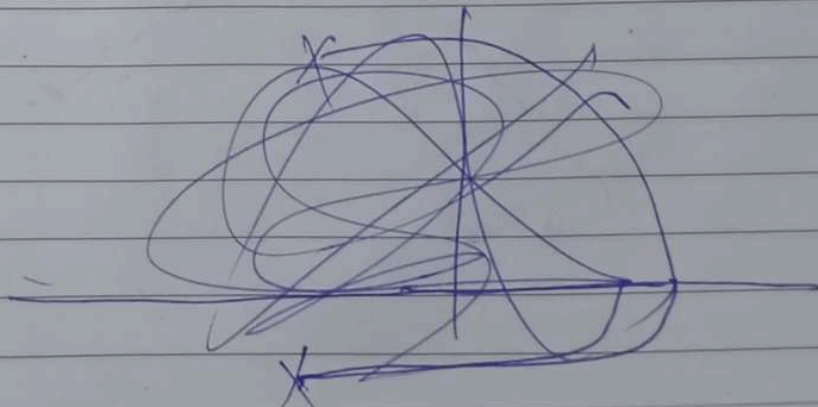
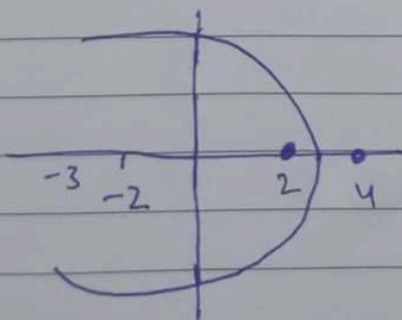
$$\text{zeros} = 2+4i, 2-4i$$



$$2) \quad G(s) = \frac{k(s-2)(s-4)}{s^2 + 6s + 25} \quad \text{Imp in final}$$

$$\text{poles} = -3 \pm j4$$

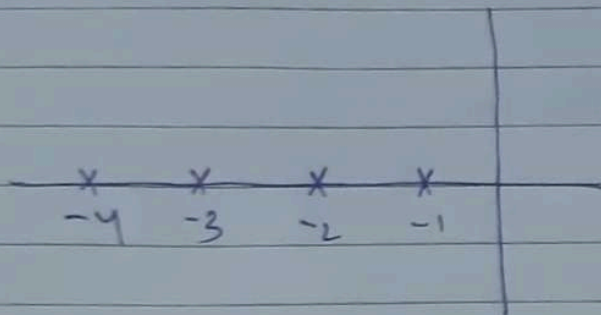
$$\text{zeros} = +2, +4$$



$$3) G(s) = \frac{k}{(s+1)(s+2)(s+3)(s+4)}$$

Poles = $-1, -2, -3, -4$

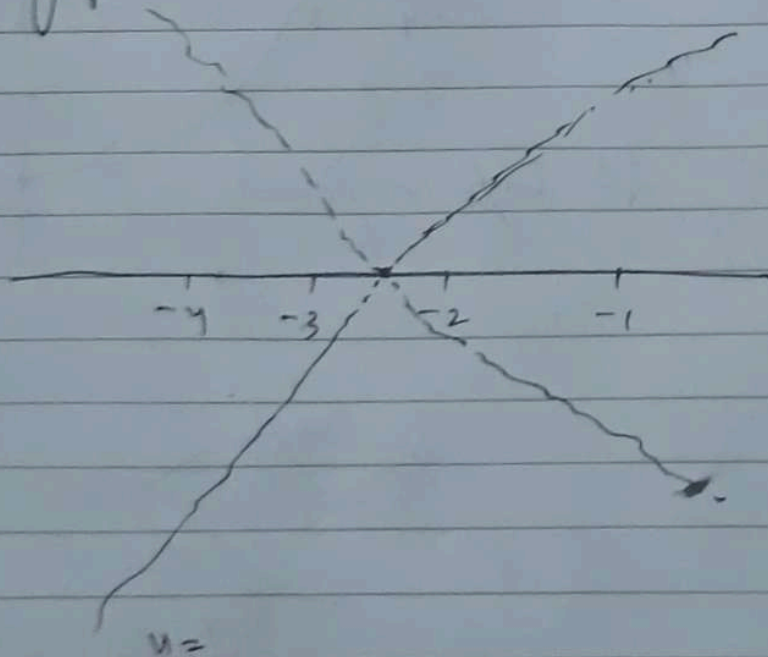
zero = null

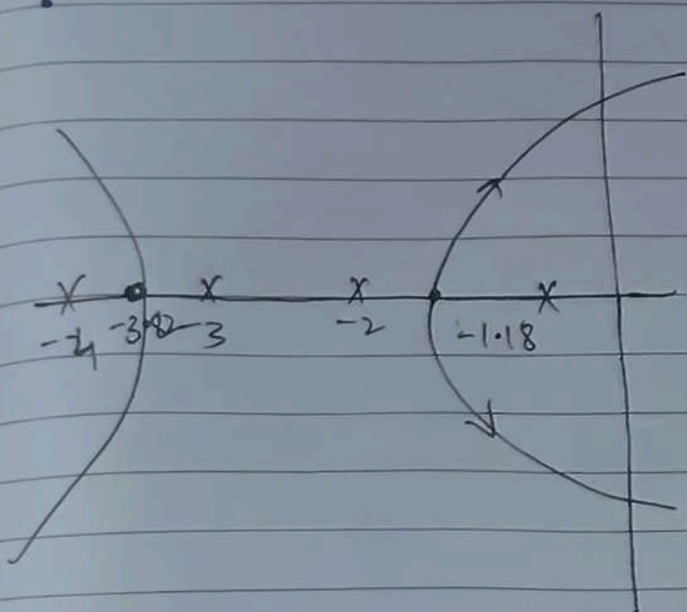


starting point = -2.5

Angle = $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \dots$

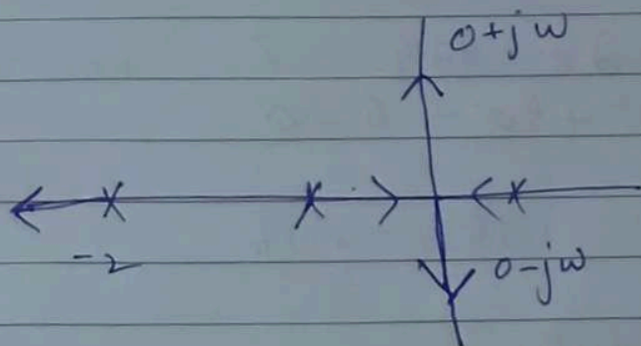
asymptotes





$$G(s) = \frac{K(s^2 + 1)}{(s-1)(s+2)(s+3)}$$

Poles = $-1, +1, -2$
 zeros = $+j, -j$



$$D = \frac{K(s+2)}{s^2+2s+10}$$

$$\text{Poles} = -1 \pm 3j$$

$$\text{zeros} = -2$$

Break-in point :-

$$\frac{1}{s+2} = \frac{1}{s-(-1+3j)} + \frac{1}{s-(-1-3j)}$$

$$\frac{1}{s+2} = \frac{s+1+3j + s+1-3j}{(s+1+3j)(s+1-3j)}$$

$$s^2 + 2s + 1 + 9 = 2s^2 + 6s + 4$$

$$s^2 + 4s - 6 = 0$$

$$s = 1.16, -5.16$$

Asymptote

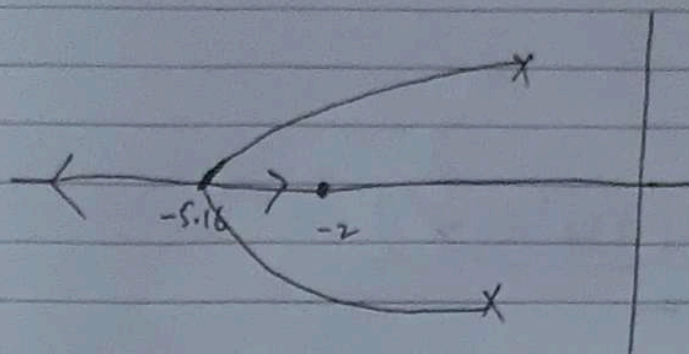
$$\sigma_a = \frac{(-1+3j) + (-1-3j) - (-2)}{2-1}$$

$$\sigma = 0$$

$$\phi_a = \frac{(2n+1)\pi}{2-1} = (2n+1)\pi \quad n=0,1$$

$$z_n$$

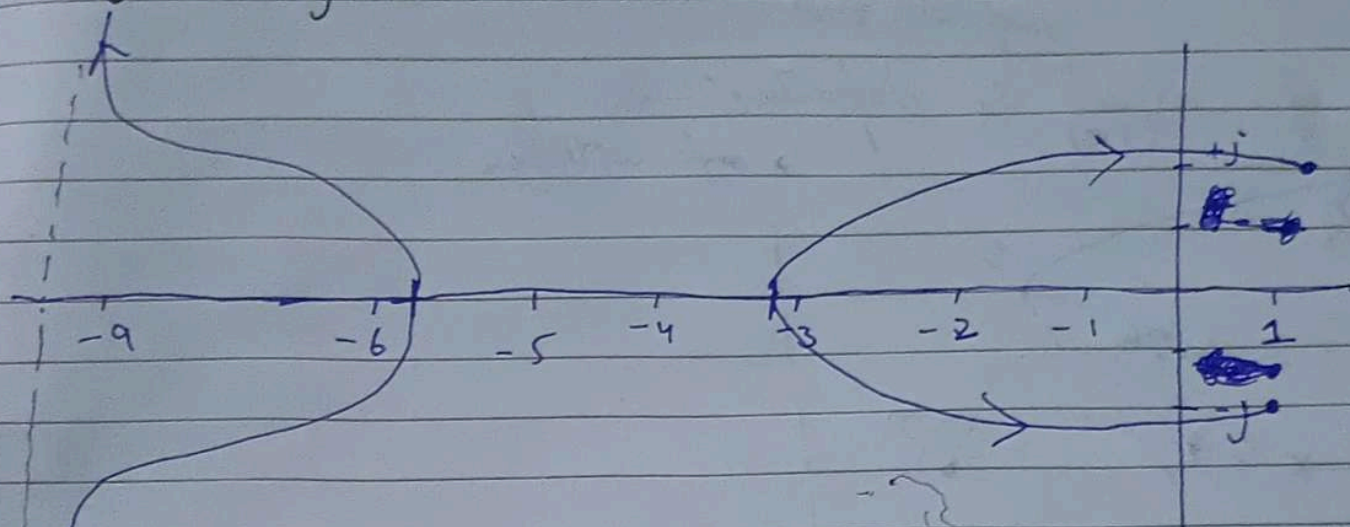
06/Mayo



$$G(s) = \frac{K(s^2 - 2s + 2)}{(s+2)(s+4)(s+5)(s+6)}$$

Poles = -2, -4, -5, -6

zeros = $1 \pm j$



Break away/in

$$\frac{1}{s+2} + \frac{1}{s+4} + \frac{1}{s+5} + \frac{1}{s+6} = \frac{1}{s-1-j} + \frac{1}{s-1+j}$$

$$(s+4)(s+5) + \dots$$

$$s = -1.36, -3.64, -15.89$$

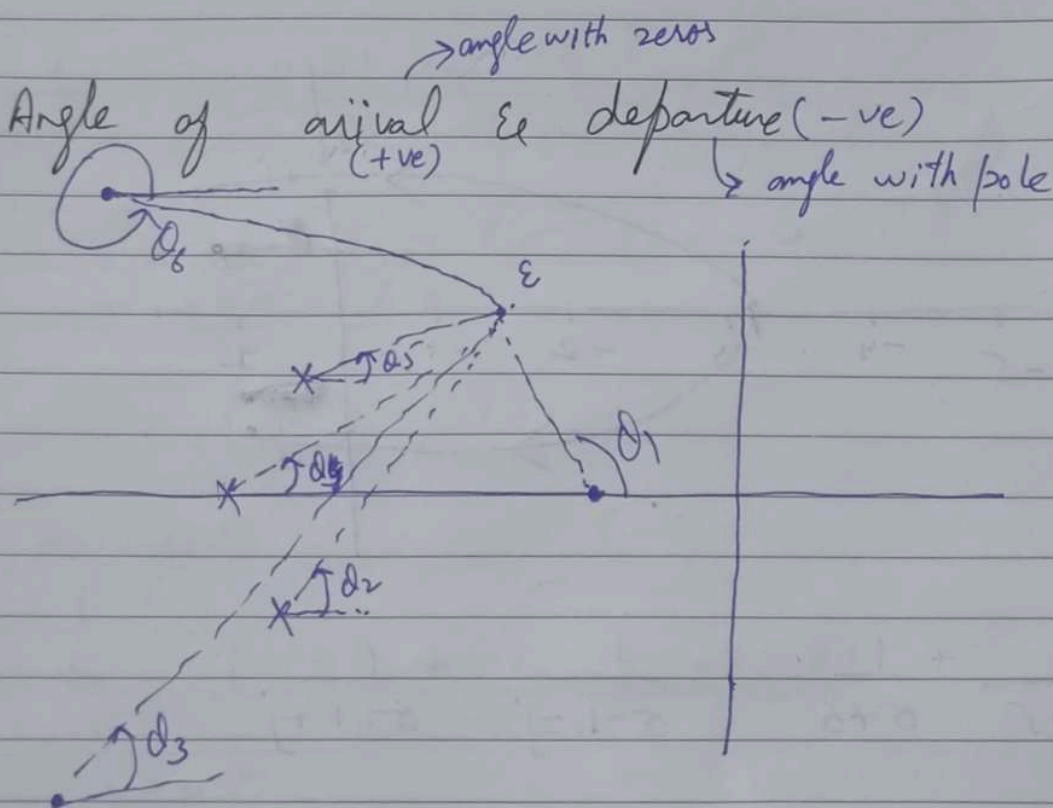
$$s = -3.16, -5.82, -19.81$$

Asymptotes: —

$$\sigma_a = \frac{(-2-4-5-6) - (1+j+1-j)}{4-2}$$

$$= \frac{-19}{2} = -9.5$$

$$\phi_a = \frac{(2n+1)\pi}{2} = \frac{\pi}{2}, \frac{3\pi}{2}$$

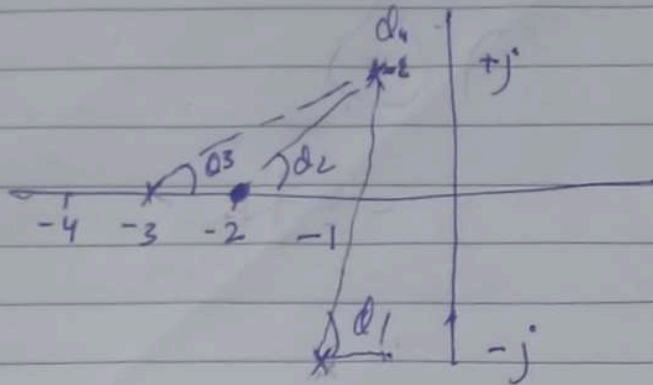


$$+\angle 1 - \angle 2 + \angle 3 - \angle 4 - \angle 5 + \angle 6 = \pm (2n+1)\pi$$

" lies if these ~~sum~~ sum equal to this then
on Root locus.

$$\frac{K(s+2)}{(s+3)(s^2+2s+2)}$$

$$\frac{1}{b}$$



$$\theta_1 = 90^\circ$$

$$\theta_1 = \tan^{-1}\left(\frac{1}{1}\right) = 45^\circ$$

$$\theta_3 = \tan^{-1}\left(\frac{1}{2}\right) = 26.5^\circ$$

$$-\theta_1 + \theta_2 - \theta_3 - \theta_4 = +180^\circ$$

$$-90 + 45 - 26.5 - \theta_4 = +180$$

$$-71.5 - \theta_4 = +180$$

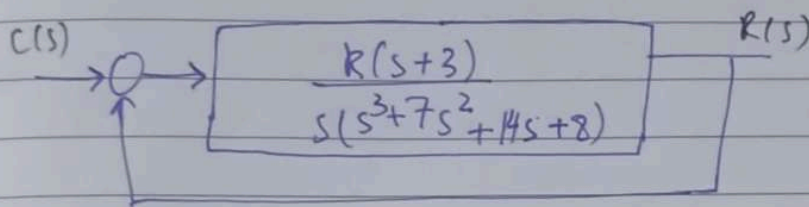
~~$$-71.5 - \theta_4 = +180$$~~

~~$$\theta_4 = -180 - 71.5$$~~

$$\theta_4 = -180 - 71.5$$

$$\theta_4 = -251.5^\circ$$

The j - ω -crossing of Root locus:-



$$T(s) = \frac{K(s+3)}{s^4 + 7s^3 + 14s^2 + 8s + Ks + 3K}$$

s^4	1	14	3K	
s^3	7	8+K	0	
s^2	$\frac{K-90}{-7}$	3K	0 $\rightarrow$ multiply by 7	$\frac{8+K-98}{-7}$
s^1	$90-K$	21K	0	$\frac{K-90}{-7}$
s^0	$-K^2-65K+720$	0	0	$\frac{21K - (8+K \times \frac{K-90}{-7})}{-7}$
	$90-K$			

For what value of "K" is the zero row

$$-K^2 - 65K + 720 = 0$$

$$90 - K$$

$$K^2 + 65K - 720 = 0$$

$$K = +9.65, -74.64$$

$$K = 9.65$$

$$(90-K)s^2 + 21K = 0$$

$$(90-9.65)s^2 + 21(9.65) = 0$$

$$80.35s^2 + 202.65 = 0$$

$$s = \pm j1.59$$

$$G(s) = \frac{K(s^2 - 4s + 20)}{(s+2)(s+4)}$$

$$\text{zeros} = (2 \pm j4)$$

$$\text{poles} = -2, -4$$

$$\frac{1}{s - (2+j4)} + \frac{1}{s - (2-j4)} = \frac{1}{s+2} + \frac{1}{s+4}$$

$$\frac{1}{s-2-j4} + \frac{1}{s-2+j4} = \frac{1}{s+2} + \frac{1}{s+4}$$

$$\frac{s-2+j4 + s-2-j4}{(s-2-j4)(s-2+j4)} = \frac{s+4 + s+2}{s^2 + 6s + 8}$$

$$\frac{2s-4}{s^2 - 4s + 4 - 16} = \frac{2s+6}{s^2 + 6s + 8}$$

$$2s-4(s^2+6s+8) = 2s+6(s^2-4s-8)$$

$$~~2s^3 + 12s~~$$

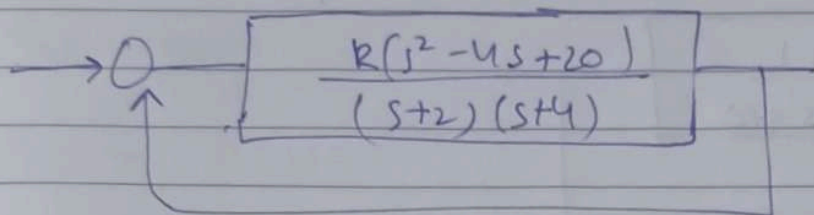
$$~~2s^3 + 120s + 160 - 4s^2~~$$

$$2s^3 + 6s^2 + 18s - 4s^2 - 24s - 32 = 2s^3 - 8s^2 - 16s + 60s^2 - 24s - 48$$

$$2s^2 - 8s - 32 = -2s^2 - 40s - 48$$

$$4s^2 + 32s + 16 = 0 \times$$

$$s = -3.16, \checkmark$$



$$T(s) = \frac{K(s^2 - 4s + 20)}{(1+K)s^2 + (6-4K)s + (8+20K)}$$

s^2	$1+K$	$8+20K$
s^1	$6-4K$	0
s^0	$8+20K$	0

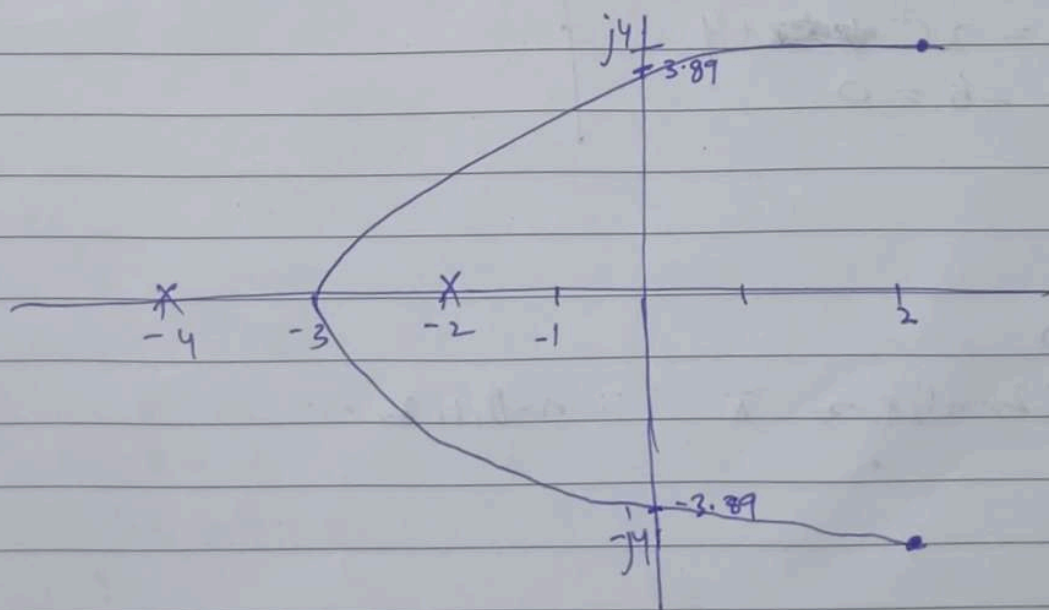
$$\begin{aligned} & -6-4K=0 \\ & K=1.5 \end{aligned}$$

$$(1+K)s^2 + 8+20K = 0$$

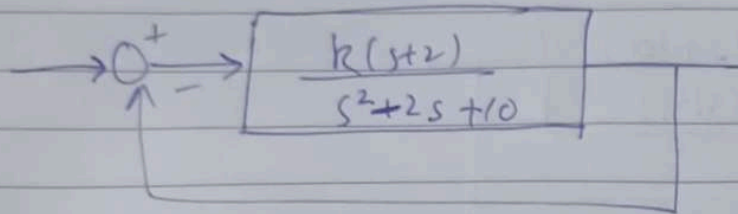
$$(1+1.5)s^2 + 8+20(1.5) = 0$$

$$2.5s^2 + 38 = 0$$

$$s = \pm j3.89$$



$$\frac{(s+2)(s-2)}{s^2+4}$$



zero = -2

poles = $+1 \pm 3j$

Break-in points:-

$$\frac{1}{s+2} = \frac{1}{s+1-3j} + \frac{1}{s+1+3j}$$

$$\frac{1}{s+2} = \frac{s+1-3j + s+1+3j}{(s+1)^2 - (-3j)^2}$$

$$\frac{1}{s+2} = \frac{2s+2}{s^2+2s+1+9}$$

X

$$s^2+2s+10 = 2s^2 + 2s + 4$$

$$s^2+2s+10 = 2s^2 + 2s + 4$$

$$s^2+2s-6 = 0$$

$$\frac{1}{s+2} = \frac{2s-2}{s^2-2s+1-9}$$

$$s^2-2s+10 = 2s^2-2s+2s-4$$

$$s^2-2s+10 = 2s^2+2s-4$$

~~s^2~~

$$s^2+8 = 0$$

X

c) Asymptotes

$$\sigma_a = 0$$

$$\sigma_a = (2n+1)\pi \sim \pi \quad \because n=0,1,2,\dots$$

d) jw crossing

$$T(s) = \frac{K(s+2)}{s^2 + (-2+K)s + (10+2K)}$$

s^2	1	$10+2K$
s^1	$-2+K$	0
s^0	$10+2K$	0

$$-2+K=0$$

$$K=+2$$

~~$$s^2 + 10 + 2K = 0$$

$$s^2 + 10 - 4 = 0$$

$$s^2 + 6 = 0$$

$$s = \pm j2.45$$~~

$$s^2 + (10+2K) = 0$$

$$s^2 + 14 = 0$$

$$s = \pm j3.74$$

$$\frac{1}{s+2} = \frac{1}{s-1+j3} + \frac{1}{s-1-j3}$$

$$\frac{1}{s+2} = \frac{s-1-j3 + s-1+j3}{(s+1)^2 - (j3)^2}$$

$$\frac{1}{s+2} = \frac{2s-2}{s^2-2s+1-9}$$

$$s^2-2s-8 = 2s^2-2s+4s-4$$

$$s^2-2s-8 = 2s^2+2s-4$$

$$s^2+4s+4 = 0$$

Control Design via Root locus:-

Poles:

$$s = -\xi \omega_n \pm j \omega_n \sqrt{1-\xi^2}$$

$$T_s = \frac{4}{\xi \omega_n} = \frac{4}{|\text{real part}|}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}} = \frac{\pi}{|\text{imag. part}|}$$

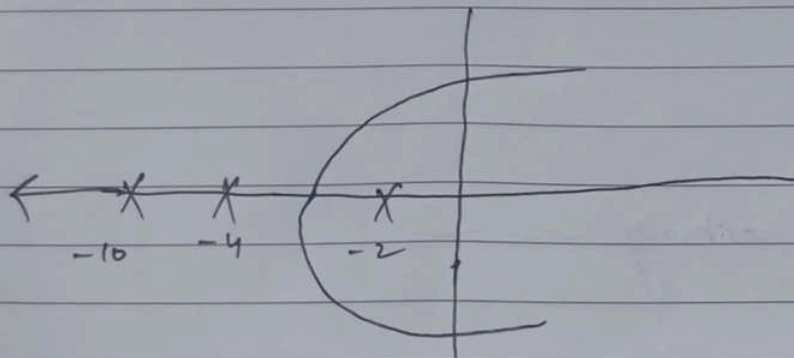
$$\cos \theta = \xi$$

$$\xi \propto \frac{1}{\theta}, \quad \xi \propto \frac{1}{\theta S}$$
$$\theta \propto OS$$

Controller design via Root Locus

- 1- Improve Transient Response (T_s , T_P , OS)
- 2- Improve steady-state response (e_{ss})

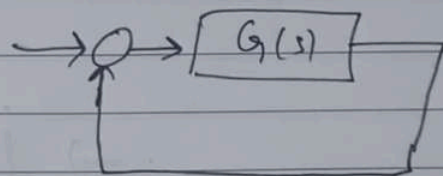
Improve transient response



Uncontrolled system

$T_s = 5s$
 $T_P = 4s$
 $OS = 10\%$ } system response

$T_s = 4s$
 $T_P = 3.2 \text{ sec}$
 $OS = 10\%$ } desired response



$$G(s) = \frac{1}{(s+2)(s+4)(s+10)}$$

$$T_s = \frac{4}{\xi \omega_n}$$

$$\xi \omega_n = \frac{4}{5} = 0.8$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}}$$

$$\omega_n \sqrt{1-\xi^2} = \frac{\pi}{4} = 0.75$$

$$0.8 \pm j0.75$$

~~~~~ derive response working ~~~~~

$$T_p = \frac{4}{\text{Real}} \Rightarrow \frac{4}{4} = \text{Real}$$

$$\text{Real} = 1.0$$

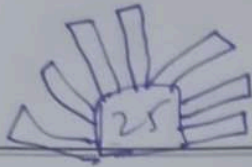
$$T_p = \frac{\pi}{|\text{imag}|}$$

$$\Rightarrow \boxed{1 \pm j1.1}$$

$$\text{imag} = \frac{\pi}{3.2} = 1.1$$

→ You have to make sure the desired pole must lie on the root locus.

→ As a control designer your task is to make sure my root locus pass through desired pole.

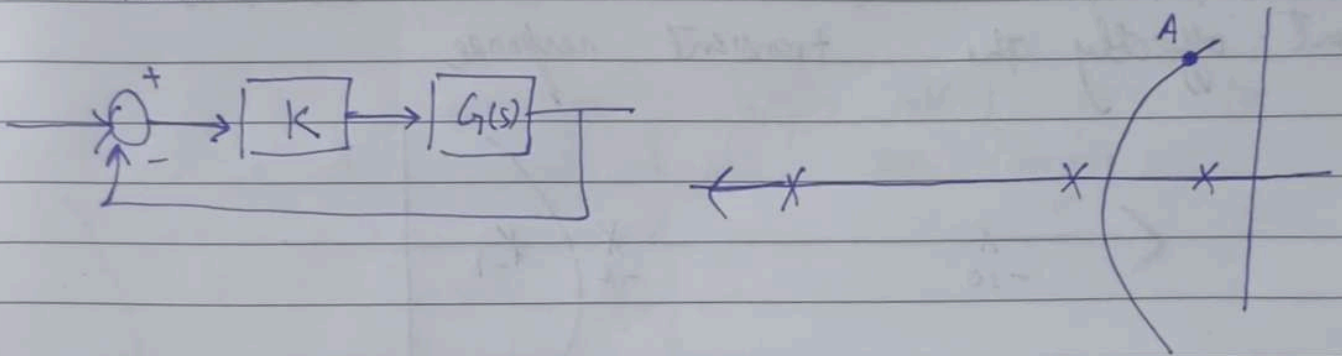


Improving the steady state error:-

Two Methods:

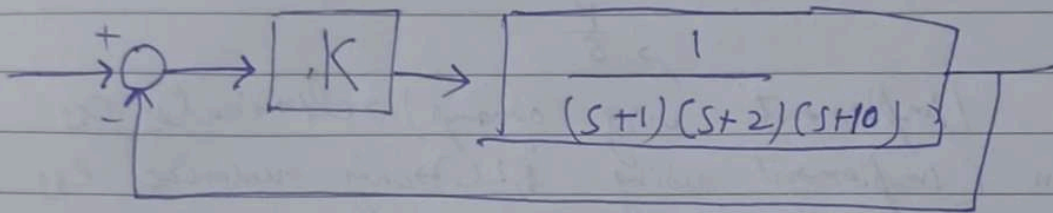
- 1- PI controller (implement using op-amp)  $\rightarrow \frac{K}{s}$  eliminates  $e_{ss}$
- 2- Lag compensation (implement using RLC)  $\rightarrow$  only minimize  $e_{ss}$   
 $\rightarrow \frac{K}{s+\alpha}$

PI Controller

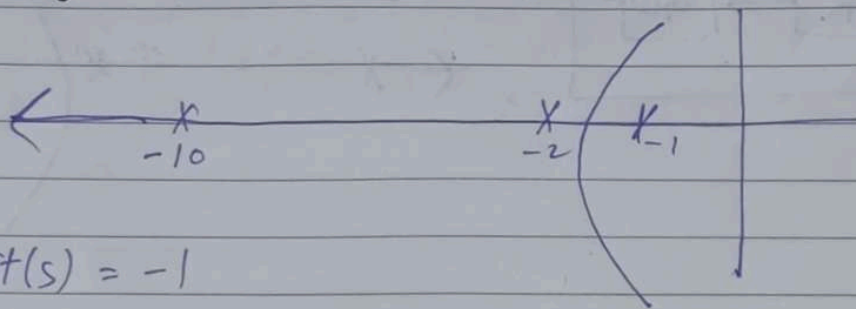




PI-controller



- Find the dominant pole for  $K = 164.6$
- Find  $T_p$ ,  $T_s$ ,  $O_s$ ,  $e_{ss}$  for this pole location
- Hence, design a controller to make  $e_{ss} = 0$  without affecting the transient response



$$a) K G(s) H(s) = -1$$

$$\frac{164.6}{(s+1)(s+2)(s+10)} = -1$$

$$(\checkmark)$$

$$s = -0.694 \pm j3.926, -11.61 (X)$$

~~cause imaginary part~~  $(T_p)$

cause  $T_p$  depends on the imaginary part

$$\text{dominant pole} = s = -0.694 \pm j3.926$$

b)

$$T_s = \frac{4}{0.694}$$

$$T_p = \frac{\pi}{3.926}$$

$$\therefore OS = \frac{-\pi \epsilon}{\sqrt{1-\epsilon^2}}$$

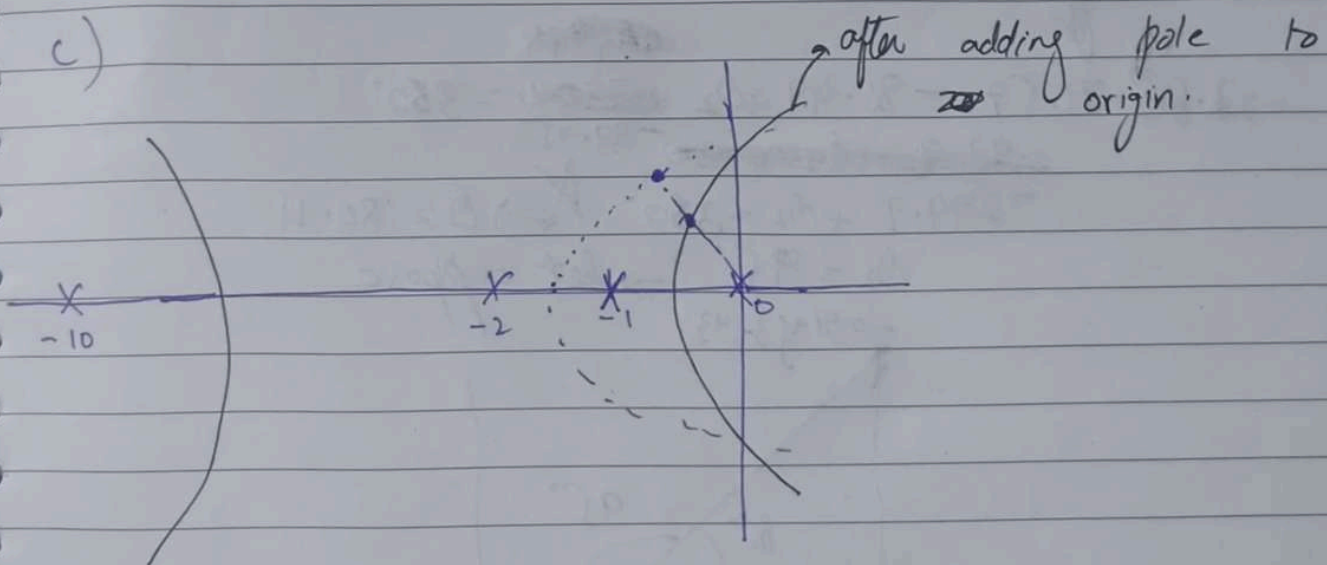
for step input

$$e_{ss} = \frac{1}{1+k_p}$$

$$\therefore R_p = \lim_{s \rightarrow 0} G(s) = \text{~~scribble~~}$$

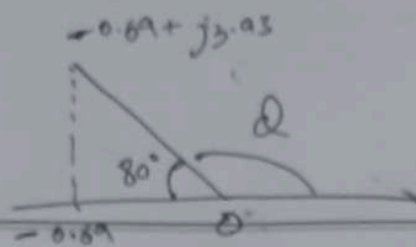
$$\epsilon = \cos \theta = \cos \left( \tan^{-1} \left( \frac{3.9}{0.694} \right) \right)$$

c)

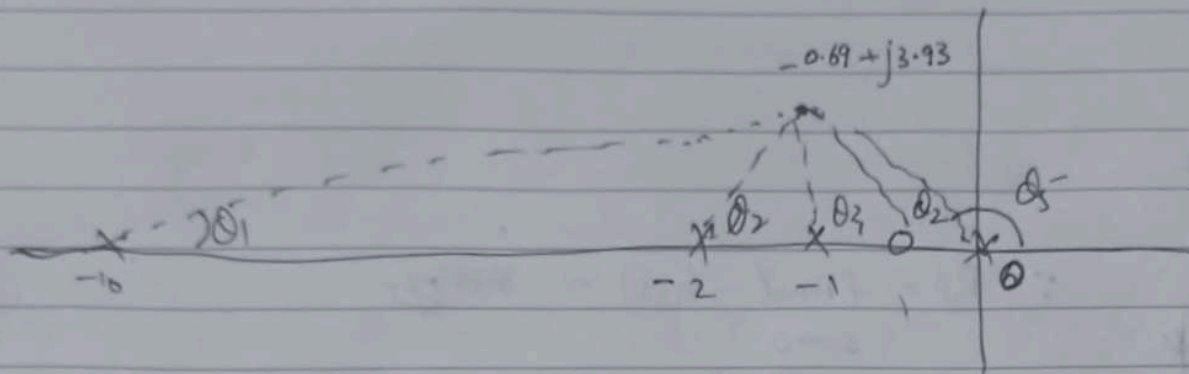


Jab bhi ~~steady~~ transient state ko improve krna  
hota hai ~~zero~~ "zero" add kro or jab  
close to origin  
steady state error improve krna to "pole" to  
add krna at origin or close to origin

$$\tan \theta = \frac{p}{b} = \frac{3.93}{0.69}$$



after adding "zero"



$$-\theta_1 - \theta_2 - \theta_3 + \theta_4 - \theta_5 = 180^\circ$$



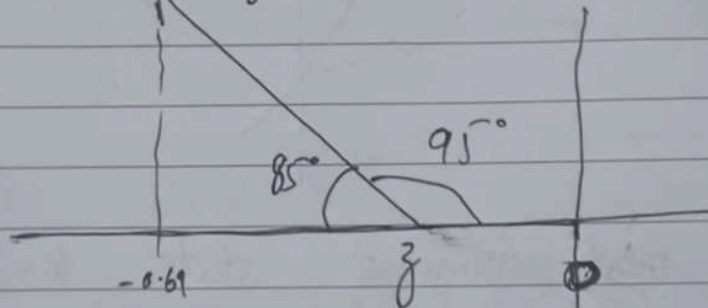
$$-22.88 - 71.57 - 85.49 + \theta_2 = 360^\circ$$

$$-279.94 + \theta_2 = 360^\circ$$

$$\theta_2 = 95^\circ \Rightarrow \theta = 80.11$$

$\theta_2 = 95^\circ \rightarrow$  let suppose

$$-0.69 + j3.93$$



$$\tan \theta = \frac{p}{b}$$

$$\tan(85^\circ) = \frac{3.926}{b}$$

$$b = 0.343$$

$$z = 0.343 - 0.694j$$



Having a pole exactly at origin is only possible by using op-amp. If we are using RLC we're not able to place at origin

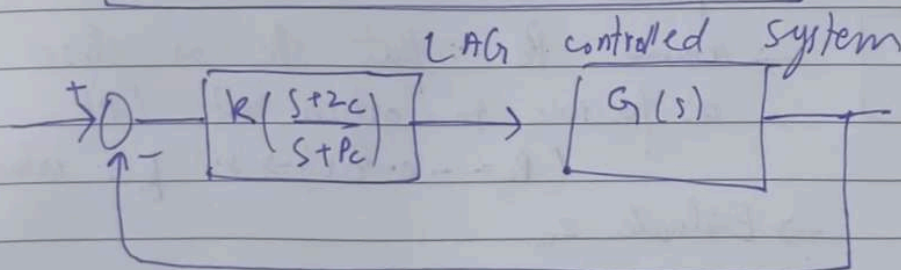
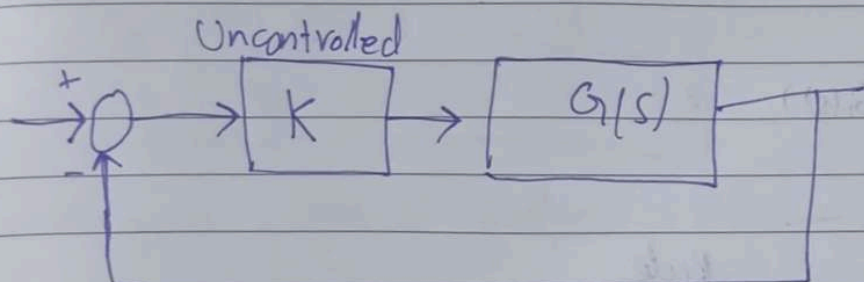
$$C(s) = \frac{k(s+0.35)}{s}$$

$$C(s) = \frac{164.6(s+0.35)}{s}$$

$$C(s) = 164.6 + \frac{57.6}{s}$$

$$L^{-1} \\ 164.6 e(t) + 57.6 \int e dt$$

LAG Compensator (FLC)



Type 1:

$$KG(s) = \frac{k(s+z_1)(s+z_2)\dots}{s(s+p_1)(s+p_2)\dots}$$

$$e(\infty) = \frac{1}{K_v}$$

$$K_v = \lim_{s \rightarrow 0} s(KG(s))$$

$$K_{V_{\text{open loop}}} = \frac{K(z_1)(z_2)}{(p_1)(p_2)}$$

CAG

Controlled system

Open loop TF

$$G(s) = \frac{K(s+z_c)(s+z_1)(s+z_2)}{s(s+p_c)(s+p_1)(s+p_2)}$$

$$e(\infty) = \frac{1}{K_V}$$

$$K_{V_c} = \lim_{s \rightarrow 0} s(G(s))$$

$$K_{V_c} = \frac{K z_c z_1 z_2}{p_c p_1 p_2}$$

$$K_{V_c} = K_{V_0} \left( \frac{z_c}{p_c} \right)$$

Rule

→ assume  $p_c$  ∴ put  $p_c$  as close as possible to origin at LHP

( $p_c = -0.01$ ) → sir bot value

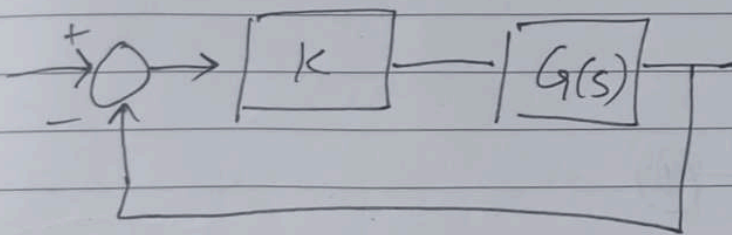
→ Evaluate  $z_c$

$$K_{p0} = K_{p0} \left( \frac{z_c}{p_c} \right) \quad \text{Type 0} \quad e(\infty) = \frac{1}{1+K_p}$$

$$K_{v0} = K_{v0} \left( \frac{z_c}{p_c} \right) \quad \text{Type 1} \quad e(\infty) = \frac{1}{K_v}$$

$$K_{a0} = K_{a0} \left( \frac{z_c}{p_c} \right) \quad \text{Type 2} \quad e(\infty) = \frac{1}{K_a}$$

Question



- Given a Type 0 system,  $e_{ss} = 0.108$  for step input
- begin a lag compensator to reduce error by 10 times.

Solution:

Uncontrolled system :  $e_{ss} = 0.108$   
Type 0

$$\frac{1}{1+K_p} = 0.108$$

$$K_{p0} = 8.23$$



Controlled system:

$$e_{ss} = \frac{0.108}{10} = 0.0108$$

$$\frac{1}{1+K_p} = 0.0108$$

$$K_p = 91.59$$

LAG component

$$K_p = K_o \left( \frac{z_c}{p_c} \right)$$

$$91.59 = 8.23 \left( \frac{z_c}{p_c} \right) \quad \text{--- (1)}$$

Assume  $p_c = -0.01$

$$z_c = -0.111 \rightarrow \text{after putting } p_c \text{ in eq (1)}$$

$$C(s) = \frac{s+0.111}{s+0.01}$$



Improving Transient Response:- ( $T_p$ ,  $T_s$ ,  $OS\%$ )

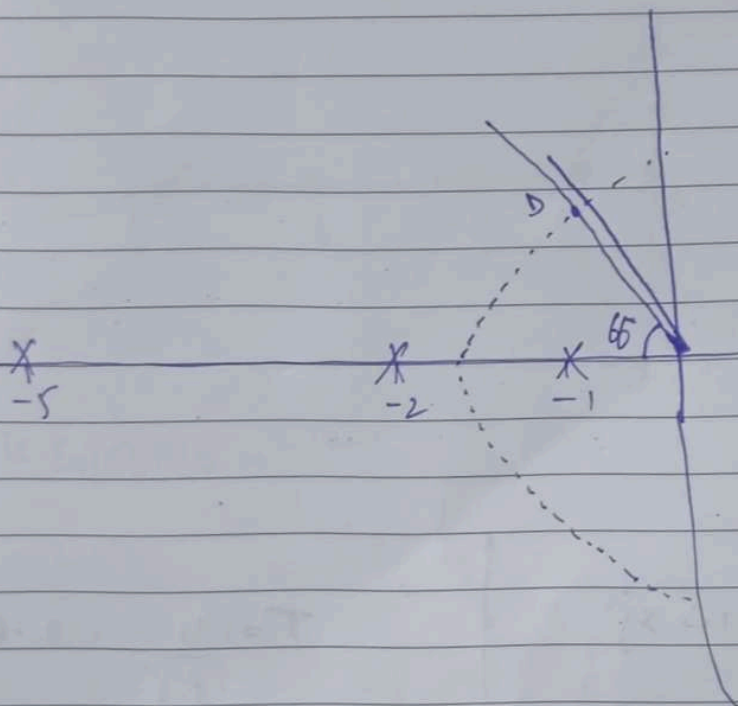
1) PD controller  $\Rightarrow C(s) = s + z_c$  < op-amp >

2) LEAD compensator  $\Rightarrow C(s) = \frac{s + z_c}{s + p_c}$  < RLC >

PD Controller :-

$$\frac{K}{(s+1)(s+2)(s+5)}$$

$$C(s) = s + z_c$$



$$z = 0.4$$

$$\cos \theta = 0.4$$

$$\theta = 66.4^\circ$$

Dominant pole

$$K = 23.72$$

$$s = -0.9 + j2.1$$

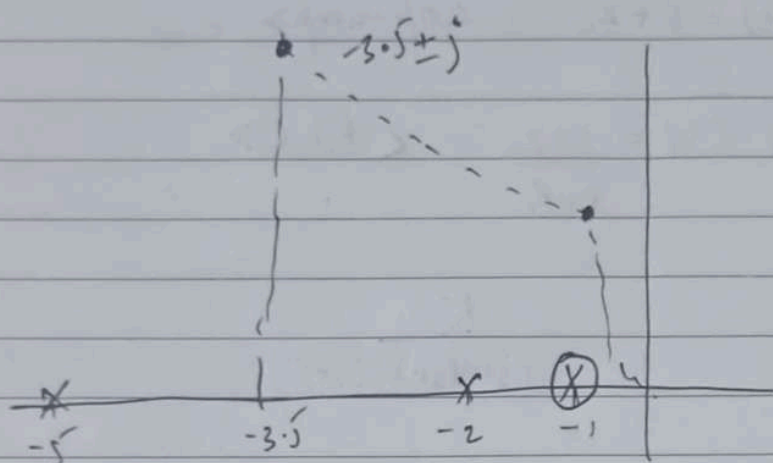
$$T_s = \frac{4}{0.9} = 4.44 \text{ sec}$$

$$T_p \approx \frac{\pi}{2.1} =$$

$$G(s) = \frac{K(s+2)}{(s+1)(s+2)(s+5)} \rightarrow \text{apply the zero.}$$

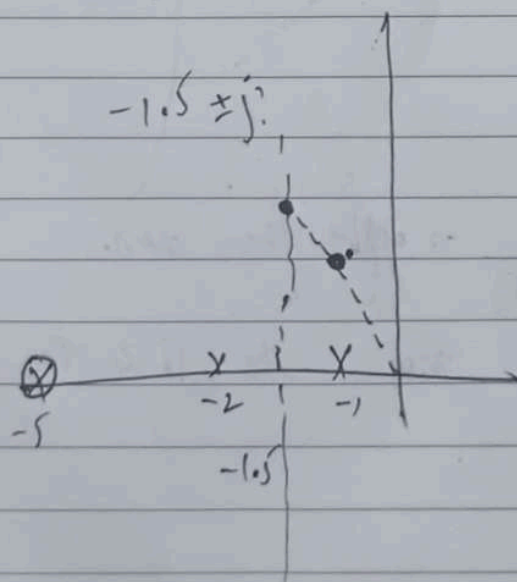
After applying zero the  $T_r$  &  $T_p$  decrease.

$$G(s) = \frac{k(s+1)}{(s+1)(s+2)(s+9)}$$



$$\zeta_s = \frac{4}{3.5}$$

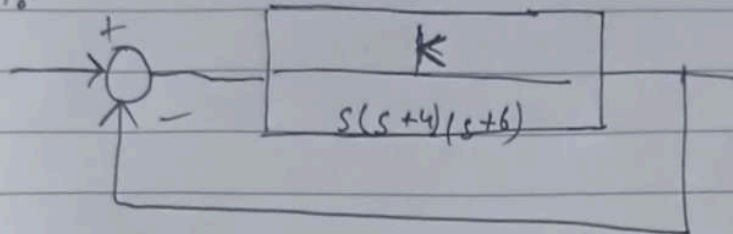
$$G(s) = \frac{k(s+5)}{(s+1)(s+2)(s+5)}$$



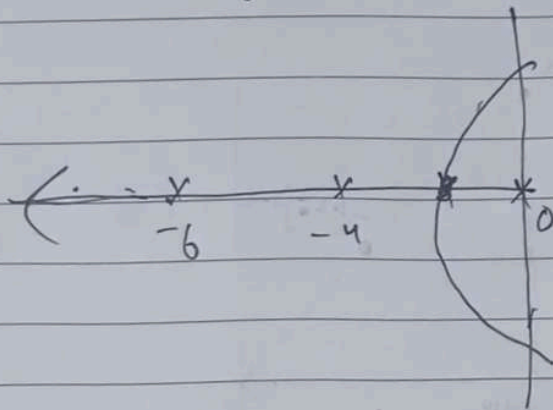
$$\zeta_s = \frac{4}{1.5} \approx 2.667$$

2/00 6 (20)  
 >1-

Question:-



- The system dominant pole occurs at  $K = 43.53$ . Find dominant pole
- Then find  $T_P$ ,  $T_s$  &  $OS = 16\%$ .
- Design a PD controller that maintain  $OS = 16\%$  but reduce  $T_s$  by 3 times.



a)  $1 + G(s)H(s) = -1$

$$\frac{43.35}{s(s+4)(s+6)} = -1$$

$$s(s+4)(s+6)$$

$$s = -1.205 \pm j 2.064, \quad \boxed{-7.59} \times$$

b)

$$T_P = \frac{4}{1.205} = 3.320 \text{ sec}$$

$$T_P = \frac{\pi}{2.064} = 1.522 \text{ sec}$$

$$\tan \theta = \left( \frac{2.064}{1.205} \right)$$

$$\theta = \tan^{-1} \left( \frac{2.064}{1.205} \right)$$

$$\theta = 59.72^\circ$$

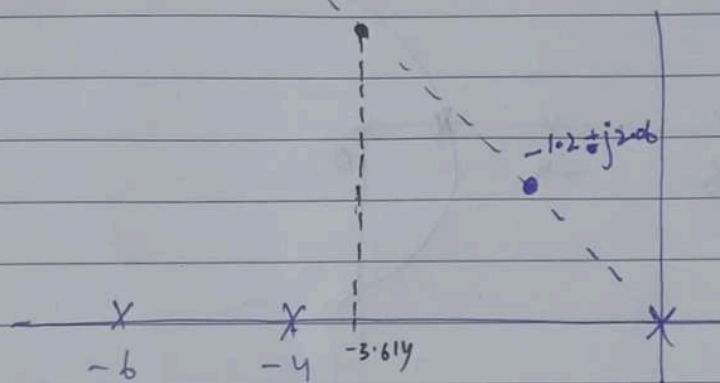
$$\xi = \cos \theta$$

$$\xi = 0.50$$

$$\%OS = e^{\frac{-\pi \cdot 0.5}{\sqrt{1-0.5^2}}}$$

$$\%OS = 0.16 \times 100\%$$

$$\%OS = 16\%$$



$$\text{New } T_s' = \frac{3.32}{3} \approx 1.106$$

$$\text{new real part} = T_s' = \frac{4}{\text{Real part}}$$

$$\text{Real part} = \frac{4}{T_s'} = \frac{4}{1.106} \\ = 3.614$$

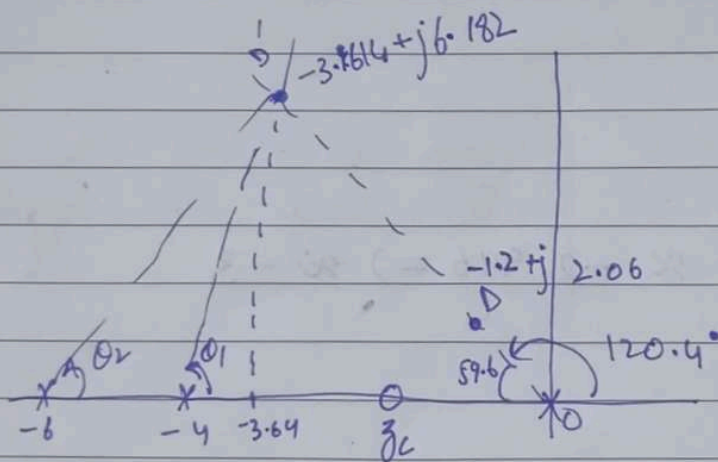


$$\tan(59.6^\circ) = \frac{P}{3.614}$$

$$P = 6.159$$

New dominant pole

$s = -3.614 + j6.159$  → This is not passing through the root locus. so we add "zero" to remove this problem.



$$-\phi_1 - \phi_2 + \phi_2 - \phi_3 = 180^\circ$$

$$-68.87 - 86.427 - 68.89 + \phi_2 - 120.4 = 180^\circ$$

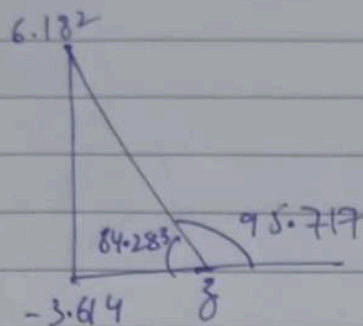
$$+\phi_2 - 275.717 = 180^\circ$$

$$\phi_2 = 180 + 275.717$$

$$\phi_2 = 455.717^\circ$$

$$\text{basic angle} = 455.717 - 360$$

$$\boxed{\phi_2 = 95.717}$$



$$\tan \theta = \frac{p}{b}$$

$$b = \frac{p}{\tan \theta}$$

$$b = \frac{6.182}{\tan(84.283^\circ)}$$

$$\boxed{b = 0.618}$$

$$Z = 0.618 - 3.614 \approx -2.996 \Rightarrow x \approx -3$$

C(S)

U(S)

$$C(s) = K(s+z)$$

$$\frac{U(s)}{E(s)} = (s+z) K$$

$$U(s) = K(s E(s)) + K_z(E(s))$$

$$u(t) = \underbrace{K \dot{e}(t)}_{D \rightarrow K_d} + \underbrace{K_z(e(t))}_{P \rightarrow K_p}$$

$$u(t) = K_p e(t) + K_d \dot{e}(t)$$

---

When we have to design PID controller we have to design PD first then PI controller.

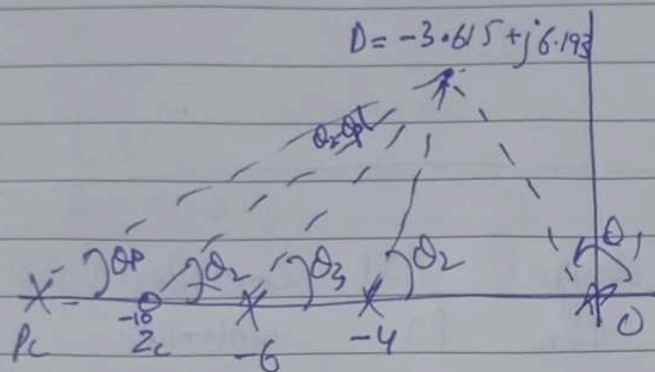


## Lead compensator

$$1) C(s) = \frac{s+2c}{s+p_c}$$

$$S(s+6)(s+4)$$

- 2) Implemented using RLC
- 3) Lesser variant of PD
- 4) Used to improve transient response.



First insert zero then pole after plotting all the zero and poles given in question.

$$-\phi_1 + \theta_2 + \theta_3 - \theta_2 - \theta_1 = 180$$

$$\theta_3 - \phi_1 = 12.3 \Rightarrow \text{assume}$$

$$\text{assume } z_c = -10$$

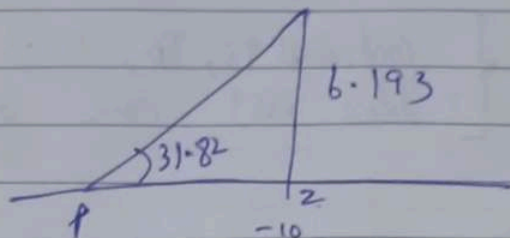
$$\tan \theta_2 = \frac{6.194}{10 - 3.615}$$

$$\theta_2 = \tan^{-1} \left( \frac{6.194}{6.385} \right)$$

$$\theta_2 = 44.125^\circ$$

$$\theta_2 - \phi_1 = 12.3^\circ$$

$$\phi_1 = 31.82^\circ$$



$$\tan(31.82^\circ) = \frac{6.193}{b}$$

$$b = \frac{6.193}{\tan(31.82^\circ)}$$

$$b = 9.98$$

$$p_c = -b - 3.615$$

$$p_c = -9.98 - 3.615$$

$$p_c = -13.15$$

$$C(s) = \frac{s+10}{s+13.15}$$

PI-controller

[ improves  $e_{ss}$   
maintain  $T_p, T_s, OS\%$  ]

$$C(s) = \frac{s+z_c}{s}$$

- 1) Identify the dominant pole (D)
- 2) Make  $e_{ss} = 0$  without compromising on  $T_s, T_p, OS$
- 3) Add a pole at origin to increase system type & ~~make~~ make  $e_{ss} = 0$
- 4) This changes R.L. The new R.L. don't pass through D
- 5) To restore R.L. passing through D, add a zero  $z_c$  - find by  $\angle_{angle} = 180^\circ$

first Lead then Lag }  
 first PD then PI }

PD - controller

{ improves  $T_s$ ,  $T_p$ ,  $OS\%$ .  
 does not care of  $ess$ .

$$C(s) = s + z_c$$

- 1) Identify dominant pole of existing system
- 2) see objectives: required  $T_s$ ,  $T_p$ ,  $OS$  (Don't improve  $ess$ !!)
- 3) Compute location of new dominant pole
- 4) To make sure R.L. passing through the new pole  $\rightarrow$  add a zero ( $z_c$ )
- 5) Find location of  $z_c$  through  $\angle_{angle} = 180^\circ$

PID controller Design

- 1) Find the present dominant pole
- 2) Evaluate the performance ( $T_s$ ,  $T_p$ ,  $OS$ ,  $ess$ ) of the controlled system
- 3) determine how much improvement is needed.
- 4) Design PD controller to meet transient responses. Find  $C(s) = s + z_c$ , & loop gain " $k$ "
- 5) PD controller would improve transient but degrade  $ess$ .
- 6) Design PI controller to make  $ess = 0$  while maintain the transient  $x$ -ties of  $k(u)$ . Find  $\frac{s + z_c}{s}$
- 7) Your PID controller is product of  

$$C(s) = k (s + z_c) \left( \frac{s + z_c}{s} \right)$$



$$= \frac{k(s+z_c)(s+z_c)}{s}$$

$$C_{PID} = k \left[ \frac{s^2 + (z_1 + z_2)s + z_1 z_2}{s} \right]$$

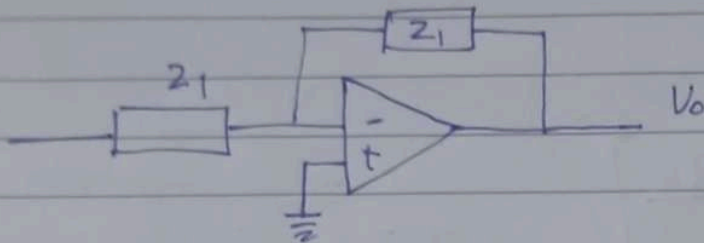
$$C_{PID} = k s + k(z_1 + z_2) + \frac{k z_1 z_2}{s}$$

$$\frac{U(s)}{E(s)} = k s + k(z_1 + z_2) + \frac{k z_1 z_2}{s}$$

$$U = k \left( \int E(s) \right) + k(z_1 + z_2)(E(s)) + k z_1 z_2 \left( \frac{E(s)}{s} \right)$$

$$\Rightarrow U(t) = \underbrace{k(z_1 + z_2)}_{K_P} e(t) + \underbrace{k}_{K_D} \dot{e}(t) + \underbrace{(k z_1 z_2)}_{K_I} \int e(t) dt$$

Controller circuit



$$T(s) = \frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

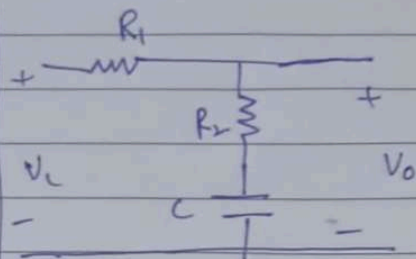
| Function     | $Z_1(s)$        | $Z_2(s)$            | $T_s = -\frac{Z_2}{Z_1}$                                                                                       |
|--------------|-----------------|---------------------|----------------------------------------------------------------------------------------------------------------|
| proportional | $\frac{R}{s}$   | $\frac{R_2}{s}$     | $-R_2/R_1$                                                                                                     |
| Integral     | $\frac{R_1}{s}$ | $\frac{C}{s}$       | $-\frac{1}{R_1 C}$                                                                                             |
| Derivative   | $\frac{C}{s}$   | $\frac{R}{s}$       | $-(R C) s$                                                                                                     |
| PI           | $\frac{R_1}{s}$ | $\frac{R_2 C}{s}$   | $-\frac{R_2}{R_1} \left( \frac{s + \frac{1}{R_2 C}}{s} \right)$                                                |
| PD           | $\frac{R_1}{s}$ | $\frac{R_2}{s}$     | $-R_2 C \left( 1 + \frac{1}{R_1 C} \right)$                                                                    |
| PID          | $\frac{R_1}{s}$ | $\frac{R_2 C_2}{s}$ | $-\left[ \left( \frac{R_2}{R_1} + \frac{C_1}{C_2} \right) + R_2 C_1 s + \frac{1}{R_1 C_2} \frac{1}{s} \right]$ |

Compensator

RLC network

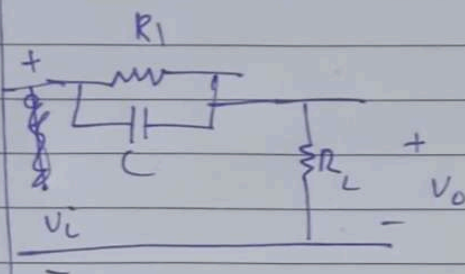
$T(s)$

LAG



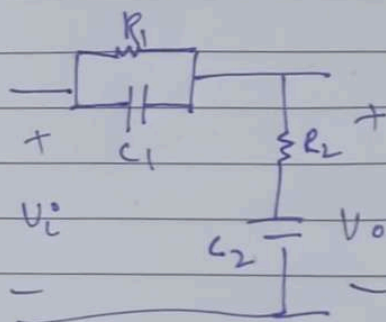
$$\frac{R_2}{R_1 + R_2} \left( \frac{s + 1/R_2 C}{s + \frac{1}{(R_1 + R_2)C}} \right)$$

LEAD



$$\frac{s + 1/R_1 C}{s + \frac{1}{R_2 C} + \frac{1}{R_1 C}}$$

LEAD-LAG



$$\frac{(s + 1/R_1 C) (s + 1/R_2 C)}{s^2 \left( \frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_2 C_1} \right) s + \frac{1}{R_1 R_2 C_1 C_2}}$$